

Integer multiplication below $n \log n$

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Modified research draft; not an OpenAI release. This derivative incorporates Douglas Colkitt’s compact-control construction and retained project refinements, prepared with assistance from OpenAI Codex, and a compressed complex network and faster Gaussian resampling prepared with assistance from Claude (Anthropic). The ternary five-subset construction, star resynthesis and paired complex producer are contributed by Zhihao Chen (jacklightChen), with substantial GPT-6 Astra assistance (as identified by the contributor). The present extension batches controlled rational projector blocks and whole complex residuals, supplies integer-width recursion with rows, and uses a dependency-path precision guard. The bulk-recursion contribution is by icekylinx, with substantial OpenAI GPT-6 Astra assistance; review corrections and integration used OpenAI Codex assistance. The dimension-thirty refinement is by Rohan Arun with substantial OpenAI Codex assistance, building on the preceding attributed work. It changes the bit producer dimension and reuses the complex network. The present combination uses auxiliary source frames and a third complex class contributed by eumemic with Claude (Anthropic) assistance in PR 13, plus two additional data-corner blocks developed by Rohan Arun with substantial OpenAI Codex assistance. The revised bound is conditional on the original manuscript’s retained algorithmic and analytic interfaces and the new written arguments. These modifications have not received independent mathematical review or formal verification; no endorsement by OpenAI or Anthropic is implied.

Abstract

We give a deterministic algorithm that multiplies two n -bit integers in $O(n(\lg n)^{1-\kappa})$ worst-case time, with $\kappa = 90799/100000000000 > 2^{-23}$, on one fixed finite-alphabet Turing machine with a fixed finite number of one-dimensional tapes. The algorithm is exact for every input length and disproves the $n \log n$ optimality conjecture of Schönhage and Strassen in this model.

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1 Introduction

Schönhage and Strassen proved in 1971 that two n -bit integers can be multiplied in $O(n \log n \log \log n)$ time on a multitape Turing machine, and proposed $n \log n$ as the optimal order of growth [25, Section 1]. After a sequence of improvements, Harvey and van der Hoeven reached the unconditional $O(n \log n)$ upper bound [15]. We obtain a strict power saving in the logarithmic factor, in the same fixed finite-tape model. The algorithm uses linear combinations of intermediate data to reduce the cost of both address rearrangement and Fourier-transform layers.

For a binary string x of length n , let $\text{val}(x)$ be its nonnegative integer value, with the leftmost bit most significant. For $0 \leq z < 2^k$, let $\text{bin}_k(z)$ denote its binary representation padded on the left to length k . We consider exact multiplication: on input $x\#y$, with $x, y \in \{0, 1\}^n$, the output must be $\text{bin}_{2n}(\text{val}(x) \text{val}(y))$. Write $\lg n = \max\{\lceil \log_2 n \rceil, 1\}$.

Theorem 1. *There is one deterministic Turing machine A , with a fixed finite alphabet and a fixed finite number of one-dimensional tapes, that computes this exact product for every $n \geq 1$ and every pair of n -bit inputs. Its worst-case running time satisfies*

$$T_A(n) = O\left(n(\lg n)^{1-\kappa}\right), \quad \kappa = 90799/100000000000 > 2^{-23}.$$

The bound is asymptotic through all input lengths. The constants and thresholds in the construction are extremely large; the algorithm is intended to establish a bit-complexity bound. The theorem refutes an eventual $\Omega(n \log n)$ lower bound for every fixed multiplication machine in this model.

1.1 The two tape operations

Splitting the inputs into radix digits reduces multiplication to convolution of short integer coefficients. Our starting point is the Gaussian-resampling reduction of Harvey and van der Hoeven [15]: it replaces the Fourier transforms needed for this convolution by multidimensional transforms whose axis lengths are powers of two. After a further conversion to convolution, one coordinate is stored as the coefficient list of a polynomial modulo $y^r + 1$, with r a power of two. Powers of y then serve as roots of unity in the remaining axes, and multiplication by such a root is a signed shift of the coefficient list. These are the synthetic polynomial transforms of Nussbaumer and Quandalle and Nussbaumer [21, 20].

In this representation, the transform calculation consists of address rearrangements, signed shifts, and two-point butterfly operations. Scanning an array of $\Theta(n)$ stored bits at each of $\Theta(\log n)$ transform levels already costs $\Theta(n \log n)$. We therefore need savings in both movement and arithmetic. Two tape procedures supply them; in both bounds, V denotes the stored bit volume.

The first procedure interchanges two disjoint contiguous address fields of u bits in an array whose addresses form a complete Cartesian product. Its cost is $O(Vu^\tau)$ for a fixed $\tau < 1$. Although its final effect is an exact permutation, its intermediate steps form pointwise XOR combinations of data. The final permutation is exact for arbitrary initial auxiliary values.

The second procedure applies

$$H_0(u, v) = ((u + v)/2, (u - v)/2)$$

simultaneously on selected coordinate bits of the numerical arrays. Each record is a polynomial with complex fixed-point coefficients, and H_0 acts coefficientwise. On the layouts used in the multiplication algorithm, one selects the same bit position in each of at most d equal-width address chunks, and the cost is $O(Vd^{\lambda'})$ for a fixed $\lambda' < 1$. Intermediate values are exact Gaussian dyadics,

whose real and imaginary parts have power-of-two denominators. A single truncation follows the completed contractive layer. Section 4 proves the address bound, and Section 5 gives the numerical layout, precision hypotheses, and layer bound.

1.2 Linear networks and the recursive saving

Each procedure arises from its own fixed linear network that exchanges two banks of values, with prescribed signs, and restores every auxiliary value. A wire's fixed place in the network is called its role. To use the network on an array, regard a row as the complete ordered suffix, including all active address fields, at a fixed prefix and row index. Distribute rows cyclically among the W roles, padding the row count for each prefix to a multiple of W . Let V now denote the node volume after padding. All role streams have the same local address set and each contains V/W bits. A scalar gate is then applied pointwise to matching addresses on the roles that it touches.

The representation of a role's array may change along the network. At a source, gate, or sink v , choose an invertible linear operator D_v on the space of arrays indexed by the common address set. We call D_v its frame: the stored array is $D_v g$ when the logical array is g . All incidences of one gate have the same frame. Along an edge from u to v , the operator $D_v D_u^{-1}$ changes the stored array from $D_u g$ to $D_v g$. At a pointwise linear gate G , the common frame commutes with the gate:

$$G(D_v g_1, \dots, D_v g_t) = D_v G(g_1, \dots, g_t).$$

The supplied input needs no preliminary change of frame: its logical interpretation is defined by applying the inverse input frame. Thus the intermediate frames cancel, and a route from input role w to output role $\rho(w)$ applies $D_{\text{sink}(\rho(w))} D_{\text{source}(w)}^{-1}$, together with its prescribed scalar sign. Undoing the role exchange and those signs leaves the endpoint operator on every role array, including the roles that carried auxiliary scalar values.

For the bit procedure, that endpoint operator is an address shear: two fields, written as vectors H, D over a common residue ring, are updated by $(H, D) \mapsto (H + D, D)$. Two further linear-cost address updates complete their interchange. For the numerical procedure, the endpoint operator becomes the simultaneous H_0 layer after diagonal phase operations and a dyadic scale. The intermediate frames are chosen so that every edge can be implemented by smaller calls to the same procedure.

The finite construction in Section 3 attaches subspaces of a fixed m -dimensional space to its gates and terminals. The subspaces at adjacent vertices are nested. Their dimension changes count the smaller operations needed on an edge, with an additional source contribution in the bit construction. The construction uses three-element subsets: gather and scatter create coefficients depending on intersection size, and auxiliary wires cancel the unwanted coefficients. Orthogonality of the corresponding indicator vectors permits the subspace assignments. The decisive property is that the total number s of smaller calls satisfies $s < Wm$.

A recursive call at parameter mf therefore uses s calls at parameter f , each on a stream of volume V/W . The parameter counts field digits for address interchange and selected axes for a numerical layer. After dividing cost by volume, multiplying the parameter by m increases the recursive contribution by $s/W < m$. This strict inequality is the source of the power saving. The later tape and arithmetic estimates preserve it, including the work of changing numerical address coordinates. Those changes are implemented by whole-chunk moves and controlled shifts, followed by a deterministic correction of the explicitly identified exceptional addresses.

1.3 From transforms to the exact product

The remaining proof places these procedures inside the multiplication reduction and accounts for every change of representation. First, the radix-digit coefficient list is placed on a product of distinct prime cyclic axes by a Chinese-remainder map that preserves convolution. Section 9 implements this map through triangular controlled shifts and charged axis moves. Gaussian resampling then expresses each source Fourier transform through a power-of-two tensor transform. Section 8 retains the source and target coordinate permutations in the transform identity, so they need not be executed as separate full sorts.

Next, Bluestein’s chirp identity converts the middle transform, including its target permutation, into normalized cyclic convolution. A phase twist changes the last cyclic coordinate into a polynomial modulo $y^r + 1$; its negative wrap gives precisely the original cyclic wrap after the twist is removed. Sections 6 and 7 compute convolution on the remaining axes over this polynomial ring. The two forward synthetic transforms retain the same frequency order through their pointwise product, and the opposite transform consumes that order. Their polynomial products use signed packing and the established $O(n \log n)$ multiplier as an ordinary subroutine.

Finally, the source permutation retained during resampling cancels across the two forward source transforms, their pointwise product, and the opposite-sign transform. Section 9 removes the known normalizations and digit scales, proves that the recovered coefficient error is strictly below $1/2$, and propagates carries to obtain the exact binary product. A compatible choice of rational parameters makes all terms in its cost table fit Theorem 1. Figure 1 summarizes these reductions.

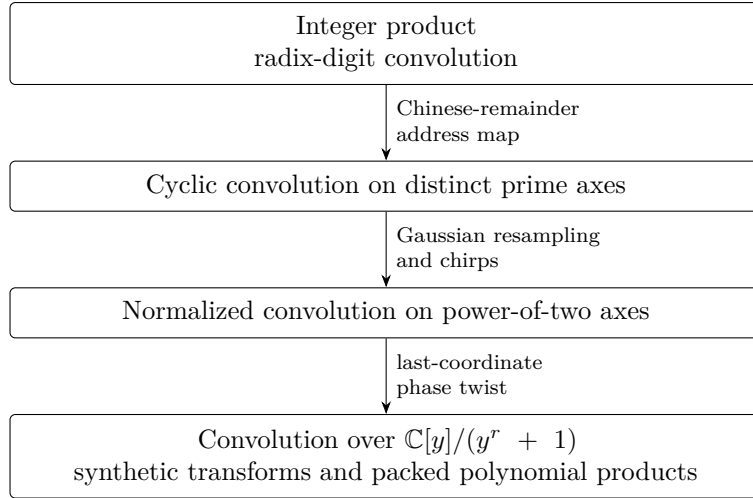


Figure 1: The reductions connecting the tape procedures to multiplication. The arrows describe changes of problem and representation. Fast address interchanges organize the layouts, and simultaneous butterfly layers accelerate the synthetic transforms. Solving the final polynomial-ring convolution evaluates the transforms needed in the preceding stages. Known rescalings, rounding, and carry propagation then recover the exact integer product.

1.4 Other arithmetic operations

Corollary 2 (Exact division and integer square root). *There are deterministic machines with fixed finite alphabets and fixed finite numbers of one-dimensional tapes performing the following tasks in $O(n(\lg n)^{1-\kappa})$ worst-case time, for every $n \geq 1$. For $x, y \in \{0, 1\}^n$, put $a = \text{val}(x)$ and $b = \text{val}(y)$.*

If $b > 0$, the division machine takes $x \# y$ and returns $\text{bin}_n(q) \# \text{bin}_n(r)$, where $q = \lfloor a/b \rfloor$ and $r = a - bq$. The square-root machine takes x and returns $\text{bin}_n(\lfloor \sqrt{a} \rfloor)$. Leading zeroes are allowed, and κ is as in Theorem 1.

The classical Newton reductions described by Brent [5, Lemmas 2.1–2.4] compute reciprocal and square-root approximations at geometrically increasing precisions. Their multiplication costs are bounded by a geometric sum controlled by the last precision, and exact remainder and square tests recover the required integer answers. Section 10 gives the complete fixed-tape implementation.

1.5 Historical context and inherited methods

Karatsuba’s recursive construction, published jointly with Ofman in 1962, gave a binary multiplication circuit of size $O(n^{\log_2 3})$ [18]. Toom’s subsequent use of polynomial evaluation and interpolation gave circuit size $n \exp(O(\sqrt{\log n}))$, and hence $O(n^{1+\eta})$ for every fixed $\eta > 0$ [26]. These early results were formulated in terms of automata and logical networks. Cook adapted Toom’s method to the Turing-machine setting; the contemporary account of Schönhage and Strassen credits this adaptation explicitly [25, Section 1, p. 282].

In the multitape Turing model, Schönhage and Strassen proved the bound $O(n \log n \log \log n)$ in 1971 [25]. Fürer improved it to $n \log n 2^{O(\log^* n)}$ in 2007, with a full journal account in 2009 [10, 11]; here $\log^* n$ counts iterated logarithms until a constant threshold is reached. De, Kurur, Saha and Saptharishi developed a modular-arithmetic route to the same scale [7]. Later improvements made the exponential factor $8^{\log^* n}$ and then $4^{\log^* n}$ [17, 14]. Harvey and van der Hoeven proved the unconditional $O(n \log n)$ bound [15].

The multitape bounds just cited concern that fixed model. Schönhage obtained linear-time multiplication on storage-modification machines [24]. Groff later gave a smaller-than- $\log n$ factor in a unit-cost RAM model with constant-time arbitrary memory access, after polynomial preprocessing whose cost is excluded from that bound [13, Sections 2 and 4]. Neither result establishes the bound in the fixed finite-tape model of Theorem 1.

The transform methods behind these bounds also underlie the present construction. Schönhage–Strassen multiplication uses roots of unity in Fermat-type residue rings, where certain multiplications become shifts. Fürer combines cheap polynomial roots of unity with more general roots, reducing the frequency of expensive multiplications. Harvey and van der Hoeven use Gaussian resampling to pass to multidimensional transforms with power-of-two dimensions, then evaluate them using polynomial transforms developed by Nussbaumer and Quandalle and subsequently by Nussbaumer [21, 20, 15]. The Gaussian-gridding work of Dutt and Rokhlin is an earlier conceptual precursor [8]; Harvey and van der Hoeven discuss the connection in [15, Section 4.4.3]. The quantitative resampling and bit-cost estimates used here are those of the latter paper.

More specifically, the proof adapts Harvey and van der Hoeven’s Gaussian-resampling identity, contraction bounds, and short-convolution procedures [15, Theorem 4.2, Proposition 4.7 and Lemmas 4.8–4.12]. Their framework already permits arbitrary dimension. Here the source and target permutations remain in the transform identity, while the faster tape procedures perform the required axis movements. Bluestein’s conversion to convolution [4] and the synthetic-root convolution algorithm are retained; the latter follows the polynomial-transform construction in [15, Section 2.4 and Section 3]. Signed coefficient packing is a form of Kronecker substitution, with the fixed-tape version of [15, Lemma 2.5] as a direct antecedent. Calls to the established $O(n \log n)$ multiplier are ordinary subroutine calls; they do not invoke the improved theorem recursively.

1.6 The finite-network ingredients

The subset-intersection construction in Section 3 has an earlier source outside multiplication. Alon’s presentation of the Frankl–Wilson construction [9] gives low-degree representations of complementary intersection graphs over different fields [2, Section 3, remark after the proof of Theorem 1.1]. At the prime $p = 2$, its subsets have size three and the two polynomial evaluations are the intersection count modulo two and the intersection count minus one. These are the two pairing formulas used here. We change the ground-set size and express the second pairing through a rational bilinear form.

The gather and scatter operations, together with the side wires, have the form of linear index-code decoding: a low-rank linear summary gives the wanted coordinate plus contributions from known neighboring coordinates, and those contributions are subtracted. Bar-Yossef, Birk, Jayram and Kol establish the fitting-matrix formulation and this decoder [3, Section 3, proof of Theorem 5]. Lubetzky and Stav treat decoding over other fields and use related intersection representations to separate their possible ranks [19, Proposition 2.1 and Section 2.3]. These comparisons explain the algebraic role of the two fields here; they do not supply the subspace labels on the subsequent network.

Restoring scratch registers with arbitrary initial contents also has a precise precedent. The transparent-computation construction of Buhrman, Cleve, Koucký, Loff and Speelman subtracts an initial register value, performs a computation, adds the resulting value, and reverses the computation [6, Section 3]. Our scratch schedule is the linear-readout version of that construction. The rank reduction used for address shears is a singular-matrix Bruhat factorization: after reversing the coordinate order, Grigor’ev’s upper-triangular factors become the two lower-triangular factors needed here [12, Appendix A.2, Proposition 14(c)].

The finite networks and their labels are proved directly in Section 3. The additional requirements are the three-stage bank exchange, nested subspaces with a strict total rank or dimension deficit, and endpoint frames acting on every role. The tape implementations then turn that deficit into the volume-normalized recurrences of Sections 4 and 5, including the costs of addressing, padding, coefficient growth, and finite precision. The cited decoding, restoration, and factorization results provide ingredients; these further properties are established in the present proof.

1.7 Lower-bound scope and matrix transposition

Schönhage and Strassen conjectured $n \log n$ as the optimal order of multiplication cost [25, Section 1]. In the present fixed-tape model, an eventual lower bound for every multiplication machine means that, for each fixed machine A , there are constants $c_A > 0$ and N_A such that $T_A(n) \geq c_A n \lg n$ whenever $n \geq N_A$. Its negation requires one machine satisfying $\liminf_{n \rightarrow \infty} T_A(n)/(n \lg n) = 0$. The power saving proved here gives a limit of zero through all input lengths, which is a stronger conclusion.

Harvey and van der Hoeven consider a binary transposition machine supplied an integer $m \geq 1$ and a $k \times k$ binary matrix, where $k = \lfloor \sqrt{m} \rfloor$. They proved that if every fixed such machine has worst-case cost $\Omega(m \lg m)$ through all sufficiently large supplied values of m , then every fixed multiplication machine has the corresponding eventual lower bound [16, Theorem 1.2]. Combining their quantitative reduction [16, Corollary 6.2] with Theorem 1 gives a deterministic fixed-tape algorithm that transposes a row-major $k \times k$ binary matrix in $O(k^2 (\lg k)^{1-\kappa})$ time, with κ as in Theorem 1. Section 11 proves the more general rectangular-matrix bound.

This comparison concerns unrestricted computation on a fixed number of one-dimensional tapes. Our address procedure forms XOR combinations of intermediate data, even though its final effect is

a permutation. It therefore lies outside models restricted to moving data without such computations; see [16, Section 1.4] for the distinction.

1.8 Companion application to exact Fourier transforms

The complex phase network of Proposition 8 is also used in the companion manuscript [22, Section 1.1, Theorem 1.1 and Corollary 1.2]. That manuscript gives a single algorithm for the exact discrete Fourier transform at every positive transform length n . As $n \rightarrow \infty$, its cost is $O(n(\log n)^{1-10^{-13}}) = o(n \log n)$.

The cost is measured in an exact complex-register model that assigns unit cost to exact complex field operations. The algorithm accepts every $x \in \mathbb{C}^n$, with no bound on its entries or on intermediate complex magnitudes. Its computation on the input uses only addition, subtraction, and multiplication by prepared input-independent scalars. The magnitudes of those scalar coefficients are unrestricted as well.

Integer operations and random access on $O(\log(n+2))$ -bit words have unit cost, with a fixed number of words for each address or integer. The model supplies one specified Fourier root ζ_{D_*} of explicitly computable order $D_* < 1024n^3$. Scalar preparation, schedule construction, and index work are charged. This is a separate application of the finite network, with its own uniform array implementation. It is neither a bit-complexity consequence nor a stable floating-point FFT consequence of the fixed-tape multiplication theorem, Theorem 1.

2 The machine model and elementary streams

A machine has one fixed finite alphabet and one fixed finite number of one-dimensional tapes. A step reads and writes at most one cell on each tape and moves each head by at most one cell. Fixed tracks and finitely many work tapes change time by a fixed factor. We distinguish moving one digit in a fixed radix, which needs a bounded number of scans, from interchanging fields whose numbers of digits grow. The latter permutation will be realized in Section 4 by computing XOR combinations of intermediate data streams. The present section supplies the elementary scans it uses. Every scan includes its counters and the return of its tape heads.

2.1 Array order and elementary scans

For a positive integer a write $[a] = \{0, \dots, a-1\}$. An array with address set $[a_1] \times \dots \times [a_t]$ is stored in lexicographic order, with the first coordinate most significant. Its entries are bits unless a record width is specified. A trailing record of B bits can equivalently be regarded as a final address field $[B]$. An address is implicit in an entry's position in this ordered stream; it is not stored beside every data bit. The *logical volume* V is the number of stored data bits. Work tapes and duplicate streams are charged in time but do not increase V in recurrences. All the fields discussed below are nonempty. A call with an empty array is handled directly.

An address permutation f moves the entry at x to $f(x)$. Thus its action on an array A is $(fA)(f(x)) = A(x)$. Adjacent fields which are irrelevant to an operation may be replaced by a single field whose length is their product. In particular, the elementary procedures below have a fixed number of formal address fields even when a surrounding algorithm has a growing list of coordinates. Lengths and positions are supplied in canonical binary on descriptor tapes; no additional tape head is attached to an address coordinate. Each elementary call receives a fixed number of collapsed canonical lengths. Constructing them from a longer surrounding list is work

of the caller; recursive calls below construct their own collapsed descriptors within their stated bounds.

When one address field is designated as a row field, collapse the fields before it to a prefix $[P]$ and those after it to a suffix $[B]$. If the row field has range $[R]$, a *row* at prefix p and row index r is the complete consecutive block $\{p\} \times \{r\} \times [B]$. Every row therefore contains B bits in the same suffix order.

For headers that concatenate a list of positive integer parameters, we use the self-delimiting code

$$b_v = \lfloor \log_2 v \rfloor + 1, \quad \Gamma(v) = 0^{b_v-1} \text{bin}_{b_v}(v), \quad |\Gamma(v)| = 2b_v - 1 \quad (v \geq 1). \quad (1)$$

The initial zeroes specify how many further bits follow the first one, so concatenated codes can be read in order. We use this code when an interface specifies a header; elementary calls still receive the fixed number of canonical binary lengths described above.

Lemma 3 (Elementary stream operations). *Fix a radix $q \geq 2$. Each of the following operations on an array of volume V takes $O(V)$ steps on a fixed number of one-dimensional tapes:*

1. *any fixed permutation of the pair of digit values at two specified radix- q positions, whether or not the positions are adjacent;*
2. *moving one specified radix- q digit to another position, preserving the order of the intervening positions;*
3. *splitting on a specified radix- q digit, applying a fixed pointwise map to aligned streams, and merging them back;*
4. *for positive integers P, R, B , within each prefix of an array on $[P] \times [R] \times [B]$, splitting whole rows cyclically into a fixed number W of streams, and, when W divides R , merging those streams in any fixed order of the row classes.*

The bounds include shape preparation, counter initialization and resets, and returning or clearing the visited portions of work tapes. Their constants may depend on the fixed radix, the fixed pointwise map, and the fixed number of row classes.

Proof. For the first operation split into q^2 streams according to the two specified digits. With these digits fixed, the order of all remaining digits is unchanged by the permutation. Enumerate output addresses in their required order and read the stream prescribed by the two preimage digits. For a digit move use q streams instead. The same observation shows that splitting on one digit gives streams with the same remaining address set in the same order. Scan their aligned entries, apply the fixed pointwise map, and merge according to the output digit; this proves the third operation. For the fourth, write $r = Wg + w$, where $0 \leq w < W$, and send the whole row to stream w . If W divides R , each stream has address set $[P] \times [R/W] \times [B]$, indexed by the prefix, g , and the unchanged suffix. Thus the streams have the same shape and each has volume V/W . The inverse merge, with any fixed permutation of the classes, reads them in the specified cyclic order.

Here are the counter details used in these scans and later ones. A binary counter is stored with its head at the least significant end. For a counter of width w , incrementing or decrementing and returning this head over L consecutive counts costs $O(L + w)$: the number of visits to position j is $O(1 + L/2^j)$. The width term is paid when initializing the counter and is absorbed by its complete range scan. A countdown of length L can be initialized, reset, and tested for underflow in $O(\log(2L)) = O(L)$ time. Keep a local copy of its length on a dedicated track, so a reset does not search a larger descriptor. For a nested scan of a fixed number of nonempty ranges, the reset

of a range is charged to the scan of that range. Induction on the number of ranges therefore gives linear total counting cost. A fixed number of empty pieces in a split can be charged to the adjacent nonempty combined range.

Only a fixed number of collapsed lengths are needed. Computing these lengths and the initial counter values by ordinary binary arithmetic takes $O((\log(2V))^C)$ steps for a fixed C . This is $O(V)$ uniformly in $V \geq 1$, after increasing a fixed constant. Each stream is scanned a fixed number of times, and every rewind, copy, or erasure is over a region whose length was charged to that scan. Delimiters and the fixed number of tracks require only a fixed alphabet enlargement. $\square$

The next operations are linear because their target field follows their control fields. This order will be preserved whenever they are invoked.

Lemma 4 (Controlled shifts and fixed rational scalings). *Fix a prime q , a fixed number of control fields, and finitely many rational coefficients whose denominators are prime to q . Whenever a coefficient is used as a unit, also require its numerator to be prime to q . Put $Q = q^b$, $b \geq 1$. Suppose a target field $y \in [Q]$ follows its control fields, each control used in an offset having $O(b)$ digits. An update*

$$y \mapsto y + a(\text{controls}) \pmod{Q}$$

costs $O(V)$ if the offset can be prepared in $b^{O(1)}$ time from the fixed number of control values. In particular, this holds for a fixed rational linear combination of them, specialized modulo Q . Multiplication of y by any of the specified rational units, and its inverse, also costs $O(V)$. Arbitrarily long intervening spectator fields and suffix records are permitted.

Proof. Fix all fields before the target, and let the suffix length be B bits. The resulting fiber consists of Q consecutive blocks of length B . For an offset $a \in [Q]$, split these blocks at $Q - a$. Append the two pieces to two tapes, doing so successively for all prefix fibers. In the merge read the second piece and then the first piece for each prefix. The pieces have become the two consecutive ranges $[0, a)$ and $[a, Q)$ of the new target coordinate. Recompute a when starting that prefix in the merge. Initializing these two piece lengths and returning the heads of the relevant control counters is included in the $b^{O(1)}$ offset work. Since

$$b^C = O(q^b) = O(Q)$$

for each fixed C , this work is absorbed by the $QB \geq Q$ bits of the fiber. Unrelated spectator counters are incremented by ripple counting; their entire contents are not consulted when computing the offset.

For positive fixed integer c prime to q , the permutation $y \mapsto cy \pmod{Q}$ has c increasing pieces, classified by

$$j = \left\lfloor \frac{cy}{Q} \right\rfloor, \quad 0 \leq j < c.$$

Prepare their endpoints once for this Q , split each fiber into these c streams, and enumerate the output coordinate $y' = 0, \dots, Q - 1$. The required stream is the unique $j \in [c]$ for which $y' + jQ \equiv 0 \pmod{c}$. Uniqueness follows from $\gcd(Q, c) = 1$; the corresponding $y = (y' + jQ)/c$ lies in $[Q]$. The stream choice is maintained by a finite residue counter. Reversing the roles of this split and merge implements multiplication by c^{-1} modulo Q .

Negation leaves the block at zero fixed and reverses the order of the other $Q - 1$ blocks. Push these blocks consecutively to a temporary tape. Popping gives their order reversed, but also reverses the bits in each block. Reverse each popped block on a second temporary tape before writing it, restoring its internal order. Every bit is moved a bounded number of times. The same procedure

is repeated for successive fibers; the temporary stack is emptied and its head returned locally. A signed rational unit is a composition of these integer multiplications, inverse multiplications, and possibly negation. Its description is fixed.

The nested scan and reset analysis of Lemma 3 applies to all these operations. In particular, copying a suffix-length counter at every target position costs $O(\log(2B)) \leq O(B)$ at that position. It does not introduce a factor $\log V$ per bit. $\square$

When coefficients have a denominator d prime to q , their reductions modulo q^b can be computed by the extended Euclidean algorithm using ordinary tape arithmetic. The dimension and all original coefficients are fixed, so these preparations have polynomial cost in b . Nothing in Lemma 4 assumes unit-cost arithmetic.

For later use, fixed-width signed fixed-point binary addition, subtraction, sign change, and multiplication or division by a fixed power of two are streaming operations of linear bit cost, provided enough integer and fractional guard bits have been allocated for exact results. If an input record is most-significant-bit first, reverse it onto the arithmetic work tape. With least significant bits first on that tape, propagate carries using finite control. Reverse the completed record if its required external order is opposite. This convention separates the order of bits inside a record from the lexicographic order of records. Each required reversal is a linear pass and is charged.

3 Two finite networks with a rank saving

We construct two fixed linear networks that exchange two banks of values while restoring every auxiliary value. When the values are replaced by arrays, a common representation on all wires at a gate lets that gate act pointwise without disturbing the representation. We explain this invariant before constructing the subspace labels that make changes of representation cheap. The first network uses bit values and rational subspaces; the second uses Gaussian dyadic values and binary subspaces. The scalar domain specifies the values combined at a gate, while the subspace labels are used to construct operators on each wire's address-indexed array.

Put

$$h = 50, \quad v = \binom{h}{3} = 19600, \quad N = v^3 = 7529536000000, \quad m = h^3 = 125000.$$

Let $\mathcal{T}$ be the set of three-element subsets of $[h]$. There are two data banks X_a, Y_a , indexed by $a = (a_1, a_2, a_3) \in \mathcal{T}^3$. A wire carries one scalar. A gate is a fixed linear map on a specified set of wires; untouched wires retain their values. The vertices on a wire are its source, the gates that touch it in time order, and its sink; its edges join consecutive vertices. We call the name of a wire its *role*, so that we can distinguish a fixed place in the network from the value currently carried there.

3.1 Scalar operations and restored auxiliary values

There are three stages. At stage j , fix the other two coordinates and operate on the v values in each bank obtained by varying coordinate j . Thus there are v^2 separate invocations at each stage. Each such invocation has its own auxiliary wires, distinct from all other invocations, including those in other stages. There are

$$I = 3v^2 = 1152480000$$

invocations. In a forward invocation, write x_T for the data used as controls and y_S for the data to be updated; $S, T \in \mathcal{T}$ are their varying j th coordinates. These are the logical source and target

banks of the invocation. For each ordered pair (S, T) that is neighboring under the rule below, there is an auxiliary wire A_{ST} . There are also central wires C_i , $i \in [h]$, and in the second construction a wire C_* . We call all these auxiliary wires *scratch wires*; their initial values may be arbitrary.

The following maps specify the construction.

- In the bit construction, scalars lie in $\mathbb{F}_2$, and S, T are neighbors when $|S \cap T| = 1$. The copy map adds x_T to A_{ST} for each neighbor pair. The side injection adds $\sum_{T: |S \cap T|=1} A_{ST}$ to y_S . The gather adds $\sum_{T \ni i} x_T$ to C_i , and the scatter adds $\sum_{i \in S} C_i$ to y_S .
- In the complex construction, scalars lie in $\mathbb{Z}[i, 1/2]$, and neighbors are distinct triples with even intersection. Copy is unchanged. Side injection adds

$$- \sum_{T \text{ neighboring } S} \frac{|S \cap T| - 1}{2} A_{ST}$$

to y_S . Gather adds $\sum_{T \ni i} x_T$ to C_i and $\sum_T x_T$ to C_* . Scatter adds $(\sum_{i \in S} C_i - C_*)/2$ to y_S .

Each update leaves its control wires unchanged. The first two rows of the following schedule remove the effect of the old scratch values; the next four introduce the source values and add their effect to the target; the last two restore the scratch values. This cancellation of arbitrary initial scratch values is a linear instance of the restoration construction for transparent computation in Buhrman, Cleve, Koucký, Loff and Speelman [6, Section 3]. The forward schedule is

row	operation	gate grouping
0	subtract side injection	one gate per target
1	subtract scatter	one gate on targets and center
2	copy	one gate per source
3	gather	one gate on sources and center
4	scatter	one gate on targets and center
5	side injection	one gate per target
6	undo gather	one gate on sources and center
7	undo copy	one gate per source.

(2)

Lemma 5. *For arbitrary initial values of its auxiliary wires, a forward invocation changes y_S to $y_S + x_S$ and leaves every other value unchanged. Performing a forward invocation from X to Y at stage 1, the inverse forward schedule from Y to X at stage 2, and a forward invocation from X to Y at stage 3 sends*

$$(X_a, Y_a) \mapsto (-Y_a, X_a)$$

and restores every auxiliary value. In the bit construction this is the permutation exchanging X_a and Y_a .

Proof. Write $\mathcal{V}$ for copy, $\mathcal{G}$ for gather, $\mathcal{J}$ for side injection and $\mathcal{R}$ for scatter. If the initial side and central vectors are a, c , the first two rows subtract $\mathcal{J}a + \mathcal{R}c$ from y . Rows 2 and 3 change them to $a + \mathcal{V}x, c + \mathcal{G}x$. Rows 4 and 5 therefore add $\mathcal{R}(c + \mathcal{G}x) + \mathcal{J}(a + \mathcal{V}x)$. Rows 6 and 7 restore c, a . The net target change is $(\mathcal{J}\mathcal{V} + \mathcal{R}\mathcal{G})x$, independently of a, c .

In the bit case, the central coefficient from x_T to y_S is $|S \cap T| \bmod 2$. The side contribution cancels exactly the intersection-one cases; distinct triples with intersection zero or two already have coefficient zero, and the self coefficient is one. In the complex case, the central coefficient is $(|S \cap T| - 1)/2$. The side cancels it for distinct triples with intersection zero or two; intersection one contributes zero, and the self coefficient is one. Thus both net maps are $y \leftarrow y + x$.

The inverse schedule with logical source Y and target X effects $X \leftarrow X - Y$. Starting from (x, y) , the three stages give $(x, y + x)$, then $(-y, y + x)$, then $(-y, x)$. Each invocation restores its own auxiliary wires. $\square$

A fixed triple has

$$z_b = 3 \binom{47}{2} = 3243, \quad z_c = \binom{47}{3} + 3 \cdot 47 = 16356$$

neighbors in the two constructions, respectively. The second count separates intersection zero and intersection two. Ordered pairs must be counted: there are vz side wires in each invocation, not $vz/2$. With $c_b = 50$ and $c_c = 51$ central wires per invocation, the total wire counts are

$$W_b = 2N + I(vz_b + c_b) = 73269972440000000, \quad (3)$$

$$W_c = 2N + I(vz_c + c_c) = 369474390296480000. \quad (4)$$

All these numbers are fixed, independently of any eventual input length.

3.2 From scalar values to arrays

We record the representation principle before choosing the labels. Let R be the scalar ring of either network and let Ω be a finite address set common to all wires. Replace each scalar by an array in R^Ω , and apply each scalar gate pointwise at every address. Write S_Ω for the resulting network on the vector of wire arrays.

At each source, gate and sink v , assign an invertible R -linear operator D_v on R^Ω , called its *frame*. A frame acts on the values indexed by addresses within one wire's array and may mix them; a gate instead combines wire values at each fixed address. All incoming and outgoing incidences of one gate use the same D_v . On an edge from u to v , apply $D_v D_u^{-1}$ to that wire's array. Define the operators on all input and output roles by

$$D_{\text{in}} = \bigoplus_w D_{\text{source}(w)}, \quad D_{\text{out}} = \bigoplus_w D_{\text{sink}(w)}.$$

For a supplied physical input f , define its logical interpretation to be $g = D_{\text{in}}^{-1} f$; this definition requires no physical preprocessing. At a vertex v , the invariant is that the stored array equals D_v applied to the logical array there. It holds at the sources by definition. On an edge whose logical array is φ , the edge operator changes $D_u \varphi$ to $D_v \varphi$. At a gate with logical input arrays φ_j , R -linearity gives $\sum_j c_j D_v \varphi_j = D_v(\sum_j c_j \varphi_j)$ for each output combination, with $c_j \in R$. Thus the common frame commutes with the pointwise gate, and the invariant continues to the sinks. The implemented operator is therefore

$$D_{\text{out}} S_\Omega D_{\text{in}}^{-1}. \quad (5)$$

In particular, if the scalar network is a signed role permutation that sends input role w to output role $\rho(w)$ with scalar sign ε_w , that route applies $\varepsilon_w D_{\text{sink}(\rho(w))} D_{\text{source}(w)}^{-1}$ to its input array. All roles, including the auxiliary roles, enter this identity. Once the scalar routing is fixed, the endpoint frames determine the array operator on each route; the intermediate frames can then be chosen to make each edge change inexpensive.

Figure 2 summarizes this invariant.

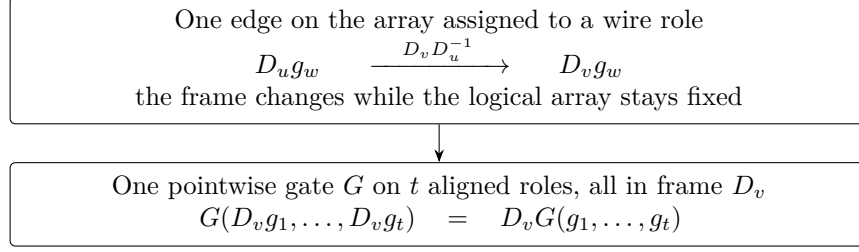


Figure 2: An edge changes the representation of one role's logical array g_w . At the next gate G , all t input arrays have the common frame D_v , so the pointwise linear gate commutes with that frame. The same D_v acts on each output component.

3.3 Subspaces attached to the gates

The scalar network already has the required signed exchange. We now choose its intermediate labels to make the changes of frame inexpensive. A label is a nondegenerate subspace of a fixed bilinear space. On every edge one label will contain the other. For such an edge, the *orthogonal residual* is the orthogonal complement of the smaller label inside the larger one; its dimension is the absolute change in label dimension. The saving will come from keeping the total dimension lost on decreasing edges small relative to N .

For the bit network, let $F = \mathbb{Q}^h$ with bilinear form

$$\langle x, y \rangle = x^\top (I - J/9)y,$$

where J is the all-one matrix. This form has eigenvalues 1 and $1 - h/9 = -41/9$, so it is nondegenerate. For a triple T let t_T be its indicator vector. Then

$$\langle t_S, t_T \rangle = |S \cap T| - 1, \quad \langle t_T, t_T \rangle = 2.$$

For the complex network, take instead $F = \mathbb{F}_2^h$ with the ordinary dot product; here $t_T \cdot t_T = 1$. Thus every triple line is nondegenerate, and neighboring triple lines are orthogonal, in the label field belonging to either construction. The scalar domain need not equal the label field. These degree-one intersection formulas specialize the construction described by Alon [2, Section 3, remark after Theorem 1.1] to the prime 2, with the ground-set size changed to $h = 50$.

The ambient label space is $\mathcal{F} = F^{\otimes 3}$ with its tensor bilinear form. It has dimension m and is nondegenerate. Write

$$u_a = t_{a_1} \otimes t_{a_2} \otimes t_{a_3}, \quad U_a = \langle u_a \rangle.$$

The prescribed terminal labels are

role	source label	sink label
X_a	U_a	$\mathcal{F}$
Y_a	0	$U_a^\perp$
each scratch role	0	$\mathcal{F}$.

Write W for the wire count in the construction under consideration. The source dimensions sum to N and the sink dimensions to $Wm - N$. Thus the signed dimension changes will sum to $Wm - 2N$, because the contributions at gates cancel. If decreasing edges lose a total dimension L , the sum of absolute changes is $Wm - 2N + 2L$. We will obtain $L < N/2$. The rational interface below will require an additional N source ranks, so this inequality will leave a strict saving in both constructions.

At stage j , fix an invocation. Put

$$A = F^{\otimes(j-1)}, \quad P = \langle t_{a_1} \otimes \cdots \otimes t_{a_{j-1}} \rangle, \quad B = P^\perp \subset A.$$

An empty tensor product is the one-dimensional ground field; thus $B = 0$ at stage 1. Let $Q = \langle t_{a_{j+1}} \otimes \cdots \otimes t_{a_3} \rangle \subset F^{\otimes(3-j)}$, again using the empty tensor product at stage 3. Since P is nondegenerate, $A = B \perp P$. We prescribe the following transition for each data wire in the stage, where t is the indicator of that wire's j th triple:

	incoming label	outgoing label
X	$A \otimes \langle t \rangle \otimes Q$	$(A \otimes F) \otimes Q$
Y	$B \otimes \langle t \rangle \otimes Q$	$((B \otimes F) \perp (P \otimes t^\perp)) \otimes Q.$

The outgoing Y label can also be written $(P \otimes \langle t \rangle)_{A \otimes F}^\perp \otimes Q$. These transitions expose one tensor factor at each stage. To see that they fit together, for $j < 3$ write $Q = \langle t_{a_{j+1}} \rangle \otimes Q'$. The next stage has

$$A' = A \otimes F, \quad P' = P \otimes \langle t \rangle, \quad B' = (P')_{A'}^\perp = (B \otimes F) \perp (P \otimes t^\perp).$$

Thus the prescribed outgoing labels are $A' \otimes \langle t_{a_{j+1}} \rangle \otimes Q'$ on X and $B' \otimes \langle t_{a_{j+1}} \rangle \otimes Q'$ on Y , exactly the next incoming labels. At stage 1 they start at U_a on X_a and zero on Y_a ; at stage 3 they end at $\mathcal{F}$ on X_a and $U_a^\perp$ on Y_a , as prescribed at the terminals.

The gate labels are organized by the decomposition

$$A \otimes F = (B \otimes F) \perp (P \otimes F),$$

with the common summand $B \otimes F$ and the remaining summand $P \otimes F$ of dimension h . Suppress the one-dimensional factor Q for the moment. On a side wire joining neighboring physical triples with indicators t_X, t_Y , the common component grows to $B \otimes F$, while the remaining component passes through

$$P \otimes \langle t_X \rangle \subset P \otimes t_Y^\perp \subset P \otimes F.$$

The first inclusion is possible because $t_X \perp t_Y$. For the central gates we assign $A \otimes F$ on the physical X bank and $B \otimes F$ on the physical Y bank. A central wire returns once from the former label to the latter, losing exactly $P \otimes F$. The gate order below confines all decreases to this return, while the other edges realize the prescribed stage transitions.

Number the successive times in an invocation by $r = 0, \dots, 7$. At stages 1 and 3, time r uses row r of the scalar schedule (2). At stage 2 it uses the inverse of row $7 - r$, with the logical banks exchanged. In either case the gates touch the physical wires in the order

Y -side, Y -center, X -side, X -center, Y -center, Y -side, X -center, X -side.

A side wire still joins neighboring physical triples in stage 2 because the neighbor relation is symmetric. The following table assigns the common gate labels. It suppresses the tensor factor Q ; for a gate indexed by a physical bank triple, t denotes that triple's indicator.

time	physical wires	common gate label
0	Y , its side wires	$B \otimes \langle t \rangle$
1	Y , all center wires	$B \otimes F$
2	X , its side wires	$(B \otimes F) \perp (P \otimes \langle t \rangle)$
3	X , all center wires	$A \otimes F$
4	Y , all center wires	$B \otimes F$
5	Y , its side wires	$(B \otimes F) \perp (P \otimes t^\perp)$
6	X , all center wires	$A \otimes F$
7	X , its side wires	$A \otimes F.$

(6)

All labels are tensored by Q and viewed as subspaces of $\mathcal{F}$. Every displayed summand is nondegenerate.

All scratch sources have label zero and all scratch sinks have label $\mathcal{F}$, as prescribed above. The next lemma verifies the stage boundaries and all intervening edge changes, including these scratch edges.

Lemma 6. *Every edge joins comparable nondegenerate labels. The only decreasing edges are the central-wire edges from time 3 to time 4; their dimension loss is h . Consequently, if c is the number of central wires per invocation and W the total number of wires, then*

$$L = Ich, \quad \sum_e |\dim U_{\text{head}(e)} - \dim U_{\text{tail}(e)}| = Wm - 2N + 2L. \quad (7)$$

For the binary label space, every nonzero orthogonal residual of an edge has an orthonormal basis.

Proof. Write $a = \dim A = h^{j-1}$ and $f = h^{3-j}$. In the table below, in denotes the preceding vertex on that data wire: its source at stage 1, and its last gate in the preceding stage otherwise. The interstage calculation above gives its label. The table lists the orthogonal residual and its dimension for every edge except the final data edges to their sinks, whose labels agree and whose residuals are zero. Tensor each residual by Q , except in the last row. The symbols t_X, t_Y are the indicators of the neighboring physical triples joined by a side wire.

wire and edge	orthogonal residual	dimension
$X : \text{in} \rightarrow 2$	$B \otimes t_X^\perp$	$(a-1)(h-1)$
$X : 2 \rightarrow 3$	$P \otimes t_X^\perp$	$h-1$
$X : 3 \rightarrow 6, 6 \rightarrow 7$	0	0
$Y : \text{in} \rightarrow 0$	0	0
$Y : 0 \rightarrow 1$	$B \otimes t_Y^\perp$	$(a-1)(h-1)$
$Y : 1 \rightarrow 4$	0	0
$Y : 4 \rightarrow 5$	$P \otimes t_Y^\perp$	$h-1$
center : source $\rightarrow 1$	$B \otimes F$	$(a-1)h$
center : $1 \rightarrow 3$	$P \otimes F$	h
center : $3 \rightarrow 4$	$P \otimes F$	h
center : $4 \rightarrow 6$	$P \otimes F$	h
side : source $\rightarrow 0$	$B \otimes \langle t_Y \rangle$	$a-1$
side : $0 \rightarrow 2$	$(B \otimes t_Y^\perp) \perp (P \otimes \langle t_X \rangle)$	$(a-1)(h-1) + 1$
side : $2 \rightarrow 5$	$P \otimes \langle t_X, t_Y \rangle^\perp$	$h-2$
side : $5 \rightarrow 7$	$P \otimes \langle t_Y \rangle$	1
scratch : last $\rightarrow$ sink	$(A \otimes F) \otimes Q^\perp$	$ah(f-1)$.

The last gate on a central wire is at time 6, and the last gate on a side wire is at time 7. Both have label $(A \otimes F) \otimes Q$. Enlarging it to $\mathcal{F}$ at the sink gives the last residual in the table, with $Q^\perp$ taken in the entire future tensor space. The side edge $2 \rightarrow 5$ uses the neighbor relation essentially. Since $t_X \perp t_Y$, we have $\langle t_X \rangle \subset t_Y^\perp$. Hence its two labels satisfy

$$(B \otimes F) \perp (P \otimes \langle t_X \rangle) \subset (B \otimes F) \perp (P \otimes t_Y^\perp).$$

The complement of $\langle t_X \rangle$ inside $t_Y^\perp$ is $\langle t_X, t_Y \rangle^\perp$, of dimension $h-2$; tensoring by P and then by Q gives the residual in the table. The other entries follow directly by splitting $A = B \perp P$ and $F = \langle t \rangle \perp t^\perp$. Label dimension is nondecreasing along every edge except center $3 \rightarrow 4$, which

removes $P \otimes F \otimes Q$. The table includes the interstage edges through in, and the initial and final data labels match their terminals, so no other dimension changes occur.

The signed sum of dimension changes telescopes at every gate: each touched wire has one incoming and one outgoing incidence with the same gate label. The sum of source dimensions is N , while the sum of sink dimensions is $Wm - N$. The signed increase is therefore $Wm - 2N$. There are Ic decreasing edges, each of loss h ; replacing signed changes by absolute changes adds $2L$.

Now work over $\mathbb{F}_2$. A nondegenerate symmetric bilinear space is nonalternating precisely when it contains a vector of norm one. The lines P, Q have such vectors. If $B \neq 0$, a coordinate unit outside the support of the earlier tensor vector belongs to B : that support has size $3^{j-1} < h^{j-1}$ when $j > 1$. The same argument gives a unit vector in $Q^\perp$ when it is nonzero. A triple complement, and the complement of two neighboring triple lines, contain coordinate units outside supports of sizes at most three and six, respectively. Full tensor spaces contain coordinate units as well. In each nonzero tensor summand, tensoring these chosen norm-one vectors gives a norm-one vector; placing it in that orthogonal summand gives one in the whole residual. Each residual is nondegenerate by the decompositions above. Thus every nonzero residual is nonalternating.

For completeness, every nondegenerate nonalternating binary symmetric space has an orthonormal basis. Split off unit lines until the remaining space is alternating. A nonzero nondegenerate alternating space splits into planes with bases a, b satisfying $a \cdot a = b \cdot b = 0$, $a \cdot b = 1$; this follows by choosing a nonzero a , choosing b with $a \cdot b = 1$, and taking the orthogonal complement of their nondegenerate plane. Retain one unit vector w from the lines already split off. A plane orthogonal to w can be absorbed into that line: the three vectors

$$w + a, \quad w + b, \quad w + a + b$$

are independent, pairwise orthogonal, and have norm one. Use one of these new unit vectors in place of w for the next plane; the remaining planes are still orthogonal to it. Repeating this operation absorbs every alternating plane. $\square$

The loss ratios have the exact values

$$\frac{L_b}{N} = \frac{75}{196}, \quad \frac{L_c}{N} = \frac{153}{392}.$$

In particular, both are smaller than $1/2$. We now turn the dimension count into the two interfaces needed later.

3.4 The rational matrix interface

For a nondegenerate rational subspace $U \subset \mathcal{F}$, write P_U for the projection onto U along $U^\perp$. Assign P_U to each gate or sink with label U . Assign the source matrix $-P_{U_a}$ to X_a and zero to every other source. These are fixed rational $m \times m$ matrices; write M_v for the matrix at vertex v . Define $\rho(X_a) = Y_a$, $\rho(Y_a) = X_a$, and let ρ fix every auxiliary wire.

In the tape application, two address chunks are written as vectors H, D of m fields. For a matrix M , the frame Φ_M moves the array entry at (H, D) to $(H + MD, D)$, leaving the other address fields unchanged. The fields use a radix coprime to the fixed denominators, and addition is modulo their common range. Hence Φ_M is an invertible $\mathbb{F}_2$ -linear permutation of array entries, and $\Phi_{M'}\Phi_M = \Phi_{M'+M}$. Using $D_v = \Phi_{M_v}$ in (5), the endpoint matrix difference on each routed role is exactly the shear applied to that role. The following identity makes it the same shear $H \leftarrow H + D$ on every role; Section 4 turns that shear into a field interchange and implements an edge change with one smaller interchange per unit of its rational rank.

Proposition 7. *The bit network has W_b wires, uses only pointwise XOR gates, and sends its input vector to the permutation ρ of that vector. Its rational matrices are common to all incidences of a gate and obey*

$$M_{\text{sink}(\rho(w))} - M_{\text{source}(w)} = I_m \quad \text{for every input role } w.$$

The sum of the rational ranks of all edge differences is

$$s_b = \sum_e \text{rank}_{\mathbb{Q}}(M_{\text{head}(e)} - M_{\text{tail}(e)}) \quad (8)$$

$$= W_b m - N + 2L_b = 9158746553232864000000 < W_b m. \quad (9)$$

Proof. Lemma 5 gives the scalar permutation, including all auxiliary roles. For $U \subset V$ nondegenerate, $P_V - P_U$ is the identity on $V \cap U^\perp$ and zero on its orthogonal complement. Its rank is $\dim V - \dim U$; reversing the edge negates the matrix without changing its rank. The only exceptions are the N negative source projections. At stage 1, $B = 0$, and the first gate on X_a has label exactly U_a . Its first edge matrix is therefore $2P_{U_a}$, of rational rank one, where the corresponding dimension change was zero. Lemma 6 consequently gives the stated rank sum. Its deficit from $W_b m$ is $N - 2L_b > 0$.

For an input X_a the endpoint difference is $P_{U_a^\perp} - (-P_{U_a}) = I_m$. Every other role has zero source matrix and identity sink matrix. This proves the interface. $\square$

3.5 The binary phase interface

Use the complex scalar network and binary labels. For a binary vector x , let $\text{wt}(x)$ denote its integer Hamming weight, and write $[b] \in \{0, 1\}$ for the integer representative of $b \in \mathbb{F}_2$. For a label U , let

$$q_U(x) = \text{wt}(P_U x) \pmod{4}.$$

The projection here is over $\mathbb{F}_2$. Let

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad C = H \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} H = aI + bX, \quad a = \frac{1+i}{2}, \quad b = \frac{1-i}{2},$$

where X interchanges the two coordinates. On functions of $x \in \mathbb{F}_2^m$ define

$$\mathcal{C}_U = H^{\otimes m} \text{diag}_x(i^{q_U(x)}) H^{\otimes m}.$$

The use of H in this definition is an identity of matrices. Each entry of $\mathcal{C}_U$ and its inverse is a Gaussian integer divided by 2^m , so both preserve arrays over $\mathbb{Z}[i, 1/2]$. The kernels implemented below have Gaussian dyadic coefficients.

To apply the scalar network to these functions, replace the scalar on each wire w by an array $f_w : \mathbb{F}_2^m \rightarrow \mathbb{Z}[i, 1/2]$. At each address, a gate applies its scalar linear map to the values on the wires it touches. The frame in (5) is $D_v = \mathcal{C}_U$ at a vertex with label U . On an edge from label U to label V , apply $\mathcal{C}_V \mathcal{C}_U^{-1}$ to that wire's array. All frames are diagonalized by the same matrix $H^{\otimes m}$. For an integer $k \geq 1$, an address for k columns is $(x^{(1)}, \dots, x^{(k)}) \in (\mathbb{F}_2^m)^k$; every frame and edge operator is then the tensor product of its one-column version over these k coordinates. Each scalar gate is still applied once at each joint address, to the vector of values on the roles it touches.

Proposition 8. *For each edge of the complex network, write U, V for its tail and head labels. Its phase-frame difference $\mathcal{C}_V \mathcal{C}_U^{-1}$ is a product of one translation kernel $aI + bX_v$, or its inverse, per*

vector in an orthonormal basis of the edge residual. Here $(X_v f)(x) = f(x + v)$. The sum of these basis lengths is

$$s_c = W_c m - 2N + 2L_c = 46184298777878576000000 < W_c m.$$

After source and sink diagonal corrections, and undoing the signed bank exchange, telescoping these phase frames gives $C^{\otimes m}$ on every role, including auxiliary roles. The same assertion holds on any positive number of columns by taking tensor products over columns.

Proof. For binary vectors z, w ,

$$\text{wt}(z + w) = \text{wt}(z) + \text{wt}(w) - 2|\text{supp}(z) \cap \text{supp}(w)|.$$

If $z \cdot w = 0$, the intersection cardinality is even, so weight is additive modulo four. If $V = U \perp E$ and $\mathcal{B}$ is an orthonormal basis of E , it follows that

$$q_V(x) - q_U(x) = \sum_{v \in \mathcal{B}} \text{wt}(v)[v \cdot x] \pmod{4}.$$

For a decreasing edge the right side is negated. Each v has norm one, so $\text{wt}(v)$ is odd. Its coefficient in the phase, including the sign for the edge direction, is therefore some $\varepsilon \in \{1, -1\}$ modulo four. The function $i^{\varepsilon[v \cdot x]}$ equals

$$\frac{1 + i^\varepsilon}{2} + \frac{1 - i^\varepsilon}{2}(-1)^{v \cdot x}.$$

The identity $H \text{diag}(1, -1)H = X$, tensored over the coordinates in $\text{supp}(v)$, sends the diagonal character $(-1)^{v \cdot x}$ to translation X_v . The displayed phase therefore becomes $aI + bX_v$ or its inverse. The nonzero residuals have such bases by Lemma 6; for a zero residual use the empty basis. The same lemma gives the count s_c . The deficit is $2N - 2L_c > 0$.

For an edge e , write $\mathcal{B}_e$ for its residual basis and $K_{e,v}$ for the forward or inverse translation kernel associated with $v \in \mathcal{B}_e$. Taking tensor products preserves their product, so on k columns

$$(\mathcal{C}_V \mathcal{C}_U^{-1})^{\otimes k} = \prod_{v \in \mathcal{B}_e} K_{e,v}^{\otimes k}.$$

Thus each residual vector contributes one factor acting on all k columns, and there are s_c such factors over the network.

Let ρ denote the underlying permutation of the complex roles: it exchanges X_a, Y_a and fixes each scratch role. By Lemma 5, the pointwise scalar network sends an input array at X_a to Y_a with sign $+1$, an input at Y_a to X_a with sign -1 , and each scratch input to its own output with sign $+1$. Applying (5) with $D_v = \mathcal{C}_{U_v}$, where U_v is the label at vertex v , shows that an input role w reaches output role $\rho(w)$ with that sign and with operator $\mathcal{C}_{U_{\text{sink}(\rho(w))}} \mathcal{C}_{U_{\text{source}(w)}}^{-1}$. For k columns, use $D_v = \mathcal{C}_{U_v}^{\otimes k}$ on the joint address set in the same identity. The route operator is then the k th tensor power of the displayed address operator, multiplied by its scalar route sign once. The zero label has frame I , while the full label has frame $C^{\otimes m}$ because $q_{\mathcal{F}}(x) = \text{wt}(x)$. Every pair except $X_a \rightarrow Y_a$ has these zero and full labels. For the remaining pair, write $u = u_a$. Its binary norm is one and its weight is $3^3 = 27$. Orthogonal weight additivity yields

$$\begin{aligned} q_{U_a^\perp}(x) - q_{U_a}(x) &= \text{wt}(x) - 2\text{wt}(u)[u \cdot x] \\ &= \text{wt}(x) + 2[u \cdot x] \pmod{4}. \end{aligned}$$

Since $2[u \cdot x] \equiv 2 \sum_{j \in \text{supp}(u)} x_j \pmod{4}$, the operator on $X_a \rightarrow Y_a$ is the tensor of C^{-1} on the 27 coordinates of $\text{supp}(u)$ and C elsewhere. Put $Z = \text{diag}(1, -1)$. Direct multiplication gives

$$C = iZC^{-1}Z, \quad C^{-1} = -iZCZ.$$

For k columns, let $Z_{a,k}$ be the tensor product of Z on those 27 coordinates in every column and the identity elsewhere. Apply $Z_{a,k}$ before the network at X_a , and apply $i^{27k}Z_{a,k}$ after the network at Y_a ; the latter scalar is applied once to the whole role. There are $27k$ inverse factors, so the identity $C = iZC^{-1}Z$ turns each of them into C . The operator on this pair is therefore $C^{\otimes mk}$ as well. Only the physical X outputs still carry the minus sign of the scalar exchange. Negate those outputs and send output role $\rho(w)$ back to role w . Every role, including every scratch role, now has the required forward tensor product. $\square$

3.6 A paired-block circuit for the bit correction

Use $h = 50$, $v = \binom{h}{3} = 19600$, $N = v^3$, and $m = h^3 = 125000$. The complex network above is retained at this ground size. For the bit network, we replace the intersection-one side correction by a smaller circuit. Fix a common point i . On the other $n = h - 1 = 49$ points, inputs x_{ab} represent source triples $\{i, a, b\}$. The required partial output is

$$z_{cd} = \sum_{\{a,b\} \cap \{c,d\} = \emptyset} x_{ab}.$$

All scalar additions are in $\mathbb{F}_2$, but the computation below always adds disjoint formal input supports.

It is useful to solve a weighted version. Give a graph an edge value e_{uv} and a vertex value w_u , with pairwise disjoint supports. Return its total value and the values left after deleting each one or two vertices, including all their incident edges. At the root, take $w_u = 0$ and $e_{uv} = x_{uv}$. Group the ordered vertices consecutively into pairs, with a final singleton when necessary. A block A becomes a vertex of a smaller graph with weight

$$\tilde{w}_A = \sum_{u \in A} w_u + \sum_{\{u,v\} \subset A} e_{uv}, \quad \tilde{e}_{AB} = \sum_{u \in A, v \in B} e_{uv}.$$

Recursively compute the smaller graph's total, its value O_A after deleting block A , and its value F_{AB} after deleting both A, B .

For $a \in A$, put

$$C_a = \sum_{u \in A \setminus \{a\}} w_u, \quad E_{a,B} = \sum_{u \in A \setminus \{a\}, v \in B} e_{uv}, \quad H_{a;B} = C_a + \sum_{C \neq A,B} E_{a,C}.$$

Compute all $H_{a;B}$ by the leave-one-out prefix/suffix routine on the vector $(C_a, (E_{a,C})_{C \neq A})$, requesting only the outputs that omit one of the E terms. The total of this vector is denoted H_a . If a, b are in different blocks A, B , their exclusion output is

$$F_{AB} + H_{a;B} + H_{b;A} + \sum_{u \in A \setminus \{a\}, v \in B \setminus \{b\}} e_{uv}.$$

For two vertices in one block it is O_A ; the single exclusion of a is $O_A + H_a$. These are disjoint partitions of the surviving weighted graph. This proves the recursion by induction. The weights created by internal edges remain inside the recursion, avoiding a separate exclusion circuit for them.

Use direct sums for at most four vertices. Compute other totals by balanced binary addition. In the four-term formula, first add $F_{AB} + H_{a;B}$, then add the other two terms, then combine those two

results. Intern equal formal supports and remove unused nodes. At $n = 49$, the exact circuit has 9813 additions and 1176 designated outputs. All 1271256 nonzero coefficients, and every required zero, are checked by comparing all output supports with their definitions.

Identify equal inputs and equal sums across the h common-point groups. A sum shared by groups $i \neq j$ consists entirely of triples $\{i, j, k\}$; its fixed pair and variable vertices identify it exactly. Keep the first encountered decomposition, in common-point and then local-node order. Retain all $q = h \binom{h-1}{2} = 58800$ partial outputs, injecting their sum into each physical target. Since every neighbor pair has a unique common point, this remains exactly the intersection-one matrix. The global graph has

$$c = 450394, \quad q = 58800, \quad c + q = 509194.$$

Here 40256 additions have been merged from the separate local graphs.

3.7 Reversible roles and frames in both directions

For every node, regard its later uses and designated outputs as outgoing edges. At an input node, allocate one role per outgoing edge and copy a pivot into the others. At an addition node, add the second incoming role into the first, reuse the first as one outgoing pivot, and copy it into fresh roles for the other outgoing edges. The other input role retires. These are invertible pointwise gates. There are $2c + q$ outgoing uses and c pivot identifications, so $R = c + q = 509194$ roles suffice. Roles are compiled from the merged graph; no simultaneously occupied physical slots are identified.

Let L be the product of these gates, V the source copy, J the injection of the designated partial outputs, and G, R_0 the original central operations. The chronological schedule is

$$L, J, L^{-1}, R_0, V, G, R_0, L, J, L^{-1}, G, V.$$

The two injections give $JLz + JL(z + Vx) = JLVx$ for arbitrary initial scratch z . The cleanup restores z and all central scratch. Since $JLV + R_0G = I$, the invocation is $y \leftarrow y + x$. Reverse and invert the schedule for stage 2. Forward, inverse, forward invocations therefore exchange the two data banks.

Use the original rational form $I - J/9$ on $F = \mathbb{Q}^{50}$. At a stage, write $A = B \perp P$ as before, suppressing the future line factor Q . Set $D_0 = B \otimes F$, $D_1 = A \otimes F$, and $D_U = (B \otimes F) \perp (P \otimes U)$. Let U_z be the span of the source indicators reaching node z . Every retained node is a sum from an original common-point group, so all its triples contain some i . For a vector u in that span,

$$\sum_j u_j = 3u_i, \quad \langle u, u \rangle = \sum_{j \neq i} u_j^2 > 0 \quad (u \neq 0).$$

Thus U_z is nondegenerate. Along every graph edge $z \rightarrow w$, $U_z \subset U_w$. Forward middle gates use D_{U_z} ; the physical roles follow paths, so these frames increase. A designated output for target T has $U_z \subset t_T^\perp$, because every contributing triple is a neighbor.

For the reversed middle computation use $D_{U_z^\perp}$. The ambient form is nondegenerate, so all these complements are nondegenerate. Reversing an edge reverses inclusion: $U_w^\perp \subset U_z^\perp$. The original output begins at the physical X line $\langle t_T \rangle \subset U_z^\perp$; an original input ends at exactly its physical Y complement $t_S^\perp$. No positivity assumption on the reachable-target spans is needed. At every gate all incident roles receive the same frame. Early mixers use D_0 , and late cleanup uses D_1 . Newly active roles enter from D_0 and retired roles grow to D_1 . Tensoring by Q gives the original data-stage boundaries in both orientations. The only decreases remain the central returns, each of dimension h .

Share the complete auxiliary bank of a stage-1 invocation (A, B) with stage 3 at $(B, \pi(A))$, matching local role indices. Here π is a bijection of triples with $|T \cap \pi(T)| = 1$: pair consecutive ground points; for a triple with a full pair and singleton, keep the singleton and cycle the full pair among pairs not containing it; otherwise keep the point in the least indexed pair and flip the other two to their partners. Each rule is bijective on its class. All invocations restore arbitrary scratch, and the joining labels are

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = \langle t_B \otimes t_{\pi(A)} \rangle^\perp \otimes F.$$

They are nondegenerate and $E \subset H$ by the chosen orthogonality. Each join removes one auxiliary role and exactly m of edge rank.

Proposition 9 (Paired-block bit interface). *The new bit network restores arbitrary scratch, exchanges the data banks, and satisfies the original all-role endpoint identity with*

$$\begin{aligned} W_b^{\text{pair}} &= 2N + 2v^2(R + h) = 406321422080000, \\ s_b^{\text{pair}} &= W_b^{\text{pair}}m - N + 6v^2h^2 = 50790175992864000000, \\ W_b^{\text{pair}}m - s_b^{\text{pair}} &= 1767136000000 > 0. \end{aligned}$$

Proof. The scalar schedule and both frame assignments were verified above. The total decreasing dimension is $3v^2h^2$. Telescoping gives total absolute label change $W_b^{\text{pair}}m - 2N + 6v^2h^2$; the unchanged negative source projections add N rational rank. Data endpoints are unchanged and every surviving auxiliary role has endpoints zero and I_m , with its scalar value restored. The original common-frame identity and rational-shear transfer apply. All circuits, rational bases, and gates are fixed finite data. $\square$

3.8 Explicit exponents at ground size fifty

Take $a = 296/10^{11}$, $a_c = 1/10^{11}$, and $L_0 = 11737/1000$. The exact deficits satisfy

$$\eta_b^{\text{pair}} = \frac{23}{661055000} > aL_0, \quad \eta_c = \frac{239}{1202215191250} > a_cL_0.$$

For $1 \leq x \leq 2$, put $z = (x-1)/(x+1)$, $S(x) = 2 \sum_{j=0}^{23} z^{2j+1}/(2j+1)$, and $E_0(x) = 2z^{49}/(49(1-z^2))$. Then $S(x) \leq \log x \leq S(x) + E_0(x)$, and exact rational comparison gives

$$\log m \leq 16(S(2) + E_0(2)) + S(m/2^{16}) + E_0(m/2^{16}) < L_0.$$

The inequality $e^{-x} > 1 - x$ therefore proves the required strict bounds with

$$\tau = 1 - \frac{296}{10^{11}}, \quad \sigma_{50} = 1 - \frac{1}{10^{11}}. \quad (10)$$

The complex network itself is unchanged apart from the ground size $h = 50$; its construction, binary residual bases, and scalar corrections remain those proved above.

3.9 An independently sized complex network

Keep the paired bit network at $h_b = 50$, with address-swap exponent $\tau = 1 - 296/10^{11}$. Instantiate the original complex construction instead at $h_c = 25$. Its constants, distinguished from those of the bit network, are

$$m_c = 15625, \quad v_c = 2300, \quad N_c = 12167000000,$$

$$W_c = 58645352620000, \quad L_c = 10315500000, \quad s_c = 916333630984500000.$$

Here $W_c = 2v_c^3 + 3v_c^2(v_c z_c + 26)$, $z_c = \binom{22}{3} + 66$, and $L_c = 3v_c^2 \cdot 25 \cdot 26$.

Proposition 10. *The original complex phase interface, including its signed scalar bank exchange, arbitrary-scratch restoration, binary residual bases and endpoint corrections, holds with these constants. Its normalized residual saving supports the strict exponent $\sigma = 1 - 418/10^{12}$.*

Proof. The original construction and residual table apply at this ground size. For an intersection of size j , the center coefficient is $(j - 1)/2$; the side correction is its negative for distinct even-intersection neighbors. Their sum is zero for $j = 0, 1, 2$ and one for $j = 3$. The same eight-step schedule therefore restores arbitrary scratch and gives the elementary bank shear, and the three stages give the signed exchange.

Binary triple indicators have norm one. The orthogonal decompositions in the residual table remain nondegenerate. Every nonzero residual has a norm-one vector: choose a coordinate outside the support of a triple or two neighboring triples (at most six coordinates), or outside an earlier/future tensor indicator (support $3^j < 25^j$); tensor with the remaining norm-one factors. Thus the original nonalternating-space argument supplies the required orthonormal bases. No even-ground-size matching or rational bit-interface positivity is needed for this unchanged complex network.

The rank accounting gives

$$s_c = W_c m_c - 2N_c + 2L_c, \quad \eta_c = \frac{W_c m_c - s_c}{W_c m_c} = \frac{14}{3464399375} > 0.$$

The needed positivity is $L_c < N_c$, not $2L_c < N_c$. Endpoint weight-27 phase corrections and the signed-exchange correction are unchanged. Exact rational logarithm bounds give $\log m_c < 966/100$ and $\eta_c > (418/10^{12})(966/100)$. The inequality $e^{-x} > 1 - x$ then yields $s_c/W_c < m_c^\sigma$. The certificate also checks $2 \leq s_c < m_c^5$. $\square$

The bit and complex dimensions need not agree. The bit interface supplies swaps at arbitrary binary widths. The complex recursion uses only that interface's exponent τ and fixed tape guarantee, while its own branching, role split and depth use m_c, W_c, s_c . All occurrences of m, W, s within the revised layer proof denote these complex constants. The retained assembly consumes their time exponents and completed-layer contract, so it introduces no equality requirement between the two arities.

With the original side wires these exponents would be

$$\tau = 1 - 296/10^{11}, \quad \sigma_{\text{orig}} = 1 - 418/10^{12}. \quad (11)$$

We replace the side wires by a shared-sum circuit.

3.10 A compressed side circuit for the complex network

Keep the complex motif of Section 3.9: $h = 25$, $v = \binom{25}{3} = 2300$, $N = v^3$, $m = h^3 = 15625$, scalars in $\mathbb{Z}[i, 1/2]$, labels in $F = \mathbb{F}_2^{25}$ with the dot product, the $h + 1$ central wires, and the gather and scatter maps. For a triple S write t_S for its indicator. The original side wires add

$$\frac{1}{2} \sum_{T \cap S = \emptyset} x_T - \frac{1}{2} \sum_{|T \cap S|=2} x_T \quad (12)$$

to y_S . With the central coefficient $(|S \cap T| - 1)/2$ this gives $y \leftarrow y + x$. They use one wire per ordered neighbor pair, that is $v z_c = 3693800$ side wires per invocation. We compute the same map with shared sums.

All sums below are formal sums of inputs x_T whose two summands have disjoint supports, so a source contributes to a node exactly when a directed path joins them. Equal supports are identified within each of the two families below, and nodes that feed no injected piece are removed.

Disjoint sums. Order the points $0, \dots, 24$. For a list of nodes indexed by distinct points and $k \leq 3$, the routine `vec` returns, for every set E of at most k of these points, the sum of the nodes whose points avoid E . It splits the list at $\lfloor n/2 \rfloor$, recurses, and forms each output by one addition of a left and a right output whose exclusion sets have total size at most k . For a point list X and weights $w(u, v)$ on its pairs, the routine `pairs` returns the sums over pairs avoiding E , $|E| \leq 3$: split X into halves L, R , recurse on both, apply `vec` to each cross row $(w(u, r))_{r \in R}$, then apply `vec` down the columns of these row outputs with the remaining exclusion budget, and add the three disjoint regions. The routine `tri` does the same for triple sums: with halves L, R of X , put

$$P_r = \text{pairs}(L, (u, v) \mapsto x_{uvr}), \quad P'_u = \text{pairs}(R, (a, b) \mapsto x_{uab}),$$

apply `vec` over $r \in R$ to $P_r[E_L]$ and over $u \in L$ to $P'_u[E_R]$, and add the four disjoint regions: triples in L , triples in R , two points in L with one in R , and one point in L with two in R .

At the top level $L = \{0, \dots, 11\}$ and $R = \{12, \dots, 24\}$. For a target S with $S_L = S \cap L$ and $S_R = S \cap R$ the disjoint pieces are

$$\text{tri}(L)[S_L], \quad \text{tri}(R)[S_R],$$

the two `vec` outputs over the halves $R_1 = \{12, \dots, 17\}$ and $R_2 = \{18, \dots, 24\}$ of the family $P_r[S_L]$, and the two `vec` outputs over the halves $L_1 = \{0, \dots, 5\}$ and $L_2 = \{6, \dots, 11\}$ of the family $P'_u[S_R]$. These six pieces partition the triples disjoint from S . They are internal nodes of the full computations because `vec` splits at the same halves.

Pair stars. For a pair $a < b$, list the other points as $u_1 < \dots < u_{23}$ and form prefix sums $\pi_i = \sum_{j \leq i} x_{abu_j}$ and suffix sums $\sigma_i = \sum_{j \geq i} x_{abu_j}$. Target $\{a, b, u_i\}$ receives π_{i-1} and σ_{i+1} . Over its three pairs, S receives exactly the triples meeting it in two points.

Injection rule. Inject every disjoint piece with coefficient $1/2$ and every pair-star piece with coefficient $-1/2$, subject to one rule: if the triples of a piece cover every point outside S , inject its two summands instead, recursively. A single triple covers three of the 22 outside points, so the rule terminates. At $h = 25$ it never splits a disjoint piece. It splits π_{22} for $i = 23$ and σ_2 for $i = 1$. Each target therefore receives twelve pieces.

The circuit has 80595 additions (68595 disjoint, 12000 pair-star) and 27600 injected pieces. Exact support comparison checks every coefficient of (12): all 3693800 nonzero entries, their signs, and all required zeros.

3.11 The transparent schedule

Allocate reversible roles as for the bit network: a node's outgoing edges are its later uses and its injected pieces; an input slot fans out, and an addition reuses its first incoming role as a pivot and retires the second. This uses $R = c + q = 80595 + 27600 = 108195$ roles. Let L be the product of the node mixers, V the copy into input slots, and J the injection, which adds $\pm \frac{1}{2}$ times a piece role to its target. With $G, \mathcal{R}$ the original gather and scatter, a forward invocation performs

$$L, -J, L^{-1}, -\mathcal{R}, V, G, \mathcal{R}, L, J, L^{-1}, G^{-1}, V^{-1}.$$

For initial side values z and central values w , the target changes by

$$-JLz - \mathcal{R}w + \mathcal{R}(w + Gx) + JL(z + Vx) = (JLV + \mathcal{R}G)x = x,$$

and the last four operations restore z and w . The inverse schedule, with logical source Y and target X , gives $X \leftarrow X - Y$; forward, inverse and forward invocations send (X, Y) to $(-Y, X)$ and restore every auxiliary value. Every gate is pointwise linear with coefficients in $\{0, \pm 1, \pm \frac{1}{2}\}$.

3.12 Binary labels

Assign each node z a subspace $U_z \subset F$:

- an input node T receives $\langle t_T \rangle$;
- a disjoint-sum node receives the coordinate space $F_X = \langle e_x : x \in X \rangle$, where X is the union of its triples;
- a pair-star node for $\{a, b\}$ with leaf set Λ receives $\langle t_{abu} : u \in \Lambda \rangle$.

The vectors t_{abu} have weight three and pairwise intersections of size two, so they form an orthonormal basis of the pair-star label. Coordinate spaces have orthonormal coordinate bases. Every label is therefore nondegenerate. Along a dataflow edge $z \rightarrow w$ the support grows, so $U_z \subset U_w$. The injection gate for target S uses the physical Y frame, whose active component is $t_S^\perp$. Every triple of a piece meets S in an even number of points, so the piece label lies in $t_S^\perp$.

Lemma 11. *If a piece leaves some point $y \notin S$ uncovered, then the residual $t_S^\perp \cap U_z^\perp$ of its injection edge has an orthonormal basis. If instead a disjoint piece covers every point outside $S = \{a, b, c\}$, its injection residual is $\langle e_a + e_b, e_b + e_c \rangle$, which has none.*

Proof. Both U_z and $t_S^\perp$ are nondegenerate, so the residual is nondegenerate. The vector e_y is orthogonal to t_S , to every coordinate vector in a disjoint label, to every t_{abu} with $u \neq y$, and to an input line t_T with $y \notin T$. Thus the residual contains a vector of norm one and is nonalternating; the manuscript's absorption argument then gives an orthonormal basis. In the second case $t_S^\perp = F_{[h] \setminus S} \perp \langle e_a + e_b, e_b + e_c \rangle$. The displayed plane is nondegenerate, and all its vectors have even weight, hence norm zero. $\square$

A nondegenerate label between $F_{[h] \setminus S}$ and $t_S^\perp$ must be one of these two spaces, since a proper nonzero part of the plane is an isotropic line. A full-coverage disjoint piece thus cannot be repaired by relabeling; the injection rule avoids it. Any partition of the disjoint triples into pieces that each miss an outside point needs at least four missed points, so four pieces per target are necessary for this label class.

The remaining residuals are covered by the following list, which the checker verifies by exact binary linear algebra for every node, edge, piece and compiled role.

- Coordinate to coordinate: $F_{X' \setminus X}$, a coordinate space.
- Input line to coordinate space: $F_X \cap t_T^\perp$ contains e_x for $x \in X \setminus T$, which is nonempty for an addition node.
- Pair-star chain: the span of the new t_{abu} .
- Entry from the early frame: the label itself.
- Exit to the late frame: $U_z^\perp$, which contains a unit coordinate vector outside the covered points; for a pair-star node, outside $\{a, b\} \cup \Lambda$, which omits at least the excluded third point.

Each nonzero residual is nondegenerate and contains a vector of odd weight, so it has an orthonormal basis.

3.13 Frames in both stage directions

Use the frames of the bit construction with these binary labels. With $A = B \perp P$ and future line factor Q , put $D_0 = B \otimes F$, $D_1 = A \otimes F$ and $D_U = (B \otimes F) \perp (P \otimes U)$, all tensored by Q . Early mixers and the early injection use D_0 ; middle node gates use D_U ; copy gates use the physical X frame $D_{\langle t_T \rangle}$ and injection gates the physical Y frame $D_{t_S^\perp}$; late cleanup uses D_1 . An edge residual is $P \otimes E \otimes Q$ for a residual $E \subset F$ from the list above. The lines P, Q are spanned by tensor products of triple indicators, which have norm one, so tensoring an orthonormal basis of E gives one of the residual. The data wires pass through the same frames as in the original motif and have the same residuals.

In the inverse schedule the nontrivial labels occur in the first L^{-1} . Assign $D_{U_z^\perp}$ to node z there, D_0 to the preceding L , and D_1 to the second L , J and L^{-1} . A forward inclusion $U_z \subset U_w$ gives $U_w^\perp \subset U_z^\perp$ with the same residual $U_w \cap U_z^\perp$. A piece role enters from the physical X line $\langle t_S \rangle \subset U_z^\perp$ with residual $t_S^\perp \cap U_z^\perp$, the forward injection residual. An input role starts at the physical Y frame $B \otimes \langle t_T \rangle$ of the first copy, passes D_0 with residual $B \otimes t_T^\perp$ as a data wire does, and ends at $t_T^\perp$, exactly its physical Y frame. Entries and exits exchange the forward residuals U_z and $U_z^\perp$. Thus every residual of the inverse stage is one already shown to have an orthonormal basis.

3.14 Rank, phases and depth

Proposition 12 (Compressed complex interface). *The complex network with this side circuit has*

$$\begin{aligned} W_c &= 2N + 3v^2(R + h + 1) = 1741801270000, \\ L_c &= 3v^2h(h + 1) = 10315500000, \\ s_c &= W_c m - 2N + 2L_c = 27215641140750000, \end{aligned}$$

satisfies the binary phase interface of Proposition 10 with these constants, and has $\eta_c = (W_c m - s_c)/(W_c m) = 28/205789375$.

Proof. The schedule gives the signed exchange and restores arbitrary scratch. Every side role has source label 0 and sink label $F^{\otimes 3}$, and its labels increase, so it contributes exactly m to the residual count. Data wires and central wires are unchanged, so the only decreases remain the h -dimensional central returns. Telescoping gives s_c . Every residual has an orthonormal basis, so each edge is a product of one translation kernel per basis vector, as in the original phase interface. The endpoint phases of the data roles and the signed-exchange correction are unchanged, and every auxiliary role still runs from frame I to $C^{\otimes m}$.

The coefficient-depth argument needs a bound on gates, not on gates per wire. An invocation has $4 \cdot 82895$ mixer gates, $2v$ copy gates, $2v$ injection gates (one per target) and four central gates. Over $3v^2$ invocations this is $5408242080000 \leq 12W_c$. Each gate has at most W_c inputs and outputs and coefficients in $\{0, \pm 1, \pm \frac{1}{2}\}$, so $E = 64(W_c + m + 1)^3$ bounds the depth of one invocation as before. Also $2 \leq s_c < m^5$, so the generalized stopped guard applies with these constants. All gates, labels and bases are fixed finite data. $\square$

The rational log bound $\log m < 966/100$ and $\eta_c > (14/10^9)(966/100)$ give, by $e^{-x} > 1 - x$,

$$s_c/W_c < m^\sigma, \quad \sigma = 1 - \frac{14}{10^9}. \quad (13)$$

This is 33.49 times the previous complex saving $418/10^{12}$.

The preceding constructions provide the retained interfaces and comparison witnesses.

3.15 A five-subset interchange motif over $\mathbb{F}_3$

Let $h = 28$, $\mathcal{T} = \binom{[h]}{5}$, $v = 98280$, $m = h^3 = 21952$, and $N = v^3$. Scalars in the interchange network lie in $\mathbb{F}_3$. Use the rational address-label space

$$F = \mathbb{Q}^h, \quad H = I - \frac{2}{25}J.$$

Its exceptional eigenvalue is $1 - 2h/25 = -31/25$, so it is nondegenerate. For the indicator t_T of a five-set,

$$\langle t_S, t_T \rangle = |S \cap T| - 2, \quad \langle t_T, t_T \rangle = 3.$$

Neighbors have intersection two. For every pair $C \in \binom{[h]}{2}$, retain

$$A_C = \sum_{T \supset C} x_T, \quad D_{C,E} = \sum_{T=C \cup U, U \cap E = \emptyset} x_T \quad \left(E \in \binom{[h] \setminus C}{3} \right).$$

Scatter $\sum_{C \subset S} A_C$ to y_S and inject $-\sum_{C \subset S} D_{C,S \setminus C}$. A source and target with intersection k have coefficient $\binom{k}{2} - [k = 2]$ modulo three. For $0 \leq k \leq 5$ this is $[k = 5]$. Thus their combined map is the identity.

Cancellation-free triple exclusion. The local producer works with a polynomial of degree at most three whose monomials have disjoint formal source supports. Pair its point indices. Aggregate a monomial according to the set of pairs it touches, and recurse on those pair indices. To exclude a set E of at most three points, let G be its touched pair indices. Begin with the coarsened sum excluding G . For every nonempty subset K of the surviving partners in the pairs G , add the contraction that selects exactly K there and avoids the remaining pairs of G . These cases partition the permitted monomials. A one-point contraction has degree at most two and uses the repository's weighted paired-exclusion recursion. A two-point contraction is a weighted vector exclusion, and a three-point contraction is one original coefficient. At at most four points direct summation supplies the base case. The exported producer checks all $\binom{26}{3} = 2600$ output coefficient vectors exactly. Retaining the total adds twelve ancestors to its 41427 active additions.

For each common pair, order the intact global pairs first and place the partners of its removed points last. Identify equal source supports across these local producers, retaining an existing decomposition, then remove unused ancestors. Source-support cardinality strictly increases at every addition, so this identification cannot create a cycle. There are 11240978 addition nodes and $q = 10v + \binom{h}{2} = 983178$ designated output uses before the next compression.

An addition shared between different common pairs has actual source intersection of size three or four. In the first case its support is specified exactly by the common three-set and a bit vector of remaining pairs; in the second by the common four-set and a bit vector of remaining points. These are exact keys, with full equality comparison after hashing. Nodes with common intersection of size two belong to a unique local context. Consequently the global enumeration does not rely on probabilistic hashes.

Resynthesizing four-point stars. For each four-set B , extract every sum of $x_{B \cup \{u\}}$ used by a node outside that star or by a designated output. Replace the entire star subgraph by a cancellation-free circuit for precisely those sums. A reproducible greedy common-subexpression rule combines the pair of current terms that occurs in most requested sums; ties are broken by increasing support cardinality, union mask, and operand masks. An already constructed union is reused. Every

maintained expression is a partition of its requested support. Thus each replacement addition is disjoint and each boundary coefficient vector is unchanged.

Canonical ordering puts intact global pairs before the remaining partners. The 20475 stars require 365 distinct templates. The independent C++ checker verifies both operands and their disjointness at every template gate and checks every required boundary sum against the independently reconstructed global DAG. This replaces 3006276 additions by at most 2623060. Hence the complete producer has at most

$$c = 10857762, \quad q = 983178, \quad R = c + q = 11840940$$

reversible auxiliary roles. The bound remains valid if a replacement creates unused nodes: delete them and after stage sharing, pad by untouched auxiliary roles with source matrix zero and sink matrix I_m when exact width is wanted.

Nondegenerate source spans. Label every producer node by the rational span U of its source indicators. Each retained node, including every replacement star node, has a common pair $\{a, b\}$. Every vector $u \in U$ satisfies $u_a = u_b = t$ and $\sum_i u_i = 5t$, so

$$u^\top H u = \sum_{i \notin \{a, b\}} u_i^2.$$

If the right side vanishes, $\sum_{i \notin \{a, b\}} u_i = 3t$ gives $t = 0$. Thus U is positive definite. Along every addition edge source supports and their spans are nested. Every side-output source meets its target in exactly two points, so $U \subset t_S^\perp$. Orthogonal complements supply the increasing labels in the reversed computation as in the shared-point construction. No binary orthonormal-basis condition is imposed on these rational labels.

The retained total A_C has source span of dimension $h - 2$. Indeed, its coordinates outside C are the indicators of all triples on $h - 2$ points, which span the whole rational coordinate space: differences of triples give all coordinate differences, and any triple has nonzero coordinate sum. The two coordinates on C are both one third of the outside coordinate sum.

Transparent retained-total schedule. The retained-total wrapper follows PRs #4 and #6; the paired input ordering follows the aligned common-point idea in PR #2. Use the outgoing-use/pivot compiler on this addition DAG. Its invertible mixer L uses one role per addition plus one per output use. Write V for source copy, R_0 for retained-total scatter, and J for the negative side injection. In chronological order apply

$$L, -R_0, -J, L^{-1}, V, L, R_0, J, L^{-1}, -V.$$

For arbitrary initial auxiliary vector z , this subtracts $(R_0 + J)Lz$, adds $(R_0 + J)L(z + Vx)$, restores z , and changes the target by x . Its inverse gives the inverse shear. The three tensor stages send (X, Y) to $(-Y, X)$; a final pointwise negation on X , at its existing full frame, produces the exact exchange and adds zero edge rank.

Use the original prefix decomposition $A = B \perp P$ and future line Q . Set $D_0 = (B \otimes F) \otimes Q$, $D_1 = (A \otimes F) \otimes Q$, and $D_U = D_0 \perp (P \otimes U \otimes Q)$. Early mixers and scatters have frame D_0 , the source copy its source-line frame, the middle mixer D_U , the retained scatter D_0 , the side injection its target-complement frame, and cleanup D_1 . Only the retained output roles decrease, from D_U to D_0 , by $h - 2$ each. In the inverse schedule the retained scatter is at D_1 and the following inverse

mixer uses $D_{U^\perp}$, giving the same loss. The source and target data boundaries are the original tensor-stage boundaries. Thus total decreasing dimension is

$$L_b = 3v^2 \binom{h}{2} (h - 2).$$

Sharing first- and third-stage banks. An explicit neighbor permutation π of the five-sets is obtained by pairing the ground points. For a set containing no full pair, keep the two singletons in its least indexed pairs and flip its other three singletons. For one full pair, keep it and flip the three singletons. For two full pairs, flip the singleton, and move the unordered pair of full-pair indices to the next edge of a fixed Euler circuit of the complete graph on the other thirteen pair indices. That graph has even degree twelve. Its Euler edges are distinct, consecutive edges meet in exactly one vertex, and their cyclic successor is a bijection. The three cases preserve their classes and all have $|T \cap \pi(T)| = 2$.

Match the stage-1 bank with fixed coordinates (A, B) to the stage-3 bank with fixed coordinates $(B, \pi(A))$, keeping local role indices. Each invocation restores arbitrary auxiliary inputs. The joining labels are

$$E = F \otimes \langle t_A \rangle \otimes \langle t_B \rangle, \quad H = (\langle t_B \otimes t_{\pi(A)} \rangle)^\perp \otimes F.$$

Orthogonality in the middle factor gives $E \subset H$. Their dimensions are h and $m - h$. Replacing the separate endpoint edges by this join removes one role and exactly m rank units, preserving the absolute deficit. All roles retain the required source/sink matrices. Therefore

$$W_b = 2v^2(v + R) = 230640858616896000, \quad s_b = W_b m - N + 2L_b = 5063027748645128371200.$$

The extra N source ranks come from the unchanged negative source projections. Consequently

$$\eta_b = \frac{W_b m - s_b}{W_b m} = \frac{39}{520019360} > \frac{3}{400000000} \frac{9997}{1000}, \quad \log(21952) < \frac{9997}{1000}.$$

Proposition 13. *The fixed $\mathbb{F}_3$ circuit exchanges the two data banks, restores every auxiliary input, and satisfies the all-role rational endpoint identity with width W_b and rank sum s_b above. Its interchange exponent is $\tau = 1 - 3/400000000$.*

Proof. The scalar schedule, nested rational frames, bank matching and negative source projections just established give the interface. The strict rank-ratio comparison and the finite-alphabet argument below give the exponent. $\square$

Finite-alphabet interchange. The address radix in the rational-shear construction is still the fixed prime chosen to avoid the rational matrix denominators; it is unrelated to the payload field $\mathbb{F}_3$. Represent original bits as payload symbols zero and one. Recursive role streams may contain symbol two. Address permutations commute with pointwise $\mathbb{F}_3$ gates exactly as they do with XOR gates, and all local gates are fixed finite tables. At completion the corrected scalar permutation restores the original payload symbols. Thus the all-role endpoint identity and the recurrence $F_k \leq (s_b/W_b)F_{k-1} + O(1)$ hold over this fixed alphabet. There is no conversion between binary and ternary numerical addresses and no multiplicative per-level encoding cost. Existing record streams may be processed bit-position by bit-position with this fixed payload alphabet.

3.16 Paired triple exclusion on the complex side

Keep the triple-based complex motif, binary labels, coefficient ring, and injection rule of eumemic's PR #3. Replace only its disjoint producer by the paired weighted-triple circuit above, now at $h = 28$. Transfer its nodes with coordinate labels, and recursively split any injected node covering all points outside its target, exactly as in PR #3. The pair-star correction and its labels are unchanged. Every addition is disjoint; the original coordinate, input-line and pair-star residual proofs apply unchanged. The full finite check gives 61022 additions, 32816 injected pieces and $R_c = 93838$ roles per invocation.

Share complete stage-1 and stage-3 auxiliary banks, including centers, as in dleen's PR #4. The triple permutation flipping every point to its partner has even intersection with its input. The join $E \subset H$ above therefore holds over $\mathbb{F}_2$. A tensor of coordinate units outside B in its first and third factors belongs to $H \cap E^\perp$ and has norm one. Its nonzero residual is nonalternating and has an orthonormal basis. The phase interface and weight-27 endpoint corrections are unchanged. With $v_c = 3276$ and $m_c = 21952$,

$$W_c = 2v_c^2(v_c + R_c + h + 1) = 2085111546336, \quad L_c = 3v_c^2h(h + 1), \quad s_c = W_cm_c - 2v_c^3 + 2L_c.$$

Here $\eta_c = 5/12693352 > (39/10^9) \cdot 10$ and $\log m_c < 10$, giving $\sigma = 1 - 39/10^9$.

Proposition 14. *The paired complex network satisfies the retained binary phase interface, including arbitrary auxiliary inputs and all endpoint corrections, with the constants m_c, W_c, s_c above and $\sigma = 1 - 39/10^9$.*

Proof. The unchanged local binary residual proofs apply to the new disjoint producer, the verified injection splits, and the pair-star correction. The preceding norm-one join witness covers bank sharing. Telescoping and the retained weight-27 corrections therefore give the interface, and the strict logarithm comparison gives the exponent. $\square$

The normalized gate coefficients remain in $\{0, \pm 1, \pm 1/2\}$. The scalar gate count is $3v_c^2(4(61022 + 3276) + 4v_c + 4)$, less than $12W_c$; the retained stopped-depth guard applies with these new constants, subject to its separately checked finite inequalities.

3.17 Conditional assembly at 2^{-28}

Combine these two finite interfaces with the compact-control movement and stopping guard and with eumemic's fast Gaussian resampling in PR #5. Take

$$\begin{aligned} \epsilon &= \frac{4999}{10000}, & c &= \frac{9999}{10000}, & \beta &= \frac{19}{25}, \\ \lambda &= 1 - \frac{749}{10^{11}}, & \lambda' &= 1 - \frac{748}{10^{11}}, & \delta &= 10^{-6}, \\ \zeta &= 10^{-4}, & C_1 &= \frac{19601}{10000}, & \kappa &= \frac{373}{10^{11}} > 2^{-28}. \end{aligned}$$

With compact-control packed and reserved-axis exponents, every constraint is strict. The smallest of the seven assembly margins is

$$g_3 = \frac{934813}{25000000000000}, \quad g_3 - \kappa = \frac{2313}{25000000000000} > 0.$$

This is a conditional asymptotic exponent claim under the same retained fixed-tape, analytic and numerical interfaces, not a full formalization of integer multiplication and not a practical running-time measurement.

The following exponents certify the retained uniform-width baseline:

$$\tau = 1 - 3/400000000, \quad \sigma = 1 - 39/10^9. \quad (14)$$

4 Faster interchange of address chunks

We use the bit network to interchange two equal address chunks on tape. An edge matrix of rank a will require a smaller interchanges. Because the sum of these ranks is less than the number of wires times m , the resulting recurrence saves a power of the chunk width.

4.1 A matrix decomposition adapted to tape order

Lemma 4 updates a later field using earlier fields in linear time. Lower triangular changes of coordinates can be performed entirely in this direction. The required factorization is the Bruhat factorization for arbitrary matrices, with coordinate order reversed; see Grigor'ev [12, Appendix A.2, Proposition 14(c)]. We give its elementary elimination proof in this order.

Lemma 15 (Lower triangular factorization). *Every rational $m \times m$ matrix A of rank a has a factorization*

$$A = E_1 \Pi E_2,$$

where E_1, E_2 are invertible lower triangular rational matrices and Π has exactly a entries equal to one, with at most one in each row and column, and all remaining entries zero.

Proof. If $A = 0$, take $E_1 = E_2 = I$ and $\Pi = 0$. Otherwise choose the topmost nonzero row and its rightmost nonzero entry. Use that pivot column to clear entries to its left by right multiplication by lower triangular elementary matrices. There are no nonzero entries to its right in the pivot row. Clear entries below the pivot by left multiplication by lower triangular elementary matrices, and scale the pivot to one. All rows above the chosen row were zero in the still active columns.

Delete the pivot row and column from the active index sets, retaining their original orders, and repeat. Elementary operations on the active indices are still lower triangular in the original order. They do not change any earlier pivot row or column, whose other entries have already been cleared. Each step isolates a rank-one block and leaves the remaining rank in the active submatrix. Thus the final partial permutation matrix has exactly a ones. If $LAR = \Pi$ is the resulting identity, then $E_1 = L^{-1}$ and $E_2 = R^{-1}$ have the required form. $\square$

Lemma 16 (The cost of a rational matrix shear). *Let $\mathcal{A}$ be a fixed finite collection of rational $m \times m$ matrices. There is a fixed odd prime q such that the following holds. Let $H = (H_1, \dots, H_m)$ and $D = (D_1, \dots, D_m)$ be two ordered groups of fields, every field ranging over $[q^b]$, with every H_i before every D_j . For $A \in \mathcal{A}$ of rational rank a , the permutation*

$$(H, D) \mapsto (H + AD, D) \quad \text{over } \mathbb{Z}/q^b\mathbb{Z}$$

can be performed with exactly a interchanges of pairs of b -digit fields and $O(V)$ further work. Every interchange is between an H field and a D field. Spectators between any of the formal fields are allowed.

Proof. Choose the factorizations of Lemma 15 for all matrices in $\mathcal{A}$ once and for all. Include their factors and inverses in a finite rational table. Choose an odd prime avoiding every denominator in the table and every numerator of a diagonal entry of an invertible factor. For example, a prime

exceeding the absolute values of all its nonzero numerators and denominators suffices. Rational elimination and trial division construct this finite table and a suitable prime in a finite computation independent of b . They may therefore be compiled into one fixed machine description. All these rational identities then specialize to $\mathbb{Z}/q^b\mathbb{Z}$, and the triangular factors remain invertible there.

A lower triangular transformation of H , or of D , costs $O(V)$: update coordinates in descending physical order. At coordinate i , multiply by its diagonal unit and add a fixed linear combination of earlier coordinates. These earlier coordinates still have their original values. Apply Lemma 4; the number of coordinates is the fixed number m . Inverses are also lower triangular.

For $A = E_1 \Pi E_2$, first transform D by E_2 and H by E_1^{-1} , then add ΠD to H , and finally transform H by E_1 and D by E_2^{-1} . Each one in position (i, j) of Π calls for $H_i \leftarrow H_i + D_j$. Implement it by

$$D_j \leftarrow D_j + H_i, \quad (H_i, D_j) \leftarrow (D_j, H_i), \quad D_j \leftarrow H_i - D_j.$$

The first and last updates target a later field controlled by an earlier field. On an initial pair (h, d) this sequence yields $(h + d, d)$. It uses just one interchange. There are a ones, which proves the claim. In particular, the number of recursive calls is the rational rank a ; temporary tapes used for the other operations do not multiply that count. $\square$

4.2 Transferring a finite circuit to address permutations

Fix integers $m, W \geq 2$. The circuit has W wires, including its scratch wires, with separate input and output terminals. Its fixed, pointwise finite-alphabet gates induce a permutation ρ of the W scalar inputs at completion: the input value at role w appears at output role $\rho(w)$. Assign a rational $m \times m$ matrix to each terminal and to each gate, the latter assignment being common to every wire at that gate. A wire segment between consecutive assigned vertices is called an edge. For an edge e put

$$A_e = M_{\text{head}(e)} - M_{\text{tail}(e)}, \quad s = \sum_e \text{rank}_{\mathbb{Q}} A_e.$$

The required conditions are

$$M_{\text{out}(\rho(w))} - M_{\text{in}(w)} = I_m \quad (1 \leq w \leq W), \quad s < Wm. \quad (15)$$

All these data are finite and fixed. Proposition 13 supplies these conditions with $W = W_b$ and $s = s_b$. Its negative source projection is essential to the endpoint identity and has already been included in the rank sum.

Proposition 17 (Power-width interchange with rows). *Assume the fixed circuit data satisfy (15), and let q be a prime as in Lemma 16 for its edge matrices, also avoiding the denominators of every assigned terminal and gate matrix M . Choose a rational number $\tau \in (0, 1)$ with $s/W < m^\tau$. For every $e = m^k$, $k \geq 0$, consider two disjoint contiguous address chunks of e radix- q digits each. Suppose a row field preceding both chunks has length divisible by W^k . There may be additional fields before and after the row field, between the chunks, and after them. The chunks can be interchanged in $O(Ve^\tau)$ time, preserving every other address field, on a fixed number of finite-alphabet tapes. The constant is independent of all field lengths and of the data.*

Proof. For $k = 0$ apply the single-digit operation of Lemma 3. For $k > 0$, write each chunk as m consecutive $b = e/m$ digit fields, giving vectors H, D over $\mathbb{Z}/q^b\mathbb{Z}$. We first construct the shear $H \leftarrow H + D$.

Split whole rows cyclically into W role streams within each prefix preceding the row field. Write the original row number as $r = Wg + (w - 1)$ with $1 \leq w \leq W$; stream w has row number g . These streams have exactly the same remaining shape, including the full ranges of H and D . Each has logical volume V/W , and its row count is divisible by W^{k-1} .

For a rational matrix M , let Φ_M act on an array by moving its entry at address (H, D) to $(H + MD, D)$, with every spectator unchanged. Thus Φ_M is the array permutation induced by that address map. Its inverse is Φ_{-M} and $\Phi_{M'}\Phi_M = \Phi_{M'+M}$. On every circuit edge apply Φ_{A_e} to that role stream; implement it by Lemma 16, recursively interchanging each pivot pair of b -digit fields. Execute each circuit gate pointwise at matching addresses of its role streams.

The row split supplies the common address set used in the frame identity (5). Here the frame at a vertex with matrix M is the address permutation Φ_M , and every edge changes it by $\Phi_{M'-M}$. The same invariant proof applies because a common address permutation commutes with every pointwise finite-alphabet gate. For the physical input arrays f_w , the logical arrays are $g_w = \Phi_{-M_{\text{in}(w)}}f_w$; this only specifies their interpretation, with no initial tape operation. The logical circuit routes g_w to role $\rho(w)$. Thus its physical output satisfies

$$f'_{\rho(w)} = \Phi_{M_{\text{out}(\rho(w))}}\Phi_{-M_{\text{in}(w)}}f_w = \Phi_{I_m}f_w,$$

by (15). Merge by returning row g of output role $\rho(w)$ to the original row $Wg + (w - 1)$ within the same preceding prefix. This undoes ρ and restores row order. The result is $H \leftarrow H + D$ on every original row, including rows assigned to scratch roles, whose initial values were arbitrary.

To interchange the two full chunks, perform

$$D \leftarrow D - H, \quad H \leftarrow H + D, \quad D \leftarrow H - D,$$

componentwise. Only the middle shear uses the circuit. The other two updates target the later group and have linear cost. The transformation sends (H, D) to (D, H) .

The number of recursive calls is the sum of the edge ranks, exactly s . Each acts on one role stream of volume V/W , retaining all other fields as spectators. Splitting, merging, gates, and triangular operations have total cost $O(V)$ because their number is fixed. Below we verify that scheduling the recursive calls on fixed tapes also costs $O(V)$ at this node. Thus if F_k is a uniform upper bound for time divided by logical volume, then

$$F_0 = O(1), \quad F_k \leq (s/W)F_{k-1} + O(1). \quad (16)$$

For $a = s/W < m^\tau$ the geometric sum $\sum_{j=0}^k a^j$ is $O(m^{k\tau})$. If $a \leq 1$, the sum is at most $k + 1 = O(m^{k\tau})$; if $a > 1$, it is $O(a^k)$. Hence $F_k = O(e^\tau)$.

A fixed-tape depth-first schedule. Here we complete the implementation assertion used in (16). Reserve W role tapes, a fixed set of I/O and elementary-operation work tapes, one parking stack tape, and a separate descriptor stack. Although W is large, it is fixed. At a call boundary all reusable work tapes, other than the current I/O and local descriptor, are empty and their heads are at their origins. The parking and descriptor heads are at the tops of their stacks.

Before a recursive call, append the $W - 1$ inactive role streams to the parking stack in a fixed order, with separators. Copy the active stream to the child I/O area, and erase the parent role tapes while traversing their current contents. Push the parent descriptor and its finite program counter to the descriptor stack. The role and work tapes are now available to the child, using the same tapes at every depth. On return, copy the child output into the designated active role. Pop the inactive streams in reverse order. To recover a stream in its original direction, prepare its

destination to the known length and write the popped symbols from the destination's right end toward its left end. Erase the popped stack cells as they are left. This uses linear time and ends with the parking head at the previously saved top. No scan passes through a parked ancestor. Cleanup is always limited to the region visited by the current call; a reused tape is not swept to the largest position ever visited by an ancestor.

Each push, pop, copy, positioning operation, or erasure moves $O(V)$ symbols at a parent node. There are only the fixed number s of child calls. This gives precisely the additional $O(V)$ term claimed above, even though each child itself has volume V/W . Temporary storage is therefore not substituted for logical volume in the factor s/W .

For completeness, descriptor processing also fits this term when the payload is one bit. Let V_0 be the volume at the root of one power-width call with width $e_0 = m^{k_0}$, and $V_j = V_0/W^j$ the volume of any depth- j child. Here $j \leq k_0 = \log_m e_0$. The root row count is at least W^{k_0} , so at depth j it is still at least one. Both original chunk ranges remain present: digits outside the current subchunks become spectators rather than being removed. Hence

$$V_j \geq q^{2e_0}, \quad j \log W \leq (\log_m e_0) \log W = O(e_0) = O(\log V_j).$$

It follows that $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. A local descriptor lists the row field, at most the current $m + m$ fields, and a fixed number of intervals formed from intervening spectator fields. Computing products of their lengths, subdividing target fields, copying lengths, and storing the current instruction take $(\log(2V_0))^{O(1)} = O(V_j)$ time at that child, since every fixed power of $\log(2V_j)$ is $O(V_j)$. Before descending, construct the child's descriptor by retaining its two target fields and combining each consecutive group of spectator fields into one interval. The ancestor's subdivision need not remain active in the child. Stack entries contain finite instructions and binary integers, not new alphabet symbols. These observations establish one finite machine for every recursion depth and every choice of spectator lengths. $\square$

We use the rational value $\tau = 1 - \frac{3}{400000000}$ established in (14). Its verification uses fixed integer comparisons and the displayed exponential inequality; it needs no real-arithmetic oracle.

4.3 Removing width and row restrictions

We have proved the fast operation for power widths with enough rows. Rows for a general array can be supplied by a few of its own high digits. Their movement will cost less than the final bound.

Lemma 18 (Arbitrary-width interchange). *There exist fixed $0 < \tau < 1$ and one fixed finite-alphabet multitape procedure with the following property. For positive integers P, G, B and $u \geq 1$, it transforms an arbitrary bit array on*

$$[P] \times [2^u] \times [G] \times [2^u] \times [B]$$

by the address permutation

$$(p, h, g, d, z) \mapsto (p, d, g, h, z)$$

in $O(Vu^\tau)$ steps, where $V = PGB2^{2u}$. The integers specifying the shape are part of the input. The time includes their processing, all padding and unpadding, and fixed-tape workspace cleanup. In particular, $B = 1$ is allowed.

Proof. First suppose the two ranges are $[q^e]$, where q is the fixed prime already chosen. Set

$$k = \lceil \log_m e \rceil, \quad \rho = \left\lceil \frac{k \log W}{2 \log q} \right\rceil.$$

For all sufficiently large e , $1 \leq \rho < e$ and $\rho = O(\log(2e))$. Handle the bounded remaining set of widths by e individual digit interchanges; the constant in $O(Ve^\tau)$ absorbs their cost.

Write each chunk as its top ρ digits followed by its remaining digits, denoted h^+, h^- and d^+, d^- , respectively. Interchange the corresponding high digits one pair at a time. The high positions now contain d^+ and h^+ ; move the digits of the latter, preserving their order, into consecutive positions immediately after d^+ . Suppressing the prefix and suffix, the field orders during the construction are

$$\begin{aligned} & (h^+, h^-, g, d^+, d^-), \\ & (d^+, h^+, h^-, g, d^-) \quad \text{after exchanging and joining the high parts,} \\ & (d^+, h^+, d^-, g, h^-) \quad \text{after interchanging the remaining parts,} \\ & (d^+, d^-, g, h^+, h^-) \quad \text{after separating the high parts again.} \end{aligned}$$

In the two middle orders, the first two fields form a row field of range

$$R = q^{2\rho} \geq W^k$$

before both remaining chunks. The joining moves preserve the order of the remaining digits and the gap field. Exchanging and joining the high parts costs $O(V\rho)$ by Lemma 3. Append zero rows, separately within each preceding prefix, until the row count is

$$R' = W^k \lceil R/W^k \rceil < 2R.$$

This changes the row field's length; it need not remain a power of q during the computation. The padded logical volume is less than $2V$.

Write the remaining width in base m :

$$e - \rho = \sum_{j=0}^k a_j m^j, \quad 0 \leq a_j < m.$$

Partition each remaining chunk into corresponding consecutive pieces, with a_j pieces of width m^j . Apply Proposition 17 to each corresponding pair. All these calls use the same row field of length R' , which is divisible by each required W^j . At completion every call preserves its row address, so it neither mixes genuine rows with padded rows nor depends on the values used for padding. The padded zero rows remain zero at these completed-call boundaries. The sum of the costs is bounded by

$$O(V) \sum_j a_j m^{j\tau} = O(Ve^\tau).$$

The fixed bound $a_j < m$ and the inequality $m^k < me$ make this a geometric sum of the stated order. Scan away the padded rows, then reverse only the moves that joined the high parts. This is the last transition in the field-order display: h^+ returns to the later high position, while d^+ stays in the earlier one. The final order is therefore the required interchange of the two full chunks. The remaining movement cost $O(V \log(2e))$ is $O(Ve^\tau)$. Computing the piece list and every new shape is polynomial in $\log(2V)$ per piece. There are at most $(m-1)(k+1) = O(\log(2e))$ pieces; since $V \geq q^{2e}$, their total descriptor cost is a fixed power of $\log(2V)$ and hence $O(V)$. Each power-width call uses the fixed schedule proved above and leaves its work tapes clean.

Finally, for binary chunks let e be the least integer with $q^e \geq 2^u$. Pad each of the two chunk ranges numerically from $[2^u]$ to $[q^e]$, putting zero in an entry if either chunk coordinate is invalid. The increase in volume is less than q^2 , a fixed factor. The padding is a nested scan through consecutive valid and invalid intervals, with their lengths prepared from the shape. It costs $O(V)$ with a constant depending on q . Lexicographic radix- q order of $[q^e]$ is its numerical order, just as binary order of $[2^u]$ is its numerical order. Padding therefore preserves the order of the existing entries without converting the digits of every address. The completed interchange preserves the set on which both chunk coordinates lie below 2^u . Scan away the other entries to recover the desired binary array. Since $e = O(u)$ for fixed q , the bound is $O(Vu^\tau)$. $\square$

The lemma permits arbitrary prefix, gap, and suffix lengths. Thus every later interchange of equal bit chunks is an instance of its stated layout, after multiplying the irrelevant contiguous lengths. For widths differing by one bit, choose the leading bit of the longer chunk as the extra bit. Suppress the outer prefix and suffix, and write g for the intervening address field. If the first chunk is longer, write it as (h_0, H) , so H and the second chunk D have equal width. The physical orders are

$$(h_0, H, g, D) \mapsto (h_0, D, g, H) \mapsto (D, g, h_0, H).$$

The first step interchanges H, D by the lemma; the second moves the single bit h_0 immediately before H . If the second chunk is longer, write it as (d_0, D) , with H the first chunk. Use

$$(H, g, d_0, D) \mapsto (d_0, H, g, D) \mapsto (d_0, D, g, H).$$

Here the first step moves d_0 immediately before H , and the second interchanges H, D . In each case g remains between the two full chunks in the final order. If H, D are empty, the indicated equal-width interchange is the identity. Each case uses one single-bit move, costing $O(V)$ by Lemma 3, in addition to the equal-width interchange. No unrestricted array transpose is being used in these operations.

4.4 Replacing the uniform recurrence by contiguous projector blocks

The preceding power-width and arbitrary-width results are retained baseline constructions with their stated uniform rank recurrence and exponent $\tau_0 = 1 - 3/400000000$. We now change the recursive compilation. The new result below does not assert the uniform inequality $s/W < m^\tau$ at its stronger exponent.

4.5 Batching a large-rank projector edge

The usual matrix-shear compiler uses one smaller interchange per rank unit. For a projector of rank close to the ambient dimension, most of these pivot interchanges can instead be one interchange of two contiguous longer fields. The construction below does not gather separated fields or reverse their digits.

A simultaneous rational change of basis. Let $\mathcal{P}$ be a fixed finite family of rational idempotents of dimension m . For $P \in \mathcal{P}$ with $m/2 < a := \text{rank } P < m$, put $r = m - a$. There is one $S \in \text{GL}_m(\mathbb{Q})$ such that, for every such P , the submatrix of SPS^{-1} on its first r rows and last r columns is invertible.

To prove this, first consider one rank- a idempotent. Every rational idempotent is rationally similar to $\text{diag}(I_a, 0)$, by concatenating rational bases of its image and kernel. Put $t = m - 2r > 0$.

On the first and last r coordinates take r disjoint rank-one blocks

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix},$$

pairing coordinate i with $m + 1 - i$, and take the identity on the middle t coordinates. This is a rational rank- a idempotent whose indicated corner is invertible. Hence the determinant of that corner of SPS^{-1} is not the zero rational function of the entries of S . Clear powers of $\det S$ from its denominator. The product of the resulting nonzero polynomials for the finite family, and $\det S$, is nonzero over $\mathbb{Q}$. A nonzero polynomial over the infinite field $\mathbb{Q}$ cannot vanish at every rational point. This proves simultaneous existence.

The choice is effective: enumerate integer matrices and retain the first invertible one satisfying all the indicated rational determinant tests. This search terminates. The family and m are fixed, so this finite calculation is part of the fixed-machine preparation. Conjugate every terminal and gate matrix by this same S . Common gate matrices, every edge rank, the scalar circuit, and the endpoint identity are preserved, since $SIS^{-1} = I$. The address prime is then chosen after this conjugation and the triangular factorizations, avoiding their finitely many denominators and invertible diagonal numerators as in the original construction.

Only the edges selected for batching need belong to $\mathcal{P}$. All other matrices can continue to use the original rank-by-rank compiler. In the present finite network, every selected edge is a nonzero scalar multiple of an idempotent: for nested nondegenerate labels $U \subset V$,

$$(P_V - P_U)^2 = P_V - P_U,$$

and reversing the edge only changes its sign. Scalar multiples do not change the pivot locations and are absorbed into the triangular factors.

The contiguous middle pivots. Let $P^2 = P$ have rank $a > m/2$, let $r = m - a > 0$, and assume its first- r -row, last- r -column corner is invertible. Split the ordered coordinates into

$$L = [1, r], \quad M = [r + 1, m - r], \quad R = [m - r + 1, m].$$

The complement $Q = I - P$ has rank r . Write $Q = UV$, where U has r columns, V has r rows, and $VU = I_r$. Such a factorization comes from a basis of $\text{im } Q$. As L and R are disjoint,

$$P_{L,R} = -U_L V_R.$$

Its invertibility makes both square matrices U_L, V_R invertible.

Run the lower-lower elimination in the retained physical order. Its first r pivots occupy the rows L and the columns R , in some permutation. Indeed their corner is invertible, so at each of these steps the next row has a nonzero entry in an as-yet active column of R , which precedes all other candidates in the rightmost-pivot rule. The active Schur complement on rows $M \cup R$ and columns $L \cup M$ has middle block

$$\begin{aligned} P_{M,M} - P_{M,R} P_{L,R}^{-1} P_{L,M} &= I - U_M V_M - (-U_M V_R)(-V_R^{-1} U_L^{-1})(-U_L V_M) \\ &= I_{m-2r}. \end{aligned}$$

Thus the next $m - 2r$ pivots are precisely the diagonal positions (i, i) with $i \in M$. Columns L lie to their left and are cleared by the allowed lower-triangular column operations. The remaining block has rank zero, because the number of isolated pivots is already $r + (m - 2r) = m - r = a$.

Consequently the factorization $P = E_1 \Pi E_2$, with E_1, E_2 invertible lower triangular, has exactly r corner pivots and one contiguous identity block of size

$$t = m - 2r = 2a - m.$$

The same conclusion holds for cP with any fixed nonzero rational c . No assertion that an arbitrary rank- a matrix has this pivot profile is needed; the idempotent identity is what makes the middle block exactly an identity matrix.

One call for the identity block. Subdivide each original contiguous chunk into m consecutive fields of width b . The triangular factors still cost $O(V)$ in the retained later-field affine primitive. Each of the r corner pivots uses one ordinary width- b interchange. For all $i \in M$ together, perform

$$D_i \leftarrow D_i + H_i, \quad (H_M, D_M) \leftarrow (D_M, H_M), \quad D_i \leftarrow H_i - D_i,$$

over $\mathbb{Z}/q^b\mathbb{Z}$. The first and last maps are coordinatewise later-field updates, each using a fixed number of linear scans. The middle map is a single interchange of the two contiguous width- tb chunks H_M, D_M . It preserves the order of their b -digit fields and hence implements all diagonal pivots exactly. The intervening fields are spectators. In particular, addition is componentwise modulo q^b , not addition with carries across the concatenated width- tb chunks.

The edge cost therefore has r recursive calls of width b and one of width tb , in place of a calls of width b , with the same $O(V)$ nonrecursive work. If $a = m$, one may leave the edge in its original m -singleton form. This avoids any same-width recursion. For every batched edge in this paragraph, $0 < t \leq m - 2$, so every child contracts its width by a fixed factor less than one.

The resulting characteristic expression. Let $\mathcal{B}$ be the selected large-rank projector edges, let $r_e = m - a_e$ and $t_e = m - 2r_e$ for $e \in \mathcal{B}$, and retain the total rank sum s . For an exponent $0 < \tau < 1$, the volume-weighted recursive contribution is

$$\mathcal{C}(\tau) = \frac{1}{W} \left[\left(s - \sum_{e \in \mathcal{B}} t_e \right) m^{-\tau} + \sum_{e \in \mathcal{B}} \left(\frac{t_e}{m} \right)^\tau \right]. \quad (17)$$

At $\tau = 1$ this is exactly $s/(Wm)$: batching preserves the existing rank deficit. The strict inequality $\mathcal{C}(\tau) < 1$ is the new time-exponent comparison. It uses only the original rank sum and the count of selected ranks; no complete edge-rank histogram is needed. Non-power widths use equal fields of width $\lfloor e/m \rfloor$, at most $m - 1$ separately handled high digits, and the same fixed-tape row recursion. With sufficient rows the nonrecursive work is $O(V)$, so the usual strong induction gives $O(Ve^\tau)$ whenever (17) is strict. The separate row and arbitrary-width construction specifies this rounding step and removes the initial row divisibility restriction.

4.6 Three geometric rank classes in the retained ternary network

The five-subset producer of PR #7 is retained. Put

$$h = 28, \quad v = \binom{28}{5} = 98280, \quad m = h^3 = 21952, \quad R = 11840940, \quad N = v^3,$$

$$W = 2v^2(v + R), \quad L = 3v^2 \binom{h}{2} (h - 2), \quad s = Wm - N + 2L.$$

This subsection needs the producer's proved width upper bound and frame schedule; it does not enumerate its internal addition graph again. If the producer uses fewer than R roles, insert dummy auxiliary roles with identity gates at the ordinary low and high frames of every invocation. Do this before identifying the first- and third-stage banks. Every dummy role restores its arbitrary scalar, has no decreasing edge, and contributes exactly m to the final rank sum. Thus the stated width and the rank classes below are realized exactly, including any padding.

Write $a = h^{j-1}$ at tensor stage j . With the notation of the retained frame proof, the early auxiliary frame is

$$D_0 = (B \otimes F) \otimes Q, \quad \dim D_0 = (a-1)h,$$

and the final auxiliary frame is

$$D_1 = (A \otimes F) \otimes Q, \quad \dim D_1 = ah.$$

The two auxiliary banks give the following edges.

1. Each role joining stages one and three passes from a frame of dimension h to a containing frame of dimension $m-h$. This is a projector edge of rank $m-2h$. There are $B = v^2R$ such roles.
2. Each stage-two auxiliary role passes from its final frame of dimension h^2 to the full sink frame of dimension m . Its rank is $m-h^2$, again with $B = v^2R$ copies.

There is one further large class on the data wires. The incoming physical X and Y dimensions at stage j are a and $a-1$. Their first relevant frames have dimensions $(a-1)h+1$ and $(a-1)h$, respectively. Hence both entrance edges have rank

$$d_j = (a-1)(h-1).$$

The remaining growth of each data role in that stage is $h-1$. These dimensions hold for the reverse middle-stage schedule as well: its low source copy and its target-line injection give precisely the same physical frames. At stage three,

$$d_3 = (h^2-1)(h-1) = 21141 > m/2,$$

and there are $2N$ copies. The special negative source matrix contributes one rank-one edge on each physical X source, as in PR #7. Consequently the complete data rank sum is

$$2N \sum_{j=1}^3 (d_j + h - 1) + N = 2Nm - N.$$

All remaining auxiliary changes inside an invocation have rank at most h , and a stage-two auxiliary entrance has rank h^2-h . Thus the three classes listed above exhaust the ranks exceeding $m/2$, although exhaustion is not needed for the certificate: leaving any other edge unbatched can only weaken the claimed upper bound.

Apply the large-projector batching lemma to these classes. A rank- a_i projector with $a_i > m/2$ uses $m-a_i$ singleton pivot interchanges and one contiguous interchange of $(2a_i-m)$ subfields. The resulting table is

class	copies B_i	rank a_i	singleton pivots	batched width k_i
stage 1–3 join	v^2R	21896	56	21840
stage 2 sink	v^2R	21168	784	20384
stage 3 data entrance	$2v^3$	21141	811	20330

Treat every rank unit outside these three middle blocks as a singleton interchange. The number of singleton calls is therefore exactly

$$S = s - \sum_{i=1}^3 B_i k_i.$$

At a parent width e , the child widths are e/m with multiplicity S and $k_i e/m$ with multiplicity B_i ; all children operate on streams of volume V/W . The finite mixed-width recurrence consequently has normalized moment

$$\Psi(\tau) = \frac{S}{W m^\tau} + \sum_{i=1}^3 \frac{B_i}{W} \left(\frac{k_i}{m} \right)^\tau.$$

At $\tau = 1$ this is $s/(Wm) = 1 - \eta$, where $\eta = 39/520019360$.

The intermediate middle-block certificate. This paragraph records the first batching step. The controlled basis in the following subsection increases its saving from $177/10^9$ to $246/10^9$, which is the value used in the final assembly. Set

$$a_b = \frac{177}{10^9}, \quad \tau = 1 - a_b.$$

Let $r_0 = 1/m$, $r_i = k_i/m$, and

$$w_0 = \frac{S}{Wm}, \quad w_i = \frac{B_i k_i}{Wm}.$$

Then $\sum_i w_i = 1 - \eta$ and $\Psi(1 - a_b) = \sum_i w_i \exp(a_b \log(1/r_i))$. Use the rational logarithm upper bounds

$$\log m < \frac{9997}{1000}, \quad \log \frac{196}{195} < \frac{512}{100000}, \quad \log \frac{14}{13} < \frac{7411}{100000}, \quad \log \frac{21952}{20330} < \frac{768}{10000}.$$

They follow from the positive series

$$\log x = 2 \sum_{j \geq 0} \frac{z^{2j+1}}{2j+1}, \quad z = \frac{x-1}{x+1},$$

using sixteen terms and remainder at most $2z^{33}/(33(1-z^2))$; for m , first extract its power of two. For $0 \leq t < 1$, $e^t \leq (1-t)^{-1}$. Calling the four displayed upper bounds ℓ_i , exact rational arithmetic gives

$$\Psi(1 - a_b) < \sum_{i=0}^3 \frac{w_i}{1 - a_b \ell_i} < 1.$$

The strict gap in this last bound is greater than 1.78×10^{-10} . All integers, rational logarithm enclosures, weights and the exact gap are supplied by `scripts/batched_bit_rank_moment.py` and its JSON certificate. Under the accompanying arbitrary-width recurrence lemma, this proves the interchange saving $a_b = 177/10^9$ for the unchanged finite scalar producer. A smaller role upper bound can be inserted directly into the same three-class formula.

4.7 A simultaneous basis with one additional contiguous pivot block

The three-class rank bound can be strengthened while keeping the finite scalar producer unchanged. The additional structure is used only for the stage-two auxiliary sink edges. The same rational change of coordinates preserves the needed corner-rank conditions on the other two large classes.

Put $F = \mathbb{Q}^h$, $V = \mathbb{Q}^H$, $H = h^2$, and $m = hH$. Reorder the tensor factors so that the original third factor is F and the first two factors form V . List coordinates in the order

$$(\alpha, i), \quad 1 \leq \alpha \leq h, \quad 1 \leq i \leq H,$$

with i varying fastest. Write Π_t and Π_q for rank-one idempotents, and Π_U for a rank- h idempotent. The three large projector classes then have the following descriptions:

$$\begin{aligned} A_1 &= I - I_F \otimes \Pi_q - \Pi_t \otimes \Pi_U, & \Pi_q \Pi_U &= \Pi_U \Pi_q = 0, \\ A_2 &= (I_F - \Pi_t) \otimes I_V, \\ A_3 &= (I_F - \Pi_t) \otimes (I_V - \Pi_q). \end{aligned}$$

Their nullities are $2h$, H , and $H + h - 1$, respectively. For the first class, q spans the line associated with $t_B \otimes t_{\pi(A)}$, $U = F \otimes \langle t_A \rangle$, and $t = t_B$. The orthogonality in the bank matching gives the displayed zero products and, in particular, $q \notin U$. The second class is the stage-two sink projection. The third is the stage-three data entrance projection. Transposing any of these matrices preserves its displayed form, with the corresponding dual rank-one lines and subspaces.

Choose $K \in \text{GL}(V)$ and independent matrices $G_i \in \text{GL}(F)$ for $1 \leq i \leq H$. Define

$$T(u \otimes e_i) = G_i u \otimes e_i, \quad S = T(I_F \otimes K).$$

Thus T changes the h first-factor coordinates independently over each of the H multiplicity coordinates. The parameters are rational matrices; S is a change of basis in the finite address-label space, not a tape operation that must be performed on the input streams.

Lemma 19 (Simultaneous controlled basis). *For the finite collection of projectors in the retained $h = 28$ motif, one can choose $K, G_1, \dots, G_H$ over $\mathbb{Q}$ so that:*

1. *for every A_1 , the submatrix of SA_1S^{-1} in its first $2h$ rows and last $2h$ columns is nonsingular;*
2. *for every A_3 , the submatrix of SA_3S^{-1} in its first $H + h - 1$ rows and last $H + h - 1$ columns is nonsingular;*
3. *for every A_2 , the submatrix of SA_2S^{-1} in its first H rows and last H columns is diagonal and nonsingular.*

Proof. First consider the third assertion. The factor $I_F \otimes K$ commutes with A_2 . At multiplicity coordinate i , the transformed operator is $G_i(I_F - \Pi_t)G_i^{-1}$. Consequently its first-to-last block is diagonal in i , with diagonal entry

$$[G_i(I_F - \Pi_t)G_i^{-1}]_{1,h}.$$

For a nonzero rank-one projector and $h \geq 2$, this entry is a nonzero rational function of G_i : writing $\Pi_t = t\varphi$ with $\varphi(t) = 1$, it equals $-(e_1^T G_i t)(\varphi G_i^{-1} e_h)$, and the two factors can both be made nonzero. Thus the required diagonal entries can all be required to be nonzero.

For the first assertion, let $Q_1 = I - A_1$. Its image is the direct sum

$$\text{im } Q_1 = (F \otimes q) \oplus (t \otimes U).$$

Choose a basis matrix B of the h -space U . After applying K , write $q' = Kq$ and $B' = KB$. The nullspace vector parametrized by $u \in F$ and $c \in \mathbb{Q}^h$ has multiplicity- i coordinate

$$G_i(q'_i u + tB'_i c).$$

The first $2h$ coordinate rows therefore evaluate

$$q'_i \ell_i(u) + \ell_i(t) B'_i c, \quad \ell_i = e_1^\top G_i, \quad 1 \leq i \leq 2h.$$

We show that their determinant is not identically zero in the parameters. Choose K so that $q'_i \neq 0$ for $1 \leq i < h$ and the submatrix $[q' \ B']$ in rows $h, \dots, 2h$ is nonsingular. Such a K exists because $[q \ B]$ has column rank $h+1$ and an invertible linear map can send this ordered independent system to any other one. On the first $h-1$ rows, choose ℓ_i to be a basis of the annihilator of t . These rows determine u modulo $\langle t \rangle$. On the remaining $h+1$ rows, choose one fixed functional ℓ with $\ell(t) = 1$. After the first $h-1$ coordinates of u are determined, these rows form exactly the nonsingular matrix $[q' \ B']$ on the remaining coordinate of u and c . Every prescribed nonzero ℓ_i extends to the first row of an invertible G_i . This proves that restriction of $\text{Sim } Q_1$ to the first $2h$ coordinates can have full rank.

For the second assertion, the image of $Q_3 = I - A_3$ is

$$\text{im } Q_3 = (\langle t \rangle \otimes V) \oplus (W_t \otimes \langle q \rangle), \quad W_t = \text{im}(I_F - \Pi_t), \quad \dim W_t = h-1.$$

Since the first summand contains $t \otimes V$, its image under $I \otimes K$ is unchanged. Write a vector after that change as $tv_i + q'_i u$ at multiplicity coordinate i , where $v \in \mathbb{Q}^H$ and $u \in W_t$. The first H coordinate rows give

$$\ell_i(t)v_i + q'_i \ell_i(u), \quad 1 \leq i \leq H.$$

Choose all $\ell_i(t) \neq 0$. They determine the H independent coordinates v_i once u is known. The next $h-1$ rows are the second rows of G_i at $i = 1, \dots, h-1$. If $f_i = e_2^\top G_i$, eliminating v_i gives

$$q'_i \left(f_i - \frac{f_i(t)}{\ell_i(t)} \ell_i \right) (u).$$

Choose K with these q'_i nonzero and choose the displayed $h-1$ functionals to be a basis of the annihilator of t , restricted to W_t . This is possible because $F = \langle t \rangle \oplus W_t$: take each $\ell_i(t) = 1$, take the relevant f_i to annihilate t , and complete these independent first two rows to G_i . Thus restriction to the first $H+h-1$ coordinates can also have full rank.

Apply the same two arguments to the transposed projectors at the last coordinate rows. This is a condition within the same parameter family, because

$$S^{-\top} = T^{-\top}(I_F \otimes K^{-\top}),$$

and $T^{-\top}$ acts independently by $G_i^{-\top}$. They show that the corresponding last-coordinate restrictions of the transformed dual nullspaces can each have full rank.

To combine the requirements, express every rank condition as a nonzero minor after clearing the denominators of the parameter matrices and their inverses. Each is a polynomial that is not identically zero, by the constructions just given. Their finite product, together with the invertibility conditions on K and all G_i , is still a nonzero polynomial. A nonzero rational polynomial cannot vanish at every rational tuple, so one rational choice meets all conditions simultaneously. The individual witness choices above establish that the polynomials are nonzero; they need not themselves meet all the other conditions.

Finally, for a rank- r nullspace projector $Q = UV$ with $VU = I_r$, its first- r by last- r submatrix after conjugation factors as the product of the two square restriction matrices just considered. Both are nonsingular. The corresponding block of $I - SQS^{-1}$ is its negative, since these first and last index sets are disjoint. This proves the first two assertions. The third has already been proved directly. $\square$

The argument is constructive in the finite-data sense required by the network interface: enumerate rational parameter matrices and retain the first choice passing the finitely many specified determinant tests. The lemma proves termination. All parameters and tests depend only on the fixed motif, not on the eventual multiplication input size. Conjugate every terminal and gate matrix by this one S . Their endpoint differences remain I_m , common gate matrices remain common, and all edge ranks are preserved.

The improved stage-two pivot profile. The lower-triangular elimination for a large projector first eliminates its nonsingular corner block. For A_2 , this block is already diagonal and its H pivots lie in matching physical order in two contiguous sets of fields. After fixed diagonal normalizations, they are one $H \times H$ identity block. The Schur-complement argument in the large-projector lemma leaves the middle identity block of size $m - 2H$. Hence a stage-two sink edge is implemented by two contiguous child interchanges, of widths

$$Hb \quad \text{and} \quad (m - 2H)b,$$

with no singleton recursive calls on its corner. Its rank remains $H + (m - 2H) = m - H$, so the rank sum at exponent one is unchanged. The first and third edge classes retain their previous singleton-corner and contiguous-middle implementations by the first two parts of the lemma.

Exact strengthened moment. Let $B = v^2 R$ and let S_0 be the singleton count from the three-class certificate. Replace S_0 by $S_0 - BH$ and add B children of width H/m . For $h = 28$ the new singleton count is

$$S_1 = 105555640096680883200.$$

The normalized child widths and their rank-mass weights are respectively

$$\left(\frac{1}{21952}, \frac{195}{196}, \frac{13}{14}, \frac{10165}{10976}, \frac{1}{28} \right)$$

and

$$\left(\frac{10841531}{520019360}, \frac{12827685}{26000968}, \frac{855179}{1857212}, \frac{20865}{2736944}, \frac{65783}{3714424} \right).$$

Use the four preceding rational logarithm bounds and additionally $\log 28 < 33323/10000$. With

$$a_b = \frac{246}{10^9}, \quad \tau = 1 - a_b,$$

the same bound $e^x \leq (1 - x)^{-1}$ gives a strict rank-moment gap exceeding 4.76×10^{-11} . The exact rational calculation is in `scripts/controlled_bit_rank_moment.py`. This proves the stronger interchange saving using the unchanged PR #7 scalar producer and the same arbitrary-width recurrence interface.

4.8 Integer widths, bounded row padding, and mixed contractions

The batched residual construction uses a fixed finite list of child sizes, not only powers of one radix. The following implementation supplies the corresponding integer-width recursion. All constants in this subsection are fixed finite circuit constants, independent of the input length.

A general moment condition. Let $m, W \geq 2$ be fixed. At one circuit invocation, let the list $t_1, \dots, t_J$ contain integers in $[1, m-1]$. A child of type i has width $t_i f$, where $f = \lfloor e/m \rfloor$, and operates on exactly $1/W$ of the parent's logical volume. Repetitions in the list represent separate calls. An ordinary rank-one call has $t_i = 1$; a batched block has the corresponding larger value of t_i . Full-rank edges may retain their original m rank-one calls, so no same-width recursive call is needed. Put

$$\Psi(x) = \frac{1}{W} \sum_{i=1}^J (t_i/m)^x, \quad t_* = \max_i t_i < m.$$

If $\Psi(\tau) < 1$ for some $0 < \tau < 1$, then the normalized recurrence

$$F(e) \leq \frac{1}{W} \sum_{i=1}^J F(t_i \lfloor e/m \rfloor) + C, \quad F(e) = O(1) \quad (1 \leq e < m),$$

has $F(e) = O(e^\tau)$ for every positive integer e . Indeed, substituting $F(u) \leq Au^\tau$ into an internal call gives at most $A\Psi(\tau)e^\tau + C$. Choose A to cover the finite base range and $C/(1 - \Psi(\tau))$. Floors can only decrease the right-hand side.

A finite integer depth bound. Define

$$D(e) = 0 \quad (e < m), \quad D(e) = 1 + D(t_* \lfloor e/m \rfloor) \quad (e \geq m).$$

This is an integer algorithm: repeatedly replace e by $t_* \lfloor e/m \rfloor$ until it is below m . The function D is nondecreasing, every child has depth bound at most $D(e) - 1$, and $D(e) = O(\log(2e))$, because each internal replacement is at most $(t_*/m)e$. A row count divisible by $W^{D(e)}$ is therefore sufficient for every root-to-leaf path. At each invocation write the row index as $Wg + w$ and split by w . Every child has complete within-row ranges, logical volume exactly V/W , and a remaining row count divisible by $W^{D(e)-1}$. No padding occurs inside this recursion.

Bit interchange with a remainder. At an integer-width call write $e = mf + r$, $0 \leq r < m$. If $f = 0$, interchange the bounded number of digits individually. Otherwise separate the high r digits of each of the two target chunks. With G denoting all intervening spectators, the layout is

$$(H^+, H^-, G, D^+, D^-).$$

First exchange corresponding digits of H^+, D^+ and move H^+ just before H^- . The layout becomes (D^+, H^+, H^-, G, D^-) . Apply the finite circuit to the two length- mf chunks H^-, D^- . Finally move H^+ back immediately before the final H^- , obtaining

$$(D^+, D^-, G, H^+, H^-).$$

All other fields are preserved. These moves use fewer than a fixed multiple of m single-digit operations and cost $O(V)$. The main chunks split into m equal width- f fields, so every rational shear and every batched contiguous-block shear has exactly the required field shape. A child retains the unused main digits and all remainders as spectators. Thus the preceding moment recurrence applies literally at every integer width; there is no per-level numerical padding of either chunk range.

Supplying the bit rows from existing digits. For an arbitrary width e , compute $D(e)$ and choose the least nonnegative integer ρ such that

$$q^{2\rho} \geq W^{D(e)}.$$

Both numbers can be constructed by fixed-radix integer arithmetic. Since $D(e) = O(\log e)$, we have $\rho = O(\log e)$. For sufficiently large e , $2\rho < e$. The exceptional set of smaller widths is fixed and finite; individual-digit interchange there is absorbed by the asymptotic constant.

Exchange the high ρ digits of the two chunks, then gather their two high parts into one row field before both remaining chunks, exactly as in the preceding layout operation. This row field has range $q^{2\rho}$. Within each original prefix, pad its range to

$$R' = W^{D(e)} \left\lceil q^{2\rho} / W^{D(e)} \right\rceil < 2q^{2\rho}.$$

Every added row includes the entire remaining address set. Recurse once at the remaining width $e - \rho$, which has depth bound at most $D(e)$; then remove the added rows and undo the gathering move. Each completed invocation preserves its row index, so padded zero rows are zero again when they are removed.

The possibly large exponent in $W^{D(e)} = e^{O(1)}$ does not multiply the logical volume by that polynomial. The row digits were already part of the two input chunks. Only rounding the row count adds data, by a factor strictly below two, once at the outer boundary. Exchanging and gathering $O(\log e)$ high digits costs $O(V \log(2e))$, which is $O(Ve^\tau)$. The radix conversion from binary chunk ranges to the fixed radix q retains the original constant factor, less than q^2 . Consequently the batched construction is an arbitrary-width interchange procedure with the asserted exponent τ and no additional power of its volume or width.

Fixed tapes and descriptors. Use the original depth-first scheduler: park only the other role streams at the active stack top, run the child on one complete role stream, and restore the parked streams on return. The number of child types is fixed, so parking, copying, cleanup and the finite arithmetic transforms contribute $O(V)$ per node. Before descending, retain the two current chunks and merge consecutive spectator fields; the descriptor has a fixed number of field intervals. Its integer lengths have $O(\log V_0)$ bits. All descendants retain the complete ranges of both root main chunks, even when most digits have become spectators. Their volume is therefore at least $q^{2(e-\rho)}$ in the outer construction, while $j \log W = O(\log e)$ at depth j . Hence $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. Polynomial descriptor work is absorbed by $O(V_j)$, as in the original fixed-tape proof. The larger fixed constant in the depth bound creates no new tape and no growing alphabet.

Lemma 20 (Batched arbitrary-width interchange). *Put $\tau = 1 - 246/10^9$. There is one fixed finite-alphabet multitape procedure which, for every $P, G, B \geq 1$ and $u \geq 1$, transforms an arbitrary bit array on*

$$[P] \times [2^u] \times [G] \times [2^u] \times [B]$$

by $(p, h, g, d, z) \mapsto (p, d, g, h, z)$ in $O(Vu^\tau)$ steps, where $V = PGB2^{2u}$. Shape processing, row padding and removal, address radix padding and removal, and fixed-tape workspace cleanup are included. The constant is independent of all lengths and all payload values. Widths differing by one bit cost $O(V(1 + u^\tau))$, where u is the smaller width, with the same arbitrary spectator gap.

Proof. Apply the controlled basis of Section 4.7 to the retained finite ternary network. Choose the fixed prime only after the finite rational factor tables have been formed, as in Lemma 16. Its controlled rank moment is strictly below one at the stated τ . The common-frame invariant of

Proposition 17 still applies: the common conjugation fixes the endpoint difference I , and only the compilation of individual edge shears changes. The integer-width construction in Section 4.8 then proves the required $O(Vu^\tau)$ bound, including self-supplied rows and the one-time padding factor. All intermediate role values may be arbitrary. The one-bit extension uses exactly the two layouts already displayed after Lemma 18, with this new equal-width procedure substituted in their interchange step. $\square$

4.9 A dimension-thirty continuation of controlled batching

The following continuation combines the ternary source geometry of Zhihao Chen’s PR #7, the controlled batching and complex depth guard of icekylinx’s PR #10, and the star resynthesis and matching extension of Rohan Arun’s PRs #9 and #11. It was prepared with substantial OpenAI Codex assistance. These are conditional, unreviewed mathematical extensions, not formal verification of the multiplication theorem.

The source family and finite producer. For the bit network only, take $h = 30$, $F = \mathbb{Q}^{30}$ with form $I - (2/25)J$, and the indicators t_S of all five-subsets. Then $\langle t_S, t_T \rangle = |S \cap T| - 2$ and $\langle t_S, t_S \rangle = 3$. The scalar identity over $\mathbb{F}_3$ is

$$\binom{j}{2} - [j = 2] = [j = 5] \pmod{3} \quad (0 \leq j \leq 5).$$

The common-pair retained totals and exclusion outputs therefore have the same interpretation as in the dimension-28 network. If u lies in a source span with common pair C , its two common coordinates equal a and its coordinate sum equals $5a$. Hence

$$u^\top (I - (2/25)J)u = \sum_{i \notin C} u_i^2 > 0 \quad (u \neq 0).$$

Indeed vanishing outside C forces $3a = 0$. Every such source span is nondegenerate, and a full retained-total span has dimension $h - 2$. The forward and complemented reverse frame schedules, arbitrary-scratch cancellation, and negative-source endpoint correction are unchanged.

The paired degree-three producer uses 28 points outside each common pair. Exact global support merging, pruning, and common-four-point star resynthesis give

$$c = 16089992, \quad Q = 1425495, \quad R = c + Q = 17515487.$$

There are $\binom{30}{4} = 27405$ stars and 456 canonical boundary templates. All additions have disjoint input supports; every required boundary mask is independently reconstructed and checked. Every replacement remains in its original common-four-point star and therefore retains the common-pair positivity argument. The actual compiled role-support paths and all source, output, and dirty-scratch basis directions of each selected template are checked in both orientations.

The bank permutation. The bipartite graph on two copies of the five-sets, with adjacency at intersection two, is regular of positive degree $\binom{5}{2}\binom{25}{3}$. Counting edges incident to a left subset proves Hall’s condition and gives a neighbor bijection. An explicit bijection can also retain the paired-point construction: replace the Euler tour on the remaining pair indices by a cyclic ordering of the edges of a complete graph with adjacent edges sharing one endpoint. Starting from a triangle, insert the edges incident to a new vertex between two old adjacent edges, with the first and last

inserted edges meeting the old endpoints. This inductively gives the required cycle at every order. Keep PR #7's zero- and one-full-pair cases; in its two-full-pair case flip the singleton and advance the pair of full-pair indices around this cycle. Each case is bijective and has intersection exactly two. All 142506 images are checked. Thus the first/third-stage join remains nested, removes one role and m rank units per join, and adds no loss.

Put $v = \binom{30}{5} = 142506$, $m_b = h^3 = 27000$, $N = v^3$, and

$$W = 2v^2(v + R), \quad L = 3v^2 \binom{h}{2} (h - 2), \quad s = Wm_b - N + 2L.$$

In particular $\eta = (Wm_b - s)/(Wm_b) = 3857/52973979000$.

Why the controlled basis still exists. The proof in Section 4.7 uses $H = h^2$, the independence of q from the h -space U , and freely chosen rational K, G_i . It does not use the numerical value 28 or the parity of $h/2$. At $h = 30$, the required $2h$ first rows lie in the first H rows, the first $H + h - 1$ rows comprise one whole multiplicity block and $h - 1$ rows of the next, and these first and last index sets are disjoint since $2(H + h - 1) < hH$. The same nonzero determinant witnesses and finite-product argument therefore apply literally. Orthogonality from the preceding bank permutation supplies $q \notin U$ and the zero products of the two idempotents. The controlled parameter family is closed under inverse transpose, as required in the dual argument.

Conjugate all gate and terminal matrices by the resulting single rational basis. Endpoint differences remain I , scalar gates retain common matrices, and all ranks remain unchanged. The three selected classes have the following rank and contiguous-block data, where $B = v^2 R$:

copies	rank	singleton pivots	block widths
B	26940	60	26880
B	26100	0	25200, 900
$2N$	26071	929	25142

The first and third use the large-projector Schur complement. The second uses the diagonal corner supplied by the controlled basis. The remaining singleton count is

$$S_1 = s - B(26880 + 25200 + 900) - 2N \cdot 25142 = 373570603170925665960.$$

Thus

$$\Psi(x) = \frac{S_1 + B(26880^x + 25200^x + 900^x) + 2N \cdot 25142^x}{W \cdot 27000^x}.$$

Using the positive rational logarithm series with sixteen terms, rounding each upper enclosure upward to a multiple of 10^{-9} , and $e^z \leq (1 - z)^{-1}$, exact arithmetic gives

$$\Psi(1 - 253/10^9) < 1, \quad 1 - \text{certified upper bound} > 9 \cdot 10^{-12}.$$

Every child width is strictly below the parent width. The arbitrary integer-width, row-reservation, and fixed-tape proof of Section 4.8 applies with these new fixed constants.

Lemma 21 (Dimension-thirty batched interchange). *Lemma 20 holds with $\tau = 1 - 253/10^9$ in place of $1 - 246/10^9$, using the dimension-thirty producer above. Its conclusion includes arbitrary spectator gaps, integer widths, row padding and removal, finite-alphabet preparation, and fixed-tape cleanup.*

Proof. The scalar and source-span identities, retained-total loss, matching, common conjugation, and strict moment comparison were established above. Substitute these finite data into the proof of Lemma 20. Its controlled-basis and row arguments use no additional special property of dimension 28. The alphabet is still $\mathbb{F}_3$, and the address prime is chosen after the finite rational factorizations. The one-bit unequal-width extension is unchanged. $\square$

The complex network is deliberately left at $h_c = 28$, $m_c = 21952$, with saving $7/10^7$. Its path-depth bound, precision correction, and guard $C_1 = 11999/10000$ remain those of PR #10. The bit and complex networks need not have equal finite dimensions: the compact-control interface consumes the bit interchange bound at arbitrary widths. The following stronger layer interface changes only that consumed bit exponent.

4.10 Auxiliary source frames

The transfer construction assigns a rational matrix to every source, gate and sink, common to all incidences of a gate, and requires

$$M_{\text{sink}(\rho(w))} - M_{\text{source}(w)} = I_m$$

for every input role w . The retained networks give every auxiliary source the zero matrix and every auxiliary sink the identity. The transfer does not need this choice.

Lemma 22 (Auxiliary source frames). *Let w be an auxiliary role, so that $\rho(w) = w$, and let M be a fixed rational $m \times m$ matrix. Assign M to the source of w and $I_m + M$ to its sink, and leave every other matrix unchanged. Then the scalar network, every common gate matrix and every endpoint identity are preserved, and the address map implemented on every role is unchanged. Only the first and last edges of w change.*

Proof. The endpoint identity for w reads $(I_m + M) - M = I_m$; no other endpoint identity involves this source or sink. The physical input stream f_w has the logical interpretation $g_w = \Phi_{-M} f_w$. This is a definition, not a tape operation, and g_w is arbitrary because f_w is. The scalar network restores arbitrary auxiliary values, so the logical value at the sink is again g_w . The common-frame identity gives the physical output

$$\Phi_{I_m + M} \Phi_{-M} f_w = \Phi_{I_m} f_w,$$

the same shear as on every other role. Gates on w keep their common matrices, so every interior edge of w , and every edge of every other role, is unchanged. $\square$

The lemma moves rank between the two end edges of an auxiliary role. Before batching only the total mattered. With contiguous blocks, a large edge costs much less per rank unit than a small one, so concentrating a role's rank in one edge lowers the cost.

Stage-two auxiliary roles. Use the notation of Section 4.6. At tensor stage two put $A = F$, $P = \langle t_{a_1} \rangle$, $B = P^\perp$ and $Q = \langle t_{a_3} \rangle$. Every auxiliary role of a stage-two invocation, including its central roles, has first gate frame $D_0 = (B \otimes F) \otimes Q$ of dimension $h^2 - h$, and last gate frame $D_1 = (A \otimes F) \otimes Q$ of dimension h^2 . Assign every such role the source matrix P_{D_0} , and therefore the sink matrix $I + P_{D_0}$. Then:

1. the entrance edge has matrix $P_{D_0} - P_{D_0} = 0$ and makes no call;
2. every interior edge is unchanged;

3. since $D_0 \subset D_1$ are nondegenerate, $D_0 \perp D_1^\perp$, and the exit edge has matrix

$$I + P_{D_0} - P_{D_1} = P_{D_1^\perp} + P_{D_0} = P_{D_0 \perp D_1^\perp}.$$

This is an idempotent of rank $m - h$. Its nullspace is $D_1 \cap D_0^\perp = P \otimes F \otimes Q$, of dimension h , because $A = B \perp P$.

Formerly the entrance had rank $h^2 - h$ and the exit rank $m - h^2$. Now the entrance has rank zero and the exit rank $m - h$. The total rank of each role, the rank sum s and the deficit $Wm - s$ are therefore unchanged. The stage-two sink class A_2 of Section 4.7 no longer occurs. Its replacement is

$$A_4 = I - \Pi_{P \otimes F \otimes Q},$$

where Π_U denotes the orthogonal projector onto U for the label form.

Lemma 23 (A common basis for the new exit class). *In the parameter family $S = T(I_F \otimes K)$ of Lemma 19, there are rational $K, G_1, \dots, G_H$ satisfying its first two assertions and, in addition: for every A_4 , the submatrix of SA_4S^{-1} in its first h rows and last h columns is nonsingular.*

Proof. Order coordinates as in Section 4.7: the original third factor F is slow and V , the first two factors, is fast. Then the nullspace of A_4 is $\langle t \rangle \otimes X$ with $t = t_{a_3}$ and $X = \langle t_{a_1} \rangle \otimes F \subset V$, of dimension h . Write $\Pi = UV'$ with $V'U = I_h$. Its first- h -row by last- h -column block after conjugation is the product of the first h rows of SU and the last h columns of $V'S^{-1}$. The corresponding block of SA_4S^{-1} is its negative, because these row and column index sets are disjoint. It is therefore enough that both square factors can be nonsingular. Each is then a polynomial condition that is not identically zero.

A nullspace vector $t \otimes x$, with $x \in X$, has coordinate $(G_i t)_\alpha (Kx)_i$ at (α, i) after applying S . The first h coordinates have $\alpha = 1$ and $1 \leq i \leq h$, so the first factor is the map $x \mapsto ((G_i t)_1 (Kx)_i)_{i \leq h}$. Give every G_i the first row $(1, \dots, 1)$, so that $(G_i t)_1 = |a_3| = 5$, and choose K sending a basis of X to $e_1, \dots, e_h$. The map is then nonsingular.

The rows of V' span the form dual $\Omega(\langle t \rangle \otimes X) = \langle \Omega_F t \rangle \otimes \Omega_V X$, where Ω is the tensor label form. This subspace has the same shape. Since $S^{-\top} = T^{-\top}(I_F \otimes K^{-\top})$ and $T^{-\top}$ acts by $G_i^{-\top}$, the same argument at the last h coordinates, with $\alpha = h$ and $H - h < i \leq H$, shows that the second factor can be nonsingular.

As in Lemma 19, clear denominators and multiply these nonzero polynomials over the finitely many A_4 . Include the polynomials for its first two assertions and the invertibility of K and every G_i . A nonzero rational polynomial does not vanish at every rational point. Its third assertion concerned A_2 and is no longer needed. $\square$

By the large-projector factorization of Section 4.5, every exit edge A_4 is compiled as h singleton interchanges and one contiguous interchange of its middle $m - 2h$ fields. The first and third classes of Section 4.6 keep their singleton corners and contiguous middle blocks.

4.11 The interchange rank moment with source frames

Keep the retained counts

$$h = 28, \quad v = \binom{28}{5} = 98280, \quad m = 21952, \quad R = 11840940,$$

$$W = 2v^2(v + R), \quad s = Wm - v^3 + 2L,$$

and put $B = v^2 R = 114371146876896000$, the number of stage-two auxiliary roles. The large classes are now

class	copies	rank	singleton pivots	batched width
stage 1–3 join	B	21896	56	21840
stage 3 data entrance	$2v^3$	21141	811	20330
stage 2 auxiliary exit	B	21924	28	21896

Every other rank unit is a singleton interchange. Relative to the controlled count S_1 of Section 4.7, each stage-two auxiliary role loses its entrance of $h^2 - h$ singletons and gains h exit corner pivots. Its former sink corner and middle blocks are replaced by the new exit. Hence

$$S' = S_1 - B(h^2 - 2h) = 22293445170300595200.$$

The normalized child widths are

$$r = \left(\frac{1}{21952}, \frac{195}{196}, \frac{10165}{10976}, \frac{391}{392} \right),$$

and their rank-mass weights are

$$w = \left(\frac{2289741}{520019360}, \frac{12827685}{26000968}, \frac{20865}{2736944}, \frac{25721153}{52001936} \right).$$

The weights sum to $1 - \eta$ with $\eta = 39/520019360$, so batching again preserves the rank deficit. The positive logarithm series used before give

$$\log m < \frac{9997}{1000}, \quad \log \frac{196}{195} < \frac{512}{10^5}, \quad \log \frac{10976}{10165} < \frac{768}{10^4}, \quad \log \frac{392}{391} < \frac{2555}{10^6}.$$

Put

$$a_b = \frac{154}{10^8}, \quad \tau = 1 - a_b.$$

With $e^x \leq (1 - x)^{-1}$ for $0 \leq x < 1$, exact rational arithmetic gives

$$\Psi(1 - a_b) < \sum_{i=0}^3 \frac{w_i}{1 - a_b \ell_i} < 1,$$

with gap

$$\frac{9704597579224556535778749633448639054699}{20663392005574270637817582110595278287793980851574} > 4.69 \times 10^{-10}.$$

The arbitrary-width recurrence of Section 4.8 therefore gives the interchange saving $a_b = 154/10^8$ for the unchanged PR #7 scalar producer.

4.12 Three further contiguous blocks in the controlled basis

Retain the dimension-thirty scalar producer, matching, endpoint matrices, and rank budget of Section 4.9. Write $H = h^2$, $m = hH$, and use the controlled family $S = T(I \otimes K)$, where $T(u \otimes e_i) = G_i u \otimes e_i$. The following additional conditions are imposed on this same basis. They are compatible with all previously required conditions by the finite nonzero-polynomial argument.

A stage-two auxiliary entrance. Before changing basis, this edge has projector

$$A_4 = \Pi_t \otimes \Pi_U, \quad \text{rank } \Pi_t = 1, \quad \text{rank } \Pi_U = H - h.$$

Indeed its frame is $D_0 = (B \otimes F) \otimes Q$, where at stage two $\dim B = h - 1$ and Q is a line. Reordering the third tensor factor first gives the displayed form. Every stage-two auxiliary role starts at zero, so there are $B_0 = v^2 R$ such rank- $(H - h)$ edges.

Write $\Pi_t = t\varphi$, $\varphi(t) = 1$, and

$$a_i = e_1^\top G_i t, \quad b_i = \varphi G_i^{-1} e_h.$$

The existing stage-two sink condition ensures $a_i b_i \neq 0$ for every i . The first- H -row, last- H -column corner of $SA_4 S^{-1}$ is

$$C_4 = \text{diag}(a_i)(K\Pi_U K^{-1}) \text{diag}(b_i).$$

Normalize the nonzero row and column factors. In this H -dimensional corner, $K\Pi_U K^{-1}$ is a projector of rank $H - h > H/2$. Require its first- h -row, last- h -column corner to be nonsingular. This is a nonzero rational-function condition on K by the large-projector argument, since K ranges over all invertible matrices on V . The lower/lower pivot profile inside C_4 therefore consists of h corner pivots and an identity block of length $H - 2h$ in matching order. Those $H - h$ pivots exhaust the rank of the entire m -dimensional edge. Since the selected columns are in its last H positions, all the eliminations are allowed in the original physical order. Thus this edge uses h singleton interchanges and one width- $(H - 2h)$ block, instead of $H - h$ singletons.

The stage-two data entrances use the same argument. More generally, the preceding corner argument applies to $\Pi_t \otimes \Pi_U$ with $\text{rank } \Pi_U = H - b > H/2$: require a nonsingular b -corner after conjugating Π_U by K . The resulting profile has b singleton pivots and one block of length $H - 2b$. This adds another finite nonzero-polynomial condition and requires no change to the controlled family.

At stage two, the prefix complement B has dimension $h - 1$. For either physical data wire its entrance residual is $B \otimes \ell^\perp \otimes Q$, where ℓ is its current source or target line and Q is the future line. On the source wire this is the difference between $B \otimes F + P \otimes \ell$ and $(B + P) \otimes \ell$; on the target wire it is the difference between $B \otimes F$ and $B \otimes \ell$. These are exactly the retained rank- $(h - 1)^2$ entrance edges. Reordering the third factor first therefore gives $\Pi_t \otimes \Pi_U$ with

$$\text{rank } \Pi_U = (h - 1)^2 = H - (2h - 1).$$

For $h \geq 4$ this exceeds $H/2$. Apply the generalized argument with $b = 2h - 1$. Each of these $2N$ edges uses $2h - 1$ singleton pivots and one block of length $H - 4h + 2$.

The first corner of a stage-three data entrance. Write $r = h - 1$ and retain $A_3 = (I - \Pi_t) \otimes (I - \Pi_q)$, of nullity $H + r$. Set $K\Pi_q K^{-1} = q'\psi'$, $\psi'(q') = 1$, $\ell_i = e_1^\top G_i$, and $w_j = G_j^{-1} e_h$. Its first- H -row, last- H -column corner is

$$(C_3)_{ij} = \delta_{ij} d_i - q'_i \psi'_j \ell_i (I - \Pi_t) w_j, \quad d_i = \ell_i (I - \Pi_t) w_i = -a_i b_i \neq 0.$$

After row normalization this is $I - Z$, where $\text{rank } Z \leq r$. We additionally require C_3 to be invertible and its first- r -row, last- r -column corner to be invertible. The following explicit witness shows that both requirements are nonzero polynomial conditions.

In an input basis $f_0, \dots, f_r$ of F , fixed rational changes absorbed into K, G_i let us take $t = f_0$, $\varphi = f_0^*$, $q' = (1, \dots, 1)^\top$ and $\psi' = H^{-1}(1, \dots, 1)$. Assign each multiplicity coordinate a label $k(i) \in \{1, \dots, r\}$, such that each of the first and last r coordinates contains the labels exactly once, in the same order. Assign the remaining coordinates arbitrarily. Each label then occurs n_k times with $0 < n_k < H$, since $r \geq 2$. Choose

$$\ell_i = f_0^* + f_{k(i)}^*, \quad w_i = f_0 - f_{k(i)}.$$

They satisfy $\ell_i w_i = 0$ and extend to an invertible G_i with first row ℓ_i and $G_i w_i = e_h$: use the other $h - 2$ coordinate rows between ℓ_i and the last row f_0^* . Then $d_i = -1$, and the normalized corner is

$$I - H^{-1} Z Z^\top, \quad Z_{ik} = [k(i) = k].$$

Its nontrivial eigenvalues are $1 - n_k/H > 0$, so it is invertible. Its indicated off-diagonal r -corner is $-I_r/H$, also invertible. This proves both required nonvanishing statements over $\mathbb{Q}$.

For a matrix $I - Z$ with rank- r update $Z = UV$, an invertible off-diagonal r -corner makes U_L, V_R invertible. The same Schur calculation as for a large projector makes the middle block exactly I_{H-2r} . Explicitly, partition the corner indices into first r , middle $H - 2r$, and last r indices, denoted L, M, R . Then

$$(I - Z)_{MM} - (I - Z)_{MR}(I - Z)_{LR}^{-1}(I - Z)_{LM} = I_M - U_M V_M + U_M V_M = I_M.$$

Idempotence of Z is unnecessary for this calculation. Because C_3 itself is invertible, the remaining r -by- r block also has full rank. Thus eliminating the first H rows uses $2r$ singleton pivots and one contiguous block of length $H - 2r$.

Keep the previously required nonsingular first- $(H + r)$ -row, last- $(H + r)$ -column corner of the full A_3 transform. After the first H pivots above, its Schur complement supplies r further corner pivots. Eliminating all these $H + r$ rows and columns gives the same middle identity block of length $m - 2(H + r)$ as before: the Schur complement is independent of the pivot order inside the eliminated invertible corner. All pivot columns of the first H rows lie in the last H positions, followed by the preceding r columns, so the permitted lower/lower operations suffice throughout. The complete profile is therefore

$$3r \text{ singletons}, \quad H - 2r \text{ contiguous pivots}, \quad m - 2H - 2r \text{ contiguous pivots}.$$

It has total rank $m - H - r$, as required. There are $2N$ such data edges.

One basis for the whole finite network. Each new requirement is a rational-function nonvanishing condition on the original freely varying K, G_i . Each has a rational witness as shown above. The old conditions likewise have nonzero numerators by Section 4.7. Their finite product with all denominator and invertibility factors is nonzero. Thus one rational tuple meets every old and new condition simultaneously. Witnesses for individual factors need not be the same tuple. This reasoning also covers every transposed projector, since the forms used above allow arbitrary rank-one idempotents and arbitrary rank- $(H - h)$ idempotents. Rational enumeration with determinant tests is a terminating fixed-machine preparation.

The stronger moment. For $h = 30$, $H = 900$, the auxiliary entrance profile has 30 singletons and a block of length 840. The data-entrance corner has 87 singletons and a new block of length

842, retaining its length-25142 middle block. The stage-two data entrances each have 59 singletons and a block of length 782. Let S_1 be the singleton count of Section 4.9 and $B_0 = v^2 R$. Put

$$S_2 = S_1 - 840B_0 - (842 + 782)(2N) = 65379670780393117512.$$

The new moment is

$$\Psi_2(x) = \frac{S_2 + B_0(26880^x + 25200^x + 900^x + 840^x) + 2N(25142^x + 842^x + 782^x)}{W 27000^x}.$$

The rank sum at $x = 1$ is unchanged. Sixteen-term positive logarithm enclosures, rounded upward to multiples of 10^{-10} , and $e^z \leq (1 - z)^{-1}$ give the exact strict certificate

$$\Psi_2(1 - 40315/10^{11}) < 1.$$

All child widths contract. Common gate matrices, endpoint identities, row handling and the finite tape alphabet are preserved, so the same arbitrary-width recurrence proof applies.

Lemma 24 (Controlled-corner interchange). *The arbitrary-width interchange conclusion of Lemma 21 holds with saving*

$$1 - \tau = 40315/10^{11}.$$

The scalar producer and rank deficit are unchanged; only the controlled basis and three recursive pivot profiles change.

Proof. Use the simultaneous rational basis and factorization above, with the strict moment $\Psi_2(\tau) < 1$, in the retained finite-network transfer and arbitrary-width row construction. All additional contiguous blocks use the same componentwise shear primitive and preserve their field order. $\square$

The complex producer, whole-residual child sizes, path-depth guard, and complex saving $7/10^7$ remain unchanged. Only the stronger bit interchange is consumed in the following layer interface and assembly.

4.13 Source frames with dimension-thirty data corners

We combine eumemic’s auxiliary source frames (Section 4.10, PR #13) with the dimension-thirty producer of Section 4.9 and the two data-edge refinements of Section 4.12. This combination and the additional data-corner analysis were prepared by Rohan Arun with substantial OpenAI Codex assistance. The source-frame construction and third complex class are credited to eumemic, the bulk recursion to icekylinx, and the ternary producer to Zhihao Chen.

Compatible changes and an unchanged rank deficit. Set $h = 30$, $H = 900$, $m = 27000$, and retain exactly the producer, matching, $R = 17515487$, $v = 142506$, $N = v^3$, $B = v^2 R$, W , and s of the dimension-thirty construction. The proof of Lemma 22 is dimension independent. For every stage-two auxiliary role use source P_{D_0} and sink $I + P_{D_0}$. Its entrance becomes zero, its internal edges are unchanged, and its exit is the rank- $(m - h)$ projector

$$E = I - \Pi_t \otimes \Pi_X, \quad \text{rank } \Pi_X = h.$$

This replaces the old entrance and sink profiles; none of their old width-840, width-900, or width-25200 blocks is counted. Instead the exit has $h = 30$ singleton pivots and one width- $(m - 2h) = 26940$ block. The edge-rank sum, deficit, scalar circuit, and common gate labels do not change. In particular the source-frame modification affects no data edge.

On each of the $2N$ stage-three data entrances, keep the width-25142 middle block and use the refined first corner of Section 4.12: 87 singleton pivots and one width-842 block. On each of the $2N$ stage-two data entrances, use 59 singleton pivots and a width-782 block. The first/third-stage auxiliary joins keep 60 singleton pivots and a width-26880 block, with B copies.

One controlled basis still suffices. The source-frame exit requires a nonsingular first- h -row, last- h -column corner. Lemma 23 proves this is a nonzero rational-function condition in the same free variables K, G_i . Its proof applies at $h = 30$: the first and last h coordinates are disjoint, and a rank- h subspace of V can be sent to any independent h -tuple by K . The first/third join conditions are unchanged.

Impose the additional data conditions proved in Section 4.12: nonzero $a_i b_i$, the nonsingular large stage-three corner, its nonsingular first H corner and off-diagonal $(h - 1)$ corner, and the stage-two data corner of size $2h - 1$. The nonzero $a_i b_i$ may still be imposed even though the old auxiliary sink edges no longer occur; they are conditions on the basis, not additional circuit edges or costs. Every condition individually has a nonzero numerator polynomial, and the parameter space is the same product of general linear groups. Their finite product is nonzero. Thus one rational tuple satisfies every exit, join and data condition, including the finite collection of dual projectors. Enumeration of rational tuples with the stated determinant tests terminates.

All selected blocks have matching physical order under lower/lower elimination. The two data proofs already track their Schur complements; adding nonvanishing conditions for the new exits changes neither those identities nor their ordering. This proves compatibility, rather than assuming that separately optimized bases can be used on different edges.

The combined moment. The complete list of batched widths and multiplicities is

$$(26880, B), \quad (26940, B), \quad (25142, 2N), \quad (842, 2N), \quad (782, 2N).$$

Treat every remaining rank unit as a singleton. Then

$$S = s - B(26880 + 26940) - 2N(25142 + 842 + 782) = 65379670780393117512,$$

$$\Psi(x) = \frac{S + B(26880^x + 26940^x) + 2N(25142^x + 842^x + 782^x)}{W 27000^x}.$$

At $x = 1$ the normalized mass is exactly $1 - 3857/52973979000$. For each child ratio r , use the sixteen-term positive logarithm series, extracting powers of two and rounding its upper endpoint upward to a multiple of 10^{-10} . With $e^z \leq (1 - z)^{-1}$, exact rational arithmetic certifies

$$\Psi(1 - 1816/10^9) < 1, \quad 1 - \text{certified upper bound} > 1.9154 \times 10^{-10}.$$

All child widths are strictly smaller than 27000. The retained mixed-width row construction and recurrence therefore apply with these fixed constants.

Lemma 25 (Source-frame and data-corner interchange). *The arbitrary-width finite-tape interchange interface of Lemma 20 holds with $1 - \tau = 1816/10^9$ for the dimension-thirty construction above.*

Proof. The source-frame endpoint identity, common controlled basis, ordered recursive blocks and strict moment have just been established. Apply Section 4.8, retaining its bounded remainder handling, row reservation, fixed alphabet and spectator fields. $\square$

5 Fast simultaneous synthetic butterfly layers

Throughout this section, the complex constants are $m = m_c = 21952$, $W = W_c = 2085111546336$ and $s = s_c = 45772350635112192$ from Proposition 14. The bit-swap primitive retains its separate arity $m_b = 21952$ and exponent $\tau = 1 - 3/400000000$.

We now apply the complex network to a common butterfly layer on several axes. The array has complete address set

$$[P] \times [2^K]^D \times [S], \quad 1 \leq D \leq d,$$

with $P, S \geq 1$, in lexicographic order. Its D consecutive K -bit chunks each select the bit at one common offset $0 \leq \rho < K$, counted from the least significant end. Each address carries a polynomial record with r complex coefficients, initially in the unit disk on the grid $2^{-p}\mathbb{Z}[i]$, with $\Theta(p)$ bits per real or imaginary component. The intended operation is the tensor product of $H_0(u, v) = ((u + v)/2, (u - v)/2)$ on the D selected bits, acting coefficientwise and preserving all other address fields.

The parameters will satisfy

$$d = \Theta(p^\epsilon), \quad K = \lfloor d^c \rfloor, \quad r = 2^{\Theta(p/d)}, \quad c > 0, \quad 0 < \epsilon < 1.$$

The exponents and comparison constants are fixed, and we impose their further restrictions in Proposition 29. Two consequences explain the layout. The polynomial length r grows faster than every fixed power of p , so a scan of a record pays for polynomial work on its address. Also $K/\log p \rightarrow \infty$, which will make the exceptional set in a packed change of address bits small enough to repair by sorting. We will obtain cost $O(Vd^{\lambda'})$, with $\lambda' < 1$ and V the logical bit volume, whereas separate butterfly factors would require D whole-array passes.

We first construct tensor products of the auxiliary two-point kernel C from Proposition 14. Its phase frames express a C tensor product using smaller groups of selected bits, separated by binary changes of basis. We implement those changes by moving compact dirty-control fields with exact exceptional repair. Splitting complete rows among the network's wire roles then reduces the volume of each recursive call. Finally, we bound the intermediate dyadic values and convert the completed C layer to the normalized H_0 layer. The entire layer is evaluated exactly before one final truncation.

5.1 Phase frames for a simultaneous butterfly layer

Hold the unselected address bits fixed and regard the stored complex values on one wire role as a function of the selected bits. The phase identities below express an edge's residual operator as a product of two-point kernels on selected directions.

Put

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H = \sqrt{2}H_0, \quad S = \text{diag}(1, i), \quad Z = \text{diag}(1, -1).$$

For the bit-flip matrix $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, define

$$C = HSH = aI + bX, \quad a = \frac{1+i}{2}, \quad b = \frac{1-i}{2}. \quad (18)$$

Direct multiplication gives

$$H_0 = bSCS, \quad C^2 = X, \quad C^4 = I, \quad C^{-1} = bI + aX = -iZCZ. \quad (19)$$

In particular, $C = iZC^{-1}Z$. These identities use only Gaussian dyadic coefficients. The Hadamard matrices in the definition of $\mathcal{C}_U$ are used to prove identities; they need not be executed by the machine.

Consider an edge from label U_- to label U_+ , and choose an orthonormal basis $v_1, \dots, v_{\rho_{\text{edge}}}$ of its residual. Proposition 14 gives the frame change

$$\mathcal{C}_{U_+}\mathcal{C}_{U_-}^{-1} = \prod_{j=1}^{\rho_{\text{edge}}} (aI + bX_{v_j})^{\epsilon_j}, \quad \epsilon_j \in \{1, -1\}.$$

Here $(X_v f)(x) = f(x + v)$, and we write $C_v = aI + bX_v$. The number of directional kernels is precisely ρ_{edge} , including on decreasing edges. The next step puts those directions into coordinate positions without performing a general transpose.

Implementing an arbitrary residual basis. For $M \in \text{GL}(m, \mathbb{F}_2)$, let $\mathcal{P}_M$ be the address permutation that moves the value at x to Mx . Thus $(\mathcal{P}_M f)(x) = f(M^{-1}x)$, and

$$\mathcal{P}_M X_v \mathcal{P}_M^{-1} = X_{Mv}. \quad (20)$$

Extend $v_1, \dots, v_{\rho_{\text{edge}}}$ to a basis of $\mathbb{F}_2^m$ and take these vectors as the first ρ_{edge} columns of M . The displayed edge identity then becomes

$$\mathcal{C}_{U_+}\mathcal{C}_{U_-}^{-1} = \mathcal{P}_M \left(\prod_{j=1}^{\rho_{\text{edge}}} C_{e_j}^{\epsilon_j} \right) \mathcal{P}_M^{-1}, \quad (21)$$

where e_j is the j th standard basis vector. The matrix M need not preserve the dot product: equation (20) only uses its invertibility.

For the array layout used below, take a group of $e = mf$ consecutive axis chunks and divide it into m consecutive slots of f chunks, numbered in physical order. Number the chunks within each slot by $j = 0, \dots, f-1$, starting at its least significant end. Write $x_{h,j}$ for the selected bit in chunk j of slot h , where $1 \leq h \leq m$. Column j is the vector $(x_{1,j}, \dots, x_{m,j})$. A frame on this group is the tensor product of the same phase frame $\mathcal{C}_U$ of Proposition 14 on these f columns.

To implement (21) on the group, first apply $\mathcal{P}_M^{-1}$ separately to each column. For each $1 \leq h \leq \rho_{\text{edge}}$, apply $(C^{\epsilon_h})^{\otimes f}$ to the f bits $x_{h,0}, \dots, x_{h,f-1}$. This is one child operation on the f axes in slot h . Finally apply $\mathcal{P}_M$ to every column. A negative directional kernel still requires only one forward child, because

$$(C^{-1})^{\otimes f} = (-i)^f Z^{\otimes f} C^{\otimes f} Z^{\otimes f}. \quad (22)$$

The rightmost phase is applied first. All inverse wrappers are address signs and constant fourth-root phases, so they add no recursive calls.

Gaussian elimination expresses any $M \in \text{GL}(m, \mathbb{F}_2)$ as a product of at most $m(m-1) + 3m$ elementary row additions: one clears at most $m-1$ other entries per pivot, and each row swap is three row additions. Reversing the elimination gives the required product since each row addition is an involution. The number is a constant because m is fixed. Adding row h' to row h in every column applies $x_{h,j} \leftarrow x_{h,j} \oplus x_{h',j}$ for every $0 \leq j < f$. Thus one row addition changes corresponding selected positions in two whole slots. The packed address operation proved below performs all of these XORs together.

The exact recursive contract. Apply the finite network with these column frames to a group of $e = mf$ selected axes. Summing its residual dimensions gives exactly s_c child calls, each on the $f = e/m$ axes of one slot. The source and sink $Z_{a,f}$ signs, together with the scalar i^{27f} at each affected sink, turn every endpoint kernel into a forward one. Negating the prescribed physical bank and undoing its exchange returns each result to its original role. Proposition 14 therefore gives $C^{\otimes e}$ on every role, including roles whose initial scratch values were arbitrary. Formula (22) supplies each inverse child from one forward call, and none of the corrections adds a child. The remaining operations are a fixed number of binary row additions and coefficientwise scalar operations.

Returning to normalized butterflies. For an address x in a full layer, write $x_1, \dots, x_D$ for its selected bits. The diagonal operator $\mathcal{S}_D$ multiplies the value stored at an address by the phase determined by those bits:

$$(\mathcal{S}_D z)(x) = i^{x_1 + \dots + x_D} z(x).$$

Here the unselected address bits are unchanged, and the same phase is applied to every coefficient stored at that address. Applying (19) on all selected coordinates gives

$$H_0^{\otimes D} = b^D \mathcal{S}_D C^{\otimes D} \mathcal{S}_D. \quad (23)$$

The scalar is

$$b^D = (-i/2)^{\lfloor D/2 \rfloor} b^{D \bmod 2},$$

so it is a binary shift and fourth-root phase, with at most one further factor b . All phases, inverse wrappers, and scalar corrections above are exact Gaussian dyadic operations. Thus, once the physical binary basis changes and recursive stream layout are supplied, the complete parallel butterfly layer is exact up to its prescribed final truncation.

5.2 Packed changes of selected address bits

This wide-slot implementation remains available for the grouped-rectangle appendix. The accelerated layer below instead uses the compact-control construction of Section 5.3.

We implement a binary row addition on corresponding selected bits in two address chunks. A third chunk supplies temporary address bits; its arbitrary initial value must be restored. The construction first uses a constant number of cyclic shifts of whole chunk ranges, then corrects an explicitly bounded set of addresses. The correction depends only on addresses, so the time bound is worst-case over all payloads.

We use three previously established streaming operations. Lemma 18 exchanges two disjoint address chunks of equal width L in $O(VL^\tau)$ time, where V is the current stream volume and $0 < \tau < 1$ is fixed. The construction in Lemma 4 cyclically shifts the complete range of a target chunk by an offset determined by preceding fields in $O(V)$ time apart from calculating the offsets. Finally, Lemma 3 applies any fixed permutation to the two-bit value at two specified address positions in $O(V)$ time. Each operation uses a fixed number of one-dimensional tapes and restores the order of all spectator fields.

The asymptotic simplification in the following lemma describes a fixed family of parameters as $p \rightarrow \infty$, after the exponents and comparison constants have been fixed. The statement that R exceeds every fixed polynomial in p means that $R/p^C \rightarrow \infty$ for each fixed C . Implied constants and eventual cutoffs may depend on the fixed choices and on when the family's asserted bounds begin to hold. In particular, each explicit bound is uniform over all of its declared inputs; its asymptotic simplification is uniform over the allowed row counts and payloads within a fixed family.

Lemma 26 (Packed selected-bit addition on one rectangle). *Let $R \geq 1$, $K \geq 6$, $f \geq 1$, $L = fK$, and $0 \leq \rho < K$. Let N_0, N_1, N_2, N_3 be positive integers, and let (χ_1, χ_2, χ_3) be a permutation of (x, y, z) . Consider R -bit records in lexicographic order on*

$$[N_0] \times [2^L]_{\chi_1} \times [N_1] \times [2^L]_{\chi_2} \times [N_2] \times [2^L]_{\chi_3} \times [N_3].$$

The subscript names the chunk at that position. The input supplies $N_0, \dots, N_3, R, K, f, \rho$ as a fixed number of canonical binary integers, together with the finite permutation of the chunk names, as in Section 2. Put

$$M = 2^{3L} \prod_{h=0}^3 N_h, \quad V = MR, \quad A_{\text{rect}} = \lceil \log_2(2V) \rceil.$$

At the selected positions $j_i = \rho + iK$, numbered from the least significant bit, one fixed multitape procedure performs

$$y_{j_i} \leftarrow y_{j_i} \oplus x_{j_i} \quad (0 \leq i < f).$$

It preserves the spectator coordinates, all other chunk bits, and every record payload; in particular, it restores the entire chunk z for every initial value. With $\delta_{\text{bad}} = \min\{1, 80(f-1)2^{-K}\}$, its time is

$$O\left(V(L^\tau + 1) + MA_{\text{rect}}^3 + \delta_{\text{bad}} MA_{\text{rect}}(R + A_{\text{rect}})\right). \quad (24)$$

When $A_{\text{rect}} = O(p)$, R exceeds every fixed polynomial in p , $f \leq d \leq p$, and $K/\log p \rightarrow \infty$, this is $O(V(L^\tau + 1))$, uniformly over the permitted lengths and payloads in the fixed family.

Proof. We first construct eight rotations that have the required effect away from a small set of addresses. We then correct exactly that set. For a Boolean control c and arbitrary integers u, w , perform the following eight operations in the displayed chronological order. Every predicate uses the current values of the integers.

$$\begin{aligned} u &\leftarrow u - c[w \equiv 0 \pmod{2}], & (1) \\ w &\leftarrow w - c[u \equiv 0 \pmod{2}], & (2) \\ u &\leftarrow u + c[w \equiv 0 \pmod{2}], & (3) \\ w &\leftarrow w + c[u \equiv 0 \pmod{2}], & (4) \\ w &\leftarrow w + c[u \equiv 1 \pmod{2}], & (5) \\ u &\leftarrow u + c[w \equiv 1 \pmod{2}], & (6) \\ w &\leftarrow w - c[u \equiv 1 \pmod{2}], & (7) \\ u &\leftarrow u - c[w \equiv 1 \pmod{2}]. & (8) \end{aligned} \quad (25)$$

If $c = 0$, these operations do nothing. If $c = 1$, the first four operations induce the cycle (00 01 10) in every square

$$\{2a, 2a+1\} \times \{2b, 2b+1\}, \quad a, b \in \mathbb{Z},$$

where the corner labels record the parities of (u, w) . The last four induce (01 10 11). Applying the first cycle and then the second gives (00 10)(01 11). These facts follow by substituting the four parity pairs into the displayed operations; translation by $(2a, 2b)$ does not change any predicate. Consequently the complete output is

$$(u', w') = (u + c(1 - 2(u \bmod 2)), w). \quad (26)$$

Here $u \bmod 2 \in \{0, 1\}$, also for negative u . Thus the parity of u is flipped precisely when $c = 1$, both quotients under division by two are preserved, and the arbitrary auxiliary integer w is restored.

Each coordinate is updated four times by at most one, so its displacement from its initial value is at most four, even if an intermediate point leaves its starting quotient square. The weaker bound eight will suffice below.

Omit the most significant selected position j_{f-1} temporarily. For each $0 \leq i < f-1$, the bits in positions $j_i, \dots, j_i + K - 1$ form a K -bit segment. When $f > 1$, these segments are disjoint, and the highest one ends immediately below j_{f-1} . Write their initial values in y and z as

$$u_i = 2g_i + a_i, \quad w_i = 2h_i + b_i, \quad a_i, b_i \in \{0, 1\}.$$

The values $g_i, h_i \in \{0, \dots, 2^{K-1} - 1\}$ are the guard values. The control for this pair of segments is the selected bit $c_i = x_{j_i}$.

Apply each line of (25) simultaneously to all these segment pairs. Physically, a line updating y by sign ε and testing parity b of z is the single cyclic shift

$$y \longleftarrow y + \varepsilon \sum_{i=0}^{f-2} 2^{j_i} [x_{j_i} = 1] [z_{j_i} = b] \pmod{2^L}. \quad (27)$$

For a line updating z , interchange y and z in this formula. The shift offset never depends on its target chunk, so each line is a bijection of the rectangle. Let S_0 be their composition. It fixes the spectator coordinates.

Let $\mathcal{B}$ consist of the addresses for which at least one used guard value, in either y or z , fails

$$10 \leq g \leq 2^{K-1} - 11. \quad (28)$$

At an address outside $\mathcal{B}$, each segment initially lies in $[20, 2^K - 21]$. Even displacement eight keeps it inside $[0, 2^K - 1]$. Inductively, each packed addition therefore agrees with all the independent segment additions: no segment carries into or borrows from the next segment. When $f > 1$, the bits below j_0 and above the highest used segment are untouched as well. By (26), S_0 then agrees with the ideal permutation T_0 that performs the required XORs at $j_0, \dots, j_{f-2}$ and fixes all other address coordinates.

There are exactly twenty excluded values for each guard in (28). Since its complete range has 2^{K-1} values, the union bound over the $2(f-1)$ guards gives

$$|\mathcal{B}| \leq \delta_{\text{bad}} M. \quad (29)$$

Indeed, the complete y, z ranges give the same guard distribution at each choice of spectator coordinates. No record value enters this count.

The ideal map T_0 preserves all guard values and therefore preserves $\mathcal{B}$. Both T_0 and S_0 are bijections, and they agree off $\mathcal{B}$. Taking complements in the finite rectangle gives

$$S_0(\mathcal{B}^c) = T_0(\mathcal{B}^c) = \mathcal{B}^c, \quad S_0(\mathcal{B}) = \mathcal{B}.$$

Thus $T_0 S_0^{-1}$ fixes $\mathcal{B}^c$ and permutes $\mathcal{B}$. It remains to implement the rotations and this correction on tapes.

For one rotation, move its target to the rightmost of the three chunk positions if necessary. Let P^{sw} be the product of the record field lengths before the earlier exchanged chunk, let G^{sw} be the product between the exchanged chunks, and let B^{sw} be R times the product after the later one. Empty products are one. The bit array has exactly the shape

$$[P^{\text{sw}}] \times [2^L] \times [G^{\text{sw}}] \times [2^L] \times [B^{\text{sw}}].$$

The third chunk lies inside one collapsed spectator interval. Pass only L and these three positive canonical lengths to Lemma 18. Each length is at most V , so the local descriptor has $O(A_{\text{rect}})$ bits. The swap costs $O(VL^\tau)$, including its own padding and cleanup.

The equal chunk lengths leave the field-length list unchanged by a completed swap. Keep the permutation of the logical names x, y, z in finite local state. With the target now after its two controls, collapse the remaining intervals to obtain

$$[\Lambda_0] \times [2^L]_{\text{control 1}} \times [\Lambda_1] \times [2^L]_{\text{control 2}} \times [\Lambda_2] \times [2^L]_{\text{target}} \times [B_{\text{rot}}],$$

where B_{rot} includes the payload and every displayed spectator length is positive. Their product with the chunk lengths is V . Together with K, f, ρ and the finite name permutation, these fixed canonical lengths form an $O(A_{\text{rect}})$ -bit local descriptor. The offset routine receives the two current L -bit logical control values. A scan of at most $f \leq L$ selected positions evaluates (27) in $L^{O(1)}$ time, without reading the target or spectator values. Lemma 4 therefore gives an $O(V)$ rotation: its offset work is absorbed by the 2^L nonempty target blocks in each fiber. Undo the swap after the rotation. The local descriptors and name permutation are updated at completed swap boundaries; the primitives use their own descriptors inside padded layouts. There are at most sixteen swaps and eight rotations in S_0 .

To implement the correction, scan the stream with a record-rank counter. All chunk names are back in their original positions. Dividing the current rank through the fixed seven field lengths recovers the three chunk values and their mixed-radix record strides. The strides exclude the R payload bits. All involved integers have $O(A_{\text{rect}})$ bits, so schoolbook arithmetic, descriptor copying, and resets take at most $O(A_{\text{rect}}^3)$ per record. Test membership in $\mathcal{B}$ and extract just those records. For a current exceptional address q , compute $q' = T_0(S_0^{-1}(q))$ by reversing the eight modular additions with negated offsets, then applying the selected XORs directly. Each reversed mask uses the current controls of that reversed line. There are at most $f \leq L \leq A_{\text{rect}}$ selected positions, so this calculation also fits $O(A_{\text{rect}}^3)$ time.

If q_h, q'_h are the current and destination values of chunk $h \in \{x, y, z\}$ and σ_h is its record stride, the fixed spectator coordinates give

$$\text{rank}(q') = \text{rank}(q) + \sum_{h \in \{x, y, z\}} (q'_h - q_h) \sigma_h.$$

The differences are signed. Attach this destination rank, padded to $\lceil \log_2 M \rceil \leq A_{\text{rect}}$ bits, to the extracted record, and mark its hole on a fixed track of the full stream. Hold the full stream untouched while stable binary radix sorting scans only the extracted records and keys, least significant key bit first. Since $T_0 S_0^{-1}$ permutes $\mathcal{B}$, these keys are distinct and are exactly the ranks of the marked holes. One forward full-stream scan reinserts the sorted records. This implements T_0 everywhere.

Finally apply the two-bit XOR to $x_{j_{f-1}}, y_{j_{f-1}}$ by Lemma 3, splitting those chunks at the selected positions and collapsing the other intervals. Its descriptor again has a fixed number of positive lengths, each at most V . This costs $O(V)$ and supplies the omitted selected position. When $f = 1$, S_0, T_0 are identities, $\mathcal{B}$ is empty, and this last step is the entire algorithm.

The movement costs $O(V(L^\tau + 1))$. A fixed number of full record scans costs $O(MA_{\text{rect}}^3 + V)$, and the at most A_{rect} radix passes cost $O(|\mathcal{B}|A_{\text{rect}}(R + A_{\text{rect}}))$; skip them when the extracted bank is empty. The initial descriptor has a fixed number of canonical integers of $O(A_{\text{rect}})$ bits, so its preparation is covered by $O(MA_{\text{rect}}^3)$. All counters, positioning, rewinds, and cleanup are included. Equation (29) proves (24) on a fixed number of tapes. After division by $V = MR$, its additional terms are bounded by A_{rect}^3/R and $80(f-1)2^{-K}A_{\text{rect}}(1 + A_{\text{rect}}/R)$. Under the stated family bounds the first tends to zero, and the second is $O(p^2 2^{-K}(1 + A_{\text{rect}}/R)) = o(1)$. This proves the final assertion. $\square$

Application to a fixed binary change of basis. An invertible matrix over $\mathbb{F}_2$ of fixed order $m \geq 3$ is a product of a fixed number of elementary row additions; a row exchange is three such additions. In one invocation of the finite network on $e = mf$ axes, slot h consists of f consecutive K -bit axis chunks. It therefore has address-bit width $L = fK$, and its selected bits $x_{h,j}$ occupy positions $\rho + jK$ for $0 \leq j < f$. To add row h' to row h , use slot h' as x , slot h as y , and any third, distinct slot as z in Lemma 26. The lemma performs $x_{h,j} \leftarrow x_{h,j} \oplus x_{h',j}$ for every j in one operation and restores the third slot, whatever its initial address value.

Under the asymptotic hypotheses in the final assertion of that lemma, the complete change of basis and its inverse therefore cost $O(V((eK)^\tau + 1))$ on a stream of volume V . Here e counts axis chunks, while eK counts address bits; the fixed factor m between eK and the slot width does not affect the bound. The same conclusion holds after the row padding and splitting below, because every role stream retains a complete suffix with the same fixed field lengths in every row.

5.3 Compact dirty controls for selected-bit additions

The following replacement for the wide-slot packed operation moves only compact control fields. Temporary address fields have arbitrary contents; they are complete ranges of existing address bits, not newly allocated independent coordinates. Their availability in every recursive call is proved separately below.

Lemma 27 (Controlled rotations with record-paid offset work). *Consider M records of R bits on a complete rectangular address set, of volume $V = MR$. Its fixed number of field lengths and controls have at most $A = \lceil \log_2(2V) \rceil$ bits each. A binary target field follows all its controls. Suppose its rotation offset, independent of that target, is computable from the controls and descriptors in $O(A^C)$ tape steps for some fixed C . Then the rotation can be performed on fixed tapes in $O(V + MA^{C'})$ time for a fixed C' . In particular, this is $O(V)$ when $A = O(p)$ and R exceeds every fixed power of p .*

Proof. Fix the prefix before the target. If its range is Q and each target block has B suffix bits, rotate by a through the two-piece split at $Q - a$ and merge in the opposite piece order. This is the controlled-rotation construction from the ordered-affine streaming lemma. Each data bit is copied a bounded number of times. There are at most M prefix fibers, since every fiber contains at least one complete record. At each fiber, compute the offset, initialize its two piece lengths, and reset the finite collection of counters; recompute the offset at the merge. Ordinary binary arithmetic and descriptor copies cost a fixed polynomial in A per fiber. The controls can be wider than the target: their lengths are bounded by A , not by the target width. Local suffix-length copies and ripple counters are charged as in the original streaming proof, so no growing descriptor is scanned per payload bit. Two piece tapes, output and fixed arithmetic work tapes suffice; rewinds and clearing are over the regions just written. This gives the stated bound. Finally $MA^{C'}/V = A^{C'}/R = o(1)$ in the specified record regime. $\square$

Let x, y be disjoint slots of fK bits, with corresponding selected positions $j_i = \rho + iK$, $0 \leq i < f$, $0 \leq \rho < K$. Put $n = f - 1$. For $f = 1$ use the elementary two-bit XOR. Otherwise fix a positive integer G , put $B = 2^G$, and assume two disjoint complete nG -bit fields u, t precede both slots and a third such field b follows both slots. Arbitrary spectator fields may intervene. Each compact field has n radix- B digits. The selected-bit operation will preserve these three entire fields.

Earlier source. Suppose x precedes y . The unused field u is a spectator. For an integer target segment v , a bit z , and an arbitrary integer temporary w , perform

$$v \leftarrow v + 2zw, \quad w \leftarrow w + (v \bmod 2), \quad v \leftarrow v + z(1 - 2w), \quad w \leftarrow w - ((v \bmod 2) \oplus z).$$

Writing the original parity as a , the first update leaves it unchanged. The final values are $v + z(1 - 2a)$ and the original w . Thus this toggles only the parity of v when $z = 1$, and restores the dirty temporary.

Apply these four updates simultaneously, with $z_i = x_{j_i}$, to the low n selected segments of y and the digits t_i of t . A target-slot update is one modular rotation of y , whose offsets are respectively

$$\sum_{i < n} 2x_{j_i} t_i 2^{j_i}, \quad \sum_{i < n} x_{j_i} (1 - 2t_i) 2^{j_i} \pmod{2^{fK}}.$$

Both control fields precede y . For a temporary update, swap t with b , rotate the back field by the packed selected bits of the current y , and swap back. For unloading, use minus the packed bits $y_{j_i} \oplus x_{j_i}$. These controls precede the back field. There are four width- nG swaps and four rotations. The arbitrary original contents of b return after each pair of swaps.

Later source. If x follows y , first execute the earlier-source construction with control bits $u_i \bmod 2$. Load the selected bits of x into u by swapping u, b , rotating the back field by their packed radix- B representation, and swapping back. Repeat the earlier-source construction with the new parities of u , then unload the selected bits of x from u by the inverse load. When the digit loads do not overflow, the two control parities differ by exactly x_{j_i} . The resulting target update is the required XOR, with u, t, b restored. This version uses twelve compact swaps and ten rotations. Every offset depends on a fixed number of fields preceding its target.

Guards and deterministic repair. Temporarily omit the highest selected position. The low n segments $[j_i, j_i + K)$ are disjoint and fit inside y , including when $\rho = K - 1$. Write each initial segment as $2g_i + a_i$, $a_i \in \{0, 1\}$. Declare an address bad if a t digit is $B - 1$, or a target guard fails

$$2B \leq g_i < 2^{K-1} - 2B.$$

For a later source also declare it bad if any u digit is $B - 1$. On a good address each temporary load stays within its digit, and the target displacement is at most $2B$ in either execution of the four-step identity. The conservative guard interval therefore prevents every carry or borrow between the wide segments. After each completed identity the guard is unchanged. Induction through the later-source sequence proves the desired map on the good set, including restoration of every temporary and every unselected address bit.

Uniform complete ranges give the common bound

$$\delta = \min\{1, n(2 \cdot 2^{-G} + 8 \cdot 2^{G-K})\}$$

on the bad fraction; for an earlier source the first factor 2 can be omitted. The ideal map T changes only the selected target parities, so it preserves the bad set. The actual sequence S is a bijection on the entire rectangle: each rotation is independent of its target and all swaps are bijections. Since $S = T$ off the bad set, S preserves that set too. Thus TS^{-1} is supported there, exactly as in the original exceptional-stream repair.

Extract the bad records, retaining marked holes in the full stream. For each extracted address q , reverse the swaps and modular rotations to compute $S^{-1}q$, then compute $T(S^{-1}q)$. All fields

and descriptors have $O(A)$ bits. A constant number of offset calculations each scan at most $n \leq A$ digits and combine shifted integers of $O(A)$ bits. Integer arithmetic, mixed-radix decoding, inverse offsets and the destination key can consequently be computed in $O(A^3)$ per record. Sort only the extracted records by their $O(A)$ -bit destination keys, using stable binary radix passes, then reinsert them in their marked holes. Keys are distinct and fill precisely those holes because TS^{-1} permutes the bad set. This uses fixed tapes and costs

$$O(V + MA^3 + \delta MA(R + A)).$$

Every inverse offset is evaluated on the current inverse controls; modular overflow on bad addresses is handled exactly, rather than treated as an integer identity. Finally, supply the omitted highest selected position through the elementary two-bit XOR in $O(V)$ time.

Proposition 28 (Compact-control selected-bit addition). *Under the stated complete-field hypotheses, take*

$$G = 4\lceil \log_2 p \rceil + 6, \quad f \leq p, \quad A = O(p), \quad K/\log p \rightarrow \infty,$$

and let R exceed every fixed power of p . The exact selected-bit row addition, with all temporary fields restored, costs

$$O(V((f \log p)^\tau + 1))$$

on a fixed number of tapes, for either order of the two slots. Constants and cutoffs are uniform within each fixed parameter family of the layer theorem.

Proof. Use the retained arbitrary-width chunk-swap primitive for the at most twelve swaps, each of width nG . Its padding, cleanup and all spectator intervals are covered by that interface. For every rotation, apply Lemma 27; its offset arithmetic is polynomial in A . No new record volume is introduced by describing the compact fields. For $\ell_p = \lceil \log_2 p \rceil$, eventually $K \geq G + 4\ell_p + 10$. Then

$$\delta \leq \frac{5}{128p^3}, \quad \frac{A^3}{R} = o(1), \quad \delta A(1 + A/R) = O(p^{-2}).$$

The repair is therefore $O(V)$ per call, including predicate scans and key arithmetic. The elementary highest-bit operation costs $O(V)$. All calls reuse their fixed work tapes and finish in the declared physical field order. $\square$

Application to a fixed binary change of basis. The original phase-frame construction expresses each binary basis change as a fixed number of row additions on m slots of $f = e/m$ axes. Use the two slots of each addition as x, y and the reserved front/back compact fields as controls. The complete basis change and its inverse cost $O(V((e \log p)^\tau + 1))$. The reservations below guarantee their complete ranges, independently of which slots are involved, at every recursive node.

5.4 Batching a common layer on a fixed number of tapes

We now turn the finite complex network into an algorithm for a whole layer. Throughout this subsection its constants are denoted by m, W, s : the network has W wires, its binary label space has dimension m , and the sum of the dimensions of its edge residuals is $s < Wm$. These are the constants of the complex network; they need not equal the constants used to prove the chunk-swap lemma.

Use the complete layout and parameter family specified at the start of the section, with fixed rational exponents c, ϵ to satisfy the recurrence and guard conditions below. Store each polynomial

in coefficient order with a common width $w = \Theta(p)$ per real or imaginary component, and assume the complete address and its binary shape descriptor have length $O(p)$. Thus an input record has $2rw$ bits. Write V for the current logical bit volume, excluding temporary work streams. We first count the work on $C^{\otimes D}$ assuming one common exact coefficient format of width $O(p)$; the guard calculation below then supplies that format.

An individual selected-bit kernel. Write each input coefficient once in this common exact format and retain it until the final truncation. The input width is $\Theta(p)$, so the initial copy costs $O(V)$ and the guarded stream has volume $O(V)$.

To apply one C kernel, use the split from Lemma 3 at the selected address bit. Call the two streams 0 and 1 according to that bit. Because that bit has its complete two-element range, the two streams have the same remaining address set in the same order. Their aligned record addresses therefore differ only in the selected bit, and their coefficients are still ordered by the same polynomial index. Scan each pair of records in that order. For corresponding complex coefficients u from stream 0 and v from stream 1, compute

$$C(u, v) = \left(\frac{u + v + i(u - v)}{2}, \frac{u + v - i(u - v)}{2} \right). \quad (30)$$

Addition, subtraction and halving use the fixed-point conventions of Section 2; multiplication by i swaps the real and imaginary components and changes one sign. The common guarded width makes these operations exact. The coefficient scan costs $O(rp)$ per pair of records, linear in their guarded bit length and including any reversal of bits inside a coefficient. Write the first result to stream 0 and the second to stream 1, both in coefficient order, then merge according to the selected bit. The merge restores the original lexicographic address order. The elementary split and merge include their counters and cleanup and cost $O(V)$; the paired arithmetic scan also costs $O(V)$. Thus e individual kernels cost $O(Ve)$ on any stream with these complete ranges and record format.

First consider an internal node containing $e = m^k$ consecutive active chunks, with $k \geq 1$. Split it into m consecutive slots, each containing e/m chunks. The phase construction supplies one recursive call on a slot for each vector in an edge residual basis, hence exactly s calls in one invocation of the network. If the logical volume entering this node is V , binary changes of basis, their inverses, scalar gates, endpoint corrections and reassembly of the physical banks cost

$$O(V((e \log p)^\tau + 1))$$

at this node. Each slot has $(e/m)K$ address bits, so Proposition 28 gives the stated bound for a row addition, using the complete compact fields constructed below. The network and all of its m -dimensional residual bases are fixed, so the total number of row additions is fixed as well. At the same guarded width, the scalar gates cost a constant number of scans of the current volume. Address and descriptor work bounded by any fixed polynomial in p per record is also absorbed: its ratio to the record's $\Theta(rp)$ bits tends to zero.

Rows and complete compact fields. Let m, W, s denote the complex network constants and keep the layer's global p, d, K at every node. Put

$$G = 4\lceil \log_2 p \rceil + 6, \quad H = dG, \quad q_F = \left\lceil \frac{2H}{K} \right\rceil, \quad q_B = \left\lceil \frac{H}{K} \right\rceil,$$

$$q_0 = \lceil \log_2 W \rceil \lceil \log_m(2d) \rceil, \quad k_0 = \lceil \log_m d \rceil, \quad Q_0 = q_0 + q_F + q_B.$$

If $D \leq Q_0$, apply all D selected kernels individually. Otherwise, process individually the first $q_0 + q_F$ and last q_B selected axes. This uses existing coordinates only. The remaining active axes are a consecutive middle block of length $D - Q_0$.

Regard the first q_0 chunks as the row index. Immediately after it, the next q_F chunks have at least $2H$ bits. Reinterpret these consecutive bits as two H -bit front fields and a spectator remainder; a remainder of length zero is omitted. The last q_B chunks similarly supply one H -bit back field and a spectator remainder. These reinterpretations move no records and preserve the total address space. For a call with f axes per slot, carve $nG = (f - 1)G \leq H$ consecutive bits from each of these fields; collapse their unused bits into spectators. The resulting field list has a fixed number of entries. Both front fields precede all active slots and the back field follows them, so the compact-control proposition applies to every ordered pair of slots. For $f = 1$ no compact fields are used.

The original row range has length $R_{\text{row}} = 2^{q_0 K} \geq W^{k_0}$. For each original prefix, pad this row range to the next multiple of W^{k_0} by adding zero rows. A row includes the entire complete front fields, active middle block, back field, all spectator coordinates and polynomial record suffix. Padding adds whole such rows, so increases total volume by at most two and does not change any within-row range.

At a node write the row index as $u = Wg + w$, $0 \leq w < W$, and send the entire row to role w . The index u consists solely of the first row region: *no compact-field bit is included in it*. Each role therefore has the same complete within-row address set and exactly $1/W$ of the parent volume. At depth j the remaining row count is divisible by W^{k_0-j} . This invariant survives further splits, padding, and reassembly. Pointwise gates on different roles have aligned g and every remaining coordinate. All compact operations restore the temporary fields before a scalar gate, recursive child, or returned result uses that layout.

The fixed-tape scheduler is unchanged: park only the other role streams at the active stack top, execute the child on a complete role stream, and restore the parked streams on return. Every child has its own current logical volume; copying, parking, rewinds and clearing cost a fixed multiple of that volume. The constant number of compact operations per edge use fixed work areas. Descriptors include the global p, d, K, G, H and the carved field lengths; all have $O(p)$ bits and there are a fixed number of fields. The existing record suffix absorbs polynomial setup at every node and per record.

Each completed invocation acts as the desired C tensor on its active axes and as identity on the reserved and spectator coordinates. It therefore commutes with the earlier individual kernels, including those on bits now interpreted as compact fields. Restoring the row permutation returns the result to each original row separately. A padded zero row is consequently zero at completion and can be deleted exactly as before. Neither temporary field contents nor intermediate scratch payloads were assumed to be zero.

The number of individually processed axes is at most Q_0 , including the fallback $D \leq Q_0$. For large p , $K \geq d^c/2$ and

$$q_F + q_B \leq \frac{3dG}{K} + 2 \leq 6d^{1-c}G + 2.$$

Thus preprocessing, padding and reservation cost

$$O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1)).$$

All reserved axes lie within the original D axes; in particular their total number never exceeds d . This fact also bounds their coefficient dependency depth by the same $8d$ allowance used in the guard proof.

Unrolling the compact-control recurrence. Fix rational $0 < \beta < 1$. Choose $0 < \tau < 1$ and $\max\{0, \log_m(s/W)\} < \sigma < 1$. Stop when $e < d^\beta$ and apply the remaining kernels individually. With $F(e)$ denoting time per current logical volume, the exact row split gives

$$F(e) \leq (s/W)F(e/m) + O(G^\tau e^\tau + 1), \quad F(e) = O(e) \quad (e < d^\beta).$$

Here G and the compact reservations are global parameters, not rebuilt at each node. This is why no factor K^τ appears in the overhead. Repair costs are already included in the node overhead.

Write $b = s/W$. At level j the internal contribution is bounded by $O(G^\tau e^\tau (b/m^\tau)^j)$. Summing through the stopping depth gives

$$O(G^\tau \log(2d)d^\chi), \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Indeed if $\sigma \leq \tau$ the root bounds every summand; otherwise the last internal level costs at most the root bound times a fixed constant and $d^{(1-\beta)(\sigma-\tau)}$. The additive 1 is absorbed by $G^\tau e^\tau$ at internal nodes. The leaf contribution is at most $O(d^{\sigma+\beta(1-\sigma)})$, by the original volume-weighted argument.

Partition $D - Q_0$ into consecutive powers of m , using its base- m expansion. There are $O(\log(2d))$ pieces. Include their costs and the reservation cost to obtain, per original volume,

$$O\left((\log p)^3 [d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c, 0\}} + 1]\right).$$

We used $\log d = \Theta(\log p)$ and $G = O(\log p)$ for each fixed $d = \Theta(p^\epsilon)$ family. Any fixed rational exponents satisfying

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda'$$

therefore give the complete $O(Vd^{\lambda'})$ layer bound. The strict gaps absorb the logarithms. The small- D fallback obeys the same bound. For fixed rational $\beta = u/v$, implement stopping by the integer test $e^v < d^u$; its polynomial descriptor cost is already included.

5.5 A generalized stopped-depth guard

Use the complex network's own constants m, W, s , independently of the bit network used by swaps. Assume $m \geq 3$ and $2 \leq s < m^5$, as checked below. For rational $0 < \beta < 1$ and $\zeta > 0$, put

$$E = 64(W + m + 1)^3, \quad B = s + E, \quad C_1 = 5 - 4\beta + \zeta,$$

$$C_0 = \left\lceil \max\{128mB^2, 18mB^2(1 + 1/\zeta)\} \right\rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

As in the retained exact-arithmetic guard proof, a dependency-depth upper bound is

$$A(e) \leq sA(e/m) + E \quad \text{internally}, \quad A(e) \leq 8e \quad (e < d^\beta).$$

The fixed node charge E remains additive. All new compact-control, padding and repair operations only permute complete coefficient encodings and introduce no coefficient arithmetic.

For a root $e \leq d$, the number of internal levels is at most $(1 - \beta) \log_m d + 1$. Hence

$$s^j \leq sd^{5(1-\beta)}, \quad A(e) \leq s(8 + E)d^{5-4\beta} \leq 9B^2 d^{5-4\beta}.$$

A root that is already a leaf also satisfies this bound. There are at most $m(1 + 1/\zeta)d^\zeta$ base- m pieces, using $\log d \leq d^\zeta/\zeta$ and $\log m > 1$. Individually processed row, front and back axes are disjoint and

number at most d , so their depth is at most $8d$. The outer phases and b^D conversion add at most $\lfloor D/2 \rfloor + 9$ units, including denominator shifts. Consequently

$$A_{\text{layer}} \leq 9mB^2(1 + 1/\zeta)d^{C_1} + 18d \leq C_0d^{C_1}.$$

The exact integer comparison for the final inequality is included in the certificate. The usual induction gives denominator exponent at most $p + \Delta$ and magnitude at most 2^Δ . Thus $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits suffice per component. The condition $\epsilon C_1 < 1$ makes these widths $O(p)$, uniformly over the retained parameter bands. Integer power comparisons implement the rational guard ceilings in polynomial descriptor time.

Proposition 29 (Simultaneous normalized butterfly layer). *Fix $\tau = 1 - 3/400000000$ and $\sigma = 1 - 39/10^9$. Let $c, \epsilon, \lambda, \lambda', \beta, \zeta$ be fixed positive rational numbers with $0 < \beta < 1$, and put*

$$C_1 = 5 - 4\beta + \zeta, \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Require

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda', \quad \epsilon C_1 < 1.$$

Fix positive constants $a_d \leq b_d$ and $a_r \leq b_r$, and fix positive constants $C_M, C_{\text{desc}}, C_w$ with $C_w > 1$. For these fixed choices there are a cutoff p_0 , one fixed finite-alphabet multitape procedure, and an implicit time constant with the following guarantee. The cutoff and time constant may depend on the fixed rational parameters and these constants. For every choice of positive integers p, d, r with $p \geq p_0$ and

$$a_d p^\epsilon \leq d \leq b_d p^\epsilon, \quad 2^{a_r p/d} \leq r \leq 2^{b_r p/d},$$

put $K = \lfloor d^c \rfloor$, and let $1 \leq D \leq d$, $P, S \geq 1$, and $0 \leq \rho < K$ be integers. Let $w \geq 1$ be an integer component width. The input supplies on a descriptor tape the self-delimiting header

$$\mathcal{E} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(P)\Gamma(S)\Gamma(K)\Gamma(\rho + 1)\Gamma(w),$$

using the integer code from (1). Assume that this header is valid with the stated values and that the accompanying array is in lexicographic order on the complete address set

$$[P] \times [2^K]^D \times [S], \quad M = PS 2^{KD},$$

with the following record format. For every address $x \in [P] \times [2^K]^D \times [S]$, write its record as

$$z(x) = \sum_{j=0}^{r-1} z_{x,j} y^j,$$

stored in coefficient order. For every x and $0 \leq j < r$, assume $|z_{x,j}| \leq 1$ and $\Re z_{x,j}, \Im z_{x,j} \in 2^{-p}\mathbb{Z}$. Each component is 2^{-p} times a signed w -bit two's-complement integer, written most significant bit first; the real component precedes the imaginary component. Require the pointwise bounds

$$\log_2(2M) \leq C_M p, \quad |\mathcal{E}| \leq C_{\text{desc}} p, \quad p + 2 \leq w \leq C_w p,$$

where $|\mathcal{E}|$ is the header's bit length. The condition $w \geq p + 2$ allows every permitted disk-grid component, including -1 and 1 . The logical bit volume is $V = 2Mrw = \Theta(Mrp)$.

On each of the D consecutive K -bit address chunks, select bit ρ , numbered from the least significant bit, and let $\mathcal{H}$ be the tensor product of

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

on those selected bits. It acts coefficientwise and fixes every unselected address bit and both spectator fields. Let Q_p truncate the real and imaginary parts of every coefficient toward zero to $2^{-p}\mathbb{Z}$.

The procedure returns $Q_p(\mathcal{H}z)$ in the same array, coefficient order, and component format in $O(Vd^{\lambda'})$ time, uniformly over all these choices of $p, d, r, D, P, S, \rho, w$ and z . Every returned coefficient belongs to the complex unit disk, and, in the maximum coefficient norm,

$$\|Q_p(\mathcal{H}z) - \mathcal{H}z\|_\infty < \sqrt{2}2^{-p}.$$

The procedure evaluates the entire layer exactly before this final truncation. Put

$$C_0 = \left\lceil \max\{128mB^2, 18mB^2(1 + 1/\zeta)\} \right\rceil, \quad B = s_c + 64(W_c + m + 1)^3,$$

where m, W_c, s_c are the fixed constants of Proposition 14. The procedure uses at most $p + \lceil C_0 d^{C_1} \rceil$ fractional bits and $\lceil C_0 d^{C_1} \rceil + 3$ integer and sign bits per real component internally; these widths are $O(p)$. The time includes address changes, deterministic repair, row padding and its removal, and fixed-tape workspace cleanup. Its tape count is fixed, and its time constant is independent of the allowed array lengths and coefficient values.

Proof. The procedure reads the current parameters and component width from $\mathcal{E}$. Parsing this header and forming the initial collapsed descriptors take polynomial time in p , already charged in the descriptor work above. The phase-frame construction reduces a group of e selected bits to s_c children on e/m selected bits. The row split gives each child $1/W_c$ of the parent's logical volume. The compact-control selected-bit procedure supplies the binary basis changes. We verify that its cost is uniform over the displayed bands for d, r . Put $\Delta = \lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, a common cutoff gives $\Delta \leq p$. The guarded record width is then $R_g = 2r(p + 2\Delta + 3) \leq 12rp$; also $R_g \geq 2rp$. Padding at most doubles the original record count M , and row splitting only decreases it. Thus a packed child with M_{child} records has the following value A_* of its rectangular width parameter:

$$\begin{aligned} A_* &= \lceil \log_2(2M_{\text{child}}R_g) \rceil \\ &\leq \lceil \log_2(4MR_g) \rceil \leq C_A p, \quad C_A = C_M + b_r + 7. \end{aligned}$$

The last inequality uses $\log_2(2M) \leq C_M p$, $\log_2 r \leq b_r p/d \leq b_r p$, and $\log_2 p \leq p$.

Uniformly for large p , the bands give $f \leq d \leq p$ and

$$K \geq \frac{1}{2} a_d^c p^{\epsilon c}, \quad r \geq 2^{(a_r/b_d)p^{1-\epsilon}}.$$

Thus $K/\log p \rightarrow \infty$, r dominates every fixed polynomial, and $G = 4\lceil \log_2 p \rceil + 6$ eventually obeys $K \geq G + 4\lceil \log_2 p \rceil + 10$. Every role stream retains both complete front fields and the complete back field outside its row index. The above $A_* \leq C_A p$ bound applies to each compact rectangle. For its repair,

$$\frac{A_*^3}{R_g} \leq \frac{C_A^3 p^2}{2r}, \quad \delta A_* \left(1 + \frac{A_*}{R_g}\right) \leq \frac{5C_A}{128p^2} \left(1 + \frac{C_A}{2r}\right).$$

The wider-control rotation bound is also uniform, because every polynomial offset charge divided by R_g tends to zero. Hence Proposition 28 costs $O(M_{\text{child}} R_g((f \log p)^\tau + 1))$ with a common constant. Reservation preprocessing costs $O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1))$ and is included by the strict condition $1 - c < \lambda'$. The row construction and the recurrence analysis above give total time $O(Vd^{\lambda'})$ on fixed tapes, including all padded and auxiliary streams. Equation (23) converts the resulting exact C -layer to $\mathcal{H}$.

The guard-width calculation above bounds both denominator and magnitude growth by $\lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, this is $o(p)$ uniformly over the allowed d , so every intermediate coefficient is represented exactly with the stated $O(p)$ width. The initial scan removes or adds only redundant leading sign bits and appends the reserved fractional zeros to put each supplied component in this guarded format. To truncate toward zero at the end, shift its absolute numerator right by the number of added fractional bits, restore its sign, and write the result in w bits. These scans cost $O(V)$. Finally, every row of $\mathcal{H}$ has absolute row sum one. Thus $\mathcal{H}$ maps disk arrays to disk arrays, and componentwise truncation preserves that disk and has error less than $\sqrt{2} 2^{-p}$: truncation cannot increase the absolute value of either component, and changes each by less than 2^{-p} . The truncated numerators have magnitude at most 2^p , so the supplied component width can hold them. $\square$

5.6 The active bulk-recursion construction

The uniform recursion and its guard above remain a separate baseline, with their original fixed exponents. We now use the stronger bit interchange of Lemma 20 and change the complex children to whole residuals. The selected-bit movement lemmas and Proposition 28 take the exponent of their interchange routine as a parameter: their proofs use only its $O(Vu^\tau)$ interface. They therefore apply with $\tau = 1 - 246/10^9$ in the following construction.

5.7 The weighted arithmetic-path budget

The precision analysis uses the length of an arithmetic dependency path through the framed complex network. We first measure this path in directional rank units, before grouping some units into bulk children.

A weighted path bound. Use the retained triple-based complex motif on h points, with $m = h^3$. This includes the paired complex producer and its shared first- and third-stage auxiliary banks. Regard every scalar gate as a map from saved inputs to its outputs. All of its inputs and outputs have one common subspace label. A directed dependency path can therefore enter a gate on one physical wire and leave on another without changing label dimension at the gate itself.

For a network edge a , let U_a^- and U_a^+ be its labels and write $r_a = |\dim U_a^+ - \dim U_a^-|$. The edge implementation has exactly r_a recursive calls, each implementing one directional C tensor on e/m selected axes. Binary changes of coordinates before and after these calls permute coefficients and add no arithmetic depth. An inverse direction is implemented using one forward child and its sign and fourth-root wrappers, so it has the same recursive-call count.

If P is a directed scalar dependency path, its common gate labels give the exact identity

$$\sum_{a \in P} r_a = \dim U_{\text{end}} - \dim U_{\text{start}} + 2 \sum_{\substack{a \in P \\ \dim U_a^+ < \dim U_a^-}} (\dim U_a^- - \dim U_a^+).$$

The endpoint difference is at most m .

Every side-producer edge is increasing. This follows from nested source labels for the nonconstant middle mixer, complemented labels in its reverse-direction counterpart, and constant early and late frames. The data-wire and interstage edges are increasing as in the retained motif. The stage-sharing joins are also increasing. The only decreasing edges are the retained central returns, each of dimension loss h .

There are three tensor stages. Within a stage, the v^2 invocations use disjoint data and auxiliary wires, and each invocation has a single central-return frontier. Equivalently, its point and total central wires all return between the same two blocks of the transparent schedule. A directed path crosses that frontier at most once. Gates acting on different invocations do not connect their wires. Thus a path encounters at most one central return per stage, even though the machine executes all invocations sequentially on its fixed tapes. Sharing stage-1 and stage-3 banks only links their existing stage boundaries and creates no additional return. Consequently

$$\sum_{a \in P} r_a \leq q := m + 6h. \quad (31)$$

The bound also applies to a path ending at an intermediate gate or inside an edge implementation: complete its current edge, and then any suffix to a terminal vertex. All weights are nonnegative.

From a scalar path to a coefficient path. A coefficient dependency path in the actual tape algorithm projects to a directed path of the scalar network: inside a scalar gate it follows one of the saved operands that contributes to the chosen output; inside an edge it stays on the edge's role stream. A child can permute or combine addresses within that role, but cannot move a value into a different parent role. The row split, parking, padding and reassembly operations do not create arithmetic dependencies between distinct parent roles. This projection justifies using (31) for the implemented recursion, rather than merely for a drawn finite circuit.

5.8 A whole-residual complex child

The complex phase interface admits a further batching step. On an edge with residual dimension a , the existing binary change of basis puts an orthonormal residual basis into the first a slots. If each slot has f selected axes, those first a slots form one contiguous interval of af axes. Instead of a child calls on f axes, use one child on that interval.

The directions need not all have the same sign. A basis vector's weight modulo four may make its directional kernel an inverse even on an increasing edge. Let B be the set of negative residual slots and let Z_B multiply an address by -1 precisely when the parity of its selected bits in those slots is odd. The exact identity $C^{-1} = -iZZ$ gives

$$\bigotimes_{j=1}^a (C^{\varepsilon_j})^{\otimes f} = (-i)^{f|B|} Z_B C^{\otimes af} Z_B, \quad \varepsilon_j \in \{1, -1\}.$$

Thus one forward child, two address-sign passes and one fixed fourth-root phase implement the mixed signs. The same binary basis changes before and after the child are retained. The wrappers cost $O(V)$: their sign is computed from the current address descriptor, whose polynomial work per record is absorbed by the existing superpolynomial record length. They introduce at most four elementary coefficient operations per value and no further recursive call.

We batch only the two specified auxiliary edge classes of ranks

$$a_1 = m - 2h, \quad a_2 = m - h^2.$$

All other edges continue to use their old one-direction children. The first class is the increasing join between a completed stage-1 auxiliary bank, at dimension h , and its matched stage-3 bank, at dimension $m - h$. The second is the stage-2 auxiliary exit from dimension h^2 to the full terminal frame. These are complete auxiliary banks, including the central roles when those are shared. If a width bound requires padding, add dummy auxiliary roles with identity scalar gates at the ordinary low and high frames before matching the first- and third-stage banks. Their labels only increase, their arbitrary scalar values are restored, and both counted auxiliary classes are then present for every padded role. The changed volume-weighted time expression is consequently

$$\frac{1}{W} \left[(s - n_1 a_1 - n_2 a_2) m^{-\sigma} + n_1 \left(\frac{a_1}{m} \right)^\sigma + n_2 \left(\frac{a_2}{m} \right)^\sigma \right],$$

where n_1, n_2 count the chosen edges. It has value $s/(Wm) < 1$ at $\sigma = 1$. Strict comparison below one supplies the new complex recursion exponent. The separate mixed-width row construction handles $f = \lfloor e/m \rfloor$, the at most $m - 1$ individually processed remaining axes, and a row reservation sufficient for the longer but still $O(\log d)$ recursion.

5.9 A depth guard for the unequal complex children

The dependency-path argument of Section 5.7 still bounds the sum of edge ranks on one scalar path by

$$q = m + 6h.$$

For $h = 28$ its relevant constants are

$$m = 21952, \quad q = 22120, \quad a_1 = 21896, \quad a_2 = 21168.$$

Both selected ranks exceed $q/2$. Therefore every arithmetic dependency path encounters at most one selected bulk edge. At a call on e axes, put $f = \lfloor e/m \rfloor$. Ignoring the local additive operations, the path has one of the following child-depth bounds:

$$qA(f), \quad (q - a_1)A(f) + A(a_1 f), \quad (q - a_2)A(f) + A(a_2 f).$$

The weight budget accounts for both increasing and decreasing frame edges and for paths that switch wires at a scalar gate. Inverse-child and endpoint sign corrections add no child. Individually processing the fewer than m leftover axes adds at most $8m$ elementary operations, absorbed into the local constant below.

Take $\rho = 6/5$. The following three normalized path expressions have a rational common upper bound strictly below one:

$$\max \left\{ \frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m} \right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m} \right)^\rho \right\} < \bar{\theta} := \frac{999}{1000}. \quad (32)$$

Here is an entirely rational verification. First $160000^5 < m^6$, so $m^\rho > 160000$. For $0 < t < 1$, Taylor's formula with its second derivative bounded on $[0, t]$ gives

$$(1 - t)^{6/5} \leq 1 - \frac{6}{5}t + \frac{3}{25} \frac{t^2}{1 - t}.$$

Indeed its second derivative is $(6/25)(1 - t)^{-4/5} \leq (6/25)/(1 - t)$. Putting $t = (m - a_i)/m$ gives upper bounds, in the displayed order,

$$\frac{553}{4000}, \quad \frac{47817969}{47897500}, \quad \frac{1213697}{1260000},$$

all strictly below $999/1000$.

Stopping and exact arithmetic. Retain a global stopping threshold d^β , with $0 < \beta < 1$. For sufficiently large d , this threshold exceeds $2m$. A leaf reached from an internal call has width at least $d^\beta/(2m)$: every child has width at least $f = \lfloor e/m \rfloor \geq e/(2m)$. Put

$$E = 64(W + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1).$$

The exact finite inequality $36W^3 + 4s + 4W + 8m + 4 < E$ covers all local scalar operations, the new leftover-axis work and the retained endpoint and inverse wrappers. For every descendant width e reached before or just after stopping, strong induction gives

$$A(e) \leq C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho. \quad (33)$$

At a leaf, use

$$8e \leq 8(2m)^{\rho-1} d^{-\beta(\rho-1)} e^\rho \leq 16m d^{-\beta(\rho-1)} e^\rho.$$

At an internal call, child widths are at most their stated fractions of e , so (32) bounds their total by $\bar{\theta} C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho$. Since $e \geq d^\beta$, the remaining gap is at least $(1 - \bar{\theta}) C_{\text{dep}} d^\beta \geq E$. This proves the induction. A root that is already a leaf separately satisfies $A(e) \leq 8d^\beta$, which is smaller than the final bound below.

For rational $\zeta > 0$, set

$$C_1 = \rho - (\rho - 1)\beta + \zeta, \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

The root bound from (33) is $C_{\text{dep}} d^{\rho-(\rho-1)\beta}$. This covers the direct arbitrary-width root construction; it also covers every retained base- m piece. Even retaining the older bound of $m(1 + 1/\zeta)d^\zeta$ such pieces, their concatenation, all reserved axis kernels, the outer phases and the b^D conversion have depth at most $C_0 d^{C_1}$. The original elementary-depth induction therefore supplies exact denominator and magnitude guards of $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits. The width remains $O(p)$ when $\epsilon C_1 < 1$.

In particular, $\beta = 1/1000$ and $\zeta = 1/10000$ give

$$C_1 = \frac{11999}{10000}, \quad 1 - \frac{7999999}{16000000} C_1 = \frac{64008011999}{160000000000} > 0.$$

Thus bulk complex children preserve the near-half dimension exponent while allowing the improved complex time exponent to be used with almost no loss at the stopping leaves.

5.10 The complex layer with unequal recursive axis counts

Apply the same integer-depth construction to the complex circuit's list of child sizes. For a selected residual of rank $a < m$, the residual orthonormal basis places its nontrivial coordinate slots first. Its a consecutive slots of f selected axes form one contiguous block of af axes. The residual factors can have signs $\varepsilon_v \in \{1, -1\}$. For their negative slot set B , the retained identity $C^{-1} = -iZCZ$ gives, after the residual basis map,

$$\prod_{v=1}^a (C^{\varepsilon_v})^{\otimes f} = (-i)^{f|B|} Z_B C^{\otimes af} Z_B.$$

Here Z_B applies Z on the f axes of each negative slot. Thus one recursive application of $C^{\otimes af}$, two address-sign passes, and one exact phase replace the a separate recursive calls; the basis maps before and after the operation are unchanged. The sign descriptors have polynomial size in the

global parameters and fit the existing per-record overhead allowance. Other residuals retain their original rank-one calls. Let Ψ_c and D_c denote the resulting moment and depth functions. In particular, the two bulk classes of the current motif have

$$a_1 = m - 2h = 21896, \quad a_2 = m - h^2 = 21168 \quad (m = 21952, h = 28),$$

so every recursive call is a strict contraction.

Complete rows and compact fields. Keep the original global p, d, K, G, H, q_F, q_B for compact control, and replace only the row reservation by

$$k_0 = D_c(d), \quad q_0 = \lceil \log_2 W \rceil k_0, \quad Q_0 = q_0 + q_F + q_B.$$

This conservative integer choice satisfies $2^{q_0 K} \geq W^{k_0}$ and $q_0 = O(\log(2d))$. If the available number D of selected axes is at most Q_0 , apply them individually. Otherwise process the first $q_0 + q_F$ and last q_B axes individually and use the remaining consecutive middle block for the recursive circuit. The row field consists only of the first q_0 chunks; neither compact field is included in it. Pad its count to the next multiple of W^{k_0} , by a factor below two, once before recursion. Every child therefore has all compact and spectator ranges and exactly $1/W$ of its parent's volume, with enough row divisibility for its remaining depth.

The reserved-axis bound is unchanged:

$$Q_0 = O(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1).$$

In particular, the larger fixed row-depth constant does not change any power of d or p in preprocessing. The exact bit interchange called inside a compact operation supplies its own rows from its own target chunk digits; its internal row bound is not an additional reservation of selected complex axes.

Integer axis counts. At a complex call of width $e = mf + r$, first apply the $r < m$ leftmost selected kernels individually and treat those axes as completed spectators. The remaining mf axes are consecutive and split into the required m equal slots. The extra work is $O(V)$ per node. The individual kernels on removed axes commute with the completed tensor map on the other axes, so this produces the desired map on all e axes. The root active block can be processed directly; its base- m expansion is unnecessary.

Stopping and weighted-tree bounds. Fix rational $0 < \beta < 1$ and stop at $e < d^\beta$, with normalized leaf cost $O(e)$. Suppose $\Psi_c(\sigma) < 1$ for $0 < \sigma < 1$. Each node's normalized volume weight is W^{-j} at depth j . Its children have total σ -weighted width at most $\Psi_c(\sigma)$ times the parent's weighted width. Therefore

$$\sum_{\text{leaves } u} W^{-\text{depth}(u)} e_u^\sigma \leq e_{\text{root}}^\sigma.$$

Since $e_u < d^\beta$, total normalized leaf work is at most

$$O(d^{\beta(1-\sigma)} e_{\text{root}}^\sigma) = O(d^{\sigma+\beta(1-\sigma)}).$$

This is a bound for a tree with unequal child widths; no common leaf depth is assumed.

Compact operations cost $O(G^\tau e^\tau + 1)$ per current logical volume. If $\sigma \leq \tau$, monotonicity gives $\Psi_c(\tau) \leq \Psi_c(\sigma) < 1$, and summing the τ -weighted widths by depth bounds all internal work by $O(G^\tau d^\tau)$. If $\sigma > \tau$, every internal width is at least d^β , so $e^\tau \leq d^{\beta(\tau-\sigma)} e^\sigma$; summing the strict σ -moment gives $O(G^\tau d^{\sigma+\beta(\tau-\sigma)})$. Thus in both cases, with

$$\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\},$$

the complete normalized layer cost is

$$O\left((\log p)^3 \left[d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c,0\}} + 1\right]\right).$$

The displayed logarithmic factor conservatively retains the existing layout and descriptor allowances. Hence the original strict conditions

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda'$$

continue to imply the complete $O(Vd^{\lambda'})$ layer bound.

Compatibility with the new dependency guard. The separate dependency-path estimate uses $q = m + 6h = 22120$, observes that one path crosses at most one selected bulk residual, and bounds its ρ -moment by

$$\max\left\{\frac{q}{m^\rho}, \quad \frac{q - a_1}{m^\rho} + \left(\frac{a_1}{m}\right)^\rho, \quad \frac{q - a_2}{m^\rho} + \left(\frac{a_2}{m}\right)^\rho\right\} < \frac{999}{1000} \quad \left(\rho = \frac{6}{5}\right).$$

Integer floors only shrink these children. For sufficiently large d , an internal parent has $e \geq d^\beta \geq 2m$, hence every recursive child has width at least $\lfloor e/m \rfloor \geq d^\beta/(2m)$ whenever it becomes a stopping leaf. The at most $m - 1$ individually processed remainder axes at each internal node belong to its fixed local overhead; add their bounded coefficient contribution to the guard's fixed constant E . They are not extra arbitrarily small stopping leaves. Thus the separately established guard exponent

$$C_1 = \rho - (\rho - 1)\beta + \zeta = \frac{6}{5} - \frac{\beta}{5} + \zeta$$

applies to this integer-width implementation. All stopping tests can be performed by the existing rational-power integer comparison.

5.11 An exact complex exponent after bulk recursion

Keep the paired complex producer of PR #7. Its constants are

$$h = 28, \quad v_c = \binom{28}{3} = 3276, \quad m = 21952, \quad R_c = 93838.$$

There are $R_c + h + 1 = 93867$ auxiliary roles per invocation, including the original central roles. Write

$$B_c = v_c^2(R_c + h + 1) = 1007397164592,$$

$$W_c = 2v_c^2(v_c + R_c + h + 1) = 2085111546336,$$

$$L_c = 3v_c^2h(h + 1) = 26143580736, \quad s_c = W_cm - 2v_c^3 + 2L_c = 45772350635112192.$$

The two selected whole-residual classes have ranks $a_1 = 21896$ and $a_2 = 21168$, each with B_c copies. The number of remaining singleton children is

$$S_c = s_c - B_c(a_1 + a_2) = 2389799139122304.$$

Their time moment is

$$\Xi(\sigma) = \frac{S_c}{W_c m^\sigma} + \frac{B_c}{W_c} \left[\left(\frac{a_1}{m} \right)^\sigma + \left(\frac{a_2}{m} \right)^\sigma \right].$$

At $\sigma = 1$, this equals $1 - \eta_c$ with $\eta_c = 5/12693352$.

Take

$$a_c = \frac{7}{10^7}, \quad \sigma = 1 - a_c.$$

The same positive logarithm series used in the bit certificate gives

$$\log m < \frac{9997}{1000}, \quad \log \frac{m}{a_1} = \log \frac{392}{391} < \frac{2555}{10^6}, \quad \log \frac{m}{a_2} = \log \frac{28}{27} < \frac{36368}{10^6}.$$

Let these upper bounds be ℓ_0, ℓ_1, ℓ_2 and put

$$w_0 = \frac{S_c}{W_c m}, \quad w_i = \frac{B_c a_i}{W_c m} \quad (i = 1, 2).$$

The exact rational comparison is

$$\Xi(1 - a_c) < \sum_{i=0}^2 \frac{w_i}{1 - a_c \ell_i} < 1.$$

The gap in the last inequality is

$$\frac{24132873235669673777179671369131}{1525632551364197359955324563293421369131} > 1.58 \times 10^{-8}.$$

This certifies the stronger complex exponent with the unchanged finite producer and the new whole-residual recursion.

Proposition 30 (Batched simultaneous normalized butterfly layer). *Fix $\tau = 1 - 246/10^9$ and $\sigma = 1 - 7/10^7$. Let $c, \epsilon, \lambda, \lambda', \beta, \zeta$ be fixed positive rational numbers with $0 < \beta < 1$, and put*

$$C_1 = 6/5 - \beta/5 + \zeta, \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Require

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda', \quad \epsilon C_1 < 1.$$

Fix positive constants $a_d \leq b_d$ and $a_r \leq b_r$, and fix positive constants $C_M, C_{\text{desc}}, C_w$ with $C_w > 1$. For these fixed choices there are a cutoff p_0 , one fixed finite-alphabet multitape procedure, and an implicit time constant with the following guarantee. The cutoff and time constant may depend on the fixed rational parameters and these constants. For every choice of positive integers p, d, r with $p \geq p_0$ and

$$a_d p^\epsilon \leq d \leq b_d p^\epsilon, \quad 2^{a_r p/d} \leq r \leq 2^{b_r p/d},$$

put $K = \lfloor d^c \rfloor$, and let $1 \leq D \leq d$, $P, S \geq 1$, and $0 \leq \rho < K$ be integers. Let $w \geq 1$ be an integer component width. The input supplies on a descriptor tape the self-delimiting header

$$\mathcal{E} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(P)\Gamma(S)\Gamma(K)\Gamma(\rho + 1)\Gamma(w),$$

using the integer code from (1). Assume that this header is valid with the stated values and that the accompanying array is in lexicographic order on the complete address set

$$[P] \times [2^K]^D \times [S], \quad M = PS 2^{KD},$$

with the following record format. For every address $x \in [P] \times [2^K]^D \times [S]$, write its record as

$$z(x) = \sum_{j=0}^{r-1} z_{x,j} y^j,$$

stored in coefficient order. For every x and $0 \leq j < r$, assume $|z_{x,j}| \leq 1$ and $\Re z_{x,j}, \Im z_{x,j} \in 2^{-p}\mathbb{Z}$. Each component is 2^{-p} times a signed w -bit two's-complement integer, written most significant bit first; the real component precedes the imaginary component. Require the pointwise bounds

$$\log_2(2M) \leq C_M p, \quad |\mathcal{E}| \leq C_{\text{desc}} p, \quad p+2 \leq w \leq C_w p,$$

where $|\mathcal{E}|$ is the header's bit length. The condition $w \geq p+2$ allows every permitted disk-grid component, including -1 and 1 . The logical bit volume is $V = 2Mrw = \Theta(Mrp)$.

On each of the D consecutive K -bit address chunks, select bit ρ , numbered from the least significant bit, and let $\mathcal{H}$ be the tensor product of

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

on those selected bits. It acts coefficientwise and fixes every unselected address bit and both spectator fields. Let Q_p truncate the real and imaginary parts of every coefficient toward zero to $2^{-p}\mathbb{Z}$.

The procedure returns $Q_p(\mathcal{H}z)$ in the same array, coefficient order, and component format in $O(Vd^{\lambda'})$ time, uniformly over all these choices of $p, d, r, D, P, S, \rho, w$ and z . Every returned coefficient belongs to the complex unit disk, and, in the maximum coefficient norm,

$$\|Q_p(\mathcal{H}z) - \mathcal{H}z\|_\infty < \sqrt{2} 2^{-p}.$$

The procedure evaluates the entire layer exactly before this final truncation. Put

$$E = 64(W_c + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1), \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil,$$

where m, W_c, s_c are the fixed constants of Proposition 14. The procedure uses at most $p + \lceil C_0 d^{C_1} \rceil$ fractional bits and $\lceil C_0 d^{C_1} \rceil + 3$ integer and sign bits per real component internally; these widths are $O(p)$. The time includes address changes, deterministic repair, row padding and its removal, and fixed-tape workspace cleanup. Its tape count is fixed, and its time constant is independent of the allowed array lengths and coefficient values.

Proof. The procedure reads the current parameters and component width from $\mathcal{E}$. Parsing this header and forming the initial collapsed descriptors take polynomial time in p , already charged in the descriptor work above. The whole-residual construction of Section 5.8 replaces the two selected classes by one child per edge. Its mixed-width time moment at σ is strictly below one by Section 5.11. For $e = mf + r_0$, $0 \leq r_0 < m$, handle the remainder axes individually; the true children have widths f , $21896f$, or $21168f$ according to their type. Section 5.10 supplies complete role streams of exactly $1/W_c$ of the parent's logical volume. The compact-control selected-bit construction supplies each binary basis change using Lemma 20. We verify that its cost is uniform over the displayed bands for d, r . Put $\Delta = \lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, a common cutoff gives $\Delta \leq p$. The

guarded record width is then $R_g = 2r(p + 2\Delta + 3) \leq 12rp$; also $R_g \geq 2rp$. Padding at most doubles the original record count M , and row splitting only decreases it. Thus a packed child with M_{child} records has the following value A_* of its rectangular width parameter:

$$\begin{aligned} A_* &= \lceil \log_2(2M_{\text{child}}R_g) \rceil \\ &\leq \lceil \log_2(4MR_g) \rceil \leq C_A p, \quad C_A = C_M + b_r + 7. \end{aligned}$$

The last inequality uses $\log_2(2M) \leq C_M p$, $\log_2 r \leq b_r p/d \leq b_r p$, and $\log_2 p \leq p$.

Uniformly for large p , the bands give $f \leq d \leq p$ and

$$K \geq \frac{1}{2} a_d^c p^{\epsilon c}, \quad r \geq 2^{(a_r/b_d)p^{1-\epsilon}}.$$

Thus $K/\log p \rightarrow \infty$, r dominates every fixed polynomial, and $G = 4\lceil \log_2 p \rceil + 6$ eventually obeys $K \geq G + 4\lceil \log_2 p \rceil + 10$. Every role stream retains both complete front fields and the complete back field outside its row index. The above $A_* \leq C_A p$ bound applies to each compact rectangle. For its repair,

$$\frac{A_*^3}{R_g} \leq \frac{C_A^3 p^2}{2r}, \quad \delta A_* \left(1 + \frac{A_*}{R_g}\right) \leq \frac{5C_A}{128p^2} \left(1 + \frac{C_A}{2r}\right).$$

The wider-control rotation bound is also uniform, because every polynomial offset charge divided by R_g tends to zero. Hence Proposition 28 costs $O(M_{\text{child}}R_g((f \log p)^\tau + 1))$ with a common constant. Reservation preprocessing costs $O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1))$ and is included by the strict condition $1 - c < \lambda'$. The integer row construction and the weighted-tree analysis in Section 5.10 give total time $O(Vd^{\lambda'})$ on fixed tapes, including all padded and auxiliary streams, mixed-width children, and bounded local remainders. Equation (23) converts the resulting exact C -layer to $\mathcal{H}$.

The dependency-path calculation in Section 5.9 bounds both denominator and magnitude growth by $\lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, this is $o(p)$ uniformly over the allowed d , so every intermediate coefficient is represented exactly with the stated $O(p)$ width. The initial scan removes or adds only redundant leading sign bits and appends the reserved fractional zeros to put each supplied component in this guarded format. To truncate toward zero at the end, shift its absolute numerator right by the number of added fractional bits, restore its sign, and write the result in w bits. These scans cost $O(V)$. Finally, every row of $\mathcal{H}$ has absolute row sum one. Thus $\mathcal{H}$ maps disk arrays to disk arrays, and componentwise truncation preserves that disk and has error less than $\sqrt{2}2^{-p}$: truncation cannot increase the absolute value of either component, and changes each by less than 2^{-p} . The truncated numerators have magnitude at most 2^p , so the supplied component width can hold them. $\square$

5.12 Consuming the stronger bit interchange

The complex network and guard remain at dimension 28; only the bit interchange called by the compact-control interface changes.

Proposition 31 (Layer using dimension-thirty bit interchange). *Fix $\tau = 1 - 253/10^9$ and $\sigma = 1 - 7/10^7$. Let $c, \epsilon, \lambda, \lambda', \beta, \zeta$ be fixed positive rational numbers with $0 < \beta < 1$, and put*

$$C_1 = 6/5 - \beta/5 + \zeta, \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Require

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda', \quad \epsilon C_1 < 1.$$

Fix positive constants $a_d \leq b_d$ and $a_r \leq b_r$, and fix positive constants $C_M, C_{\text{desc}}, C_w$ with $C_w > 1$. For these fixed choices there are a cutoff p_0 , one fixed finite-alphabet multitape procedure, and an implicit time constant with the following guarantee. The cutoff and time constant may depend on the fixed rational parameters and these constants. For every choice of positive integers p, d, r with $p \geq p_0$ and

$$a_d p^\epsilon \leq d \leq b_d p^\epsilon, \quad 2^{a_r p/d} \leq r \leq 2^{b_r p/d},$$

put $K = \lfloor d^c \rfloor$, and let $1 \leq D \leq d$, $P, S \geq 1$, and $0 \leq \rho < K$ be integers. Let $w \geq 1$ be an integer component width. The input supplies on a descriptor tape the self-delimiting header

$$\mathcal{E} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(P)\Gamma(S)\Gamma(K)\Gamma(\rho+1)\Gamma(w),$$

using the integer code from (1). Assume that this header is valid with the stated values and that the accompanying array is in lexicographic order on the complete address set

$$[P] \times [2^K]^D \times [S], \quad M = PS 2^{KD},$$

with the following record format. For every address $x \in [P] \times [2^K]^D \times [S]$, write its record as

$$z(x) = \sum_{j=0}^{r-1} z_{x,j} y^j,$$

stored in coefficient order. For every x and $0 \leq j < r$, assume $|z_{x,j}| \leq 1$ and $\Re z_{x,j}, \Im z_{x,j} \in 2^{-p}\mathbb{Z}$. Each component is 2^{-p} times a signed w -bit two's-complement integer, written most significant bit first; the real component precedes the imaginary component. Require the pointwise bounds

$$\log_2(2M) \leq C_M p, \quad |\mathcal{E}| \leq C_{\text{desc}} p, \quad p+2 \leq w \leq C_w p,$$

where $|\mathcal{E}|$ is the header's bit length. The condition $w \geq p+2$ allows every permitted disk-grid component, including -1 and 1 . The logical bit volume is $V = 2Mrw = \Theta(Mrp)$.

On each of the D consecutive K -bit address chunks, select bit ρ , numbered from the least significant bit, and let $\mathcal{H}$ be the tensor product of

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

on those selected bits. It acts coefficientwise and fixes every unselected address bit and both spectator fields. Let Q_p truncate the real and imaginary parts of every coefficient toward zero to $2^{-p}\mathbb{Z}$.

The procedure returns $Q_p(\mathcal{H}z)$ in the same array, coefficient order, and component format in $O(Vd^{X'})$ time, uniformly over all these choices of $p, d, r, D, P, S, \rho, w$ and z . Every returned coefficient belongs to the complex unit disk, and, in the maximum coefficient norm,

$$\|Q_p(\mathcal{H}z) - \mathcal{H}z\|_\infty < \sqrt{2} 2^{-p}.$$

The procedure evaluates the entire layer exactly before this final truncation. Put

$$E = 64(W_c + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1), \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil,$$

where m, W_c, s_c are the fixed constants of Proposition 14. The procedure uses at most $p + \lceil C_0 d^{C_1} \rceil$ fractional bits and $\lceil C_0 d^{C_1} \rceil + 3$ integer and sign bits per real component internally; these widths are $O(p)$. The time includes address changes, deterministic repair, row padding and its removal, and fixed-tape workspace cleanup. Its tape count is fixed, and its time constant is independent of the allowed array lengths and coefficient values.

Proof. The procedure reads the current parameters and component width from $\mathcal{E}$. Parsing this header and forming the initial collapsed descriptors take polynomial time in p , already charged in the descriptor work above. The whole-residual construction of Section 5.8 replaces the two selected classes by one child per edge. Its mixed-width time moment at σ is strictly below one by Section 5.11. For $e = mf + r_0$, $0 \leq r_0 < m$, handle the remainder axes individually; the true children have widths f , $21896f$, or $21168f$ according to their type. Section 5.10 supplies complete role streams of exactly $1/W_c$ of the parent's logical volume. The compact-control selected-bit construction supplies each binary basis change using Lemma 21. We verify that its cost is uniform over the displayed bands for d, r . Put $\Delta = \lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, a common cutoff gives $\Delta \leq p$. The guarded record width is then $R_g = 2r(p + 2\Delta + 3) \leq 12rp$; also $R_g \geq 2rp$. Padding at most doubles the original record count M , and row splitting only decreases it. Thus a packed child with M_{child} records has the following value A_* of its rectangular width parameter:

$$\begin{aligned} A_* &= \lceil \log_2(2M_{\text{child}}R_g) \rceil \\ &\leq \lceil \log_2(4MR_g) \rceil \leq C_A p, \quad C_A = C_M + b_r + 7. \end{aligned}$$

The last inequality uses $\log_2(2M) \leq C_M p$, $\log_2 r \leq b_r p/d \leq b_r p$, and $\log_2 p \leq p$.

Uniformly for large p , the bands give $f \leq d \leq p$ and

$$K \geq \frac{1}{2} a_d^c p^{\epsilon c}, \quad r \geq 2^{(a_r/b_d)p^{1-\epsilon}}.$$

Thus $K/\log p \rightarrow \infty$, r dominates every fixed polynomial, and $G = 4\lceil \log_2 p \rceil + 6$ eventually obeys $K \geq G + 4\lceil \log_2 p \rceil + 10$. Every role stream retains both complete front fields and the complete back field outside its row index. The above $A_* \leq C_A p$ bound applies to each compact rectangle. For its repair,

$$\frac{A_*^3}{R_g} \leq \frac{C_A^3 p^2}{2r}, \quad \delta A_* \left(1 + \frac{A_*}{R_g}\right) \leq \frac{5C_A}{128p^2} \left(1 + \frac{C_A}{2r}\right).$$

The wider-control rotation bound is also uniform, because every polynomial offset charge divided by R_g tends to zero. Hence Proposition 28 costs $O(M_{\text{child}}R_g((f \log p)^\tau + 1))$ with a common constant. Reservation preprocessing costs $O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1))$ and is included by the strict condition $1 - c < \lambda'$. The integer row construction and the weighted-tree analysis in Section 5.10 give total time $O(Vd^{\lambda'})$ on fixed tapes, including all padded and auxiliary streams, mixed-width children, and bounded local remainders. Equation (23) converts the resulting exact C -layer to $\mathcal{H}$.

The dependency-path calculation in Section 5.9 bounds both denominator and magnitude growth by $\lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, this is $o(p)$ uniformly over the allowed d , so every intermediate coefficient is represented exactly with the stated $O(p)$ width. The initial scan removes or adds only redundant leading sign bits and appends the reserved fractional zeros to put each supplied component in this guarded format. To truncate toward zero at the end, shift its absolute numerator right by the number of added fractional bits, restore its sign, and write the result in w bits. These scans cost $O(V)$. Finally, every row of $\mathcal{H}$ has absolute row sum one. Thus $\mathcal{H}$ maps disk arrays to disk arrays, and componentwise truncation preserves that disk and has error less than $\sqrt{2}2^{-p}$: truncation cannot increase the absolute value of either component, and changes each by less than 2^{-p} . The truncated numerators have magnitude at most 2^p , so the supplied component width can hold them. $\square$

5.13 A third whole-residual complex class

The whole-residual construction of Section 5.8 applies to any edge residual. We additionally select the stage-three data entrance edges. At stage three the incoming physical X and Y labels have

dimensions h^2 and $h^2 - 1$, and their first relevant frames have dimensions $(h^2 - 1)h + 1$ and $(h^2 - 1)h$. Both entrance residuals are $B \otimes t^\perp$ with $\dim B = h^2 - 1$, of rank

$$d_3 = (h^2 - 1)(h - 1) = 21141.$$

There are $2v_c^3 = 70317217152$ such edges. Each has an orthonormal residual basis by the retained binary-frame lemma. It is therefore applied as one child on $21141f$ selected axes, with the same sign wrappers.

The dependency-path guard. The path budget $q = m + 6h = 22120$ of Section 5.7 is unchanged. Since $2d_3 > q$, a dependency path still crosses at most one of the three selected classes. For the new class put $t = (m - d_3)/m = 811/21952$. With the same bound for $(1 - t)^{6/5}$,

$$\frac{q - d_3}{m^{6/5}} + \left(\frac{d_3}{m}\right)^{6/5} < \frac{979}{160000} + 1 - \frac{6}{5}t + \frac{3}{25} \frac{t^2}{1 - t} = \frac{372026142559}{386739360000} < \frac{999}{1000}.$$

Consequently the guard of Section 5.9 holds with three classes, with the same $\rho = 6/5$, $\bar{\theta} = 999/1000$, C_{dep} , C_0 and $C_1 = 6/5 - \beta/5 + \zeta$. The mixed-width layer of Section 5.10 has child widths f , $21896f$, $21168f$ and $21141f$.

The exact complex moment. With $B_c = 1007397164592$, $a_1 = 21896$ and $a_2 = 21168$ as in Section 5.11, the remaining singleton count is

$$S'_c = s_c - B_c(a_1 + a_2) - 2v_c^3 d_3 = 903222851311872.$$

The normalized widths are

$$r = \left(\frac{1}{21952}, \frac{391}{392}, \frac{27}{28}, \frac{21141}{21952} \right),$$

and their rank-mass weights are

$$w = \left(\frac{250477}{12693352}, \frac{12233999}{25386704}, \frac{844803}{1813336}, \frac{824499}{25386704} \right),$$

summing to $1 - \eta_c$ with $\eta_c = 5/12693352$. In addition to the logarithm bounds of Section 5.11, use $\log(21952/21141) < 37644/10^6$. For

$$a_c = \frac{18}{10^7}, \quad \sigma = 1 - a_c,$$

the same exact comparison gives

$$\Xi(1 - a_c) < \sum_{i=0}^3 \frac{w_i}{1 - a_c \ell_i} < 1,$$

with gap

$$\frac{12084702927461727916508294311187513281917509}{3098906698934144550424761963671962638378246719417509} > 3.89 \times 10^{-9}.$$

This certifies the complex saving $a_c = 18/10^7$. It exceeds the interchange saving, so $\sigma < \tau$ and the complex network does not bind.

5.14 A sharper certificate for the same three complex classes

Retain exactly the dimension-28 complex producer, three whole-residual classes, and dependency-path guard of Section 5.13. No source-frame modification is made to the complex network. Its weights, ratios and displayed rational logarithm upper bounds are unchanged. Evaluating the same exact bound at

$$a_c = 18179/10^{10}, \quad \sigma = 1 - a_c,$$

gives a strict gap greater than 2.1196×10^{-11} . This changes only the selected exponent, not the complex construction or its path budget. In particular $C_1 = 11999/10000$ remains certified.

For the combined bit saving $a_b = 1816/10^9$ and $\beta = 1/1000$, the leaf saving is

$$(1 - \beta)a_c = 18160821/10^{13} > 1816/10^9.$$

Consequently both $\sigma < \tau$ and $\sigma + \beta(1 - \sigma) < \tau$. This latter condition is essential: comparing the bit and complex savings alone would miss the leaf cost. Use $\lambda = \tau + 10^{-16}$ and $\lambda' = \tau + 2 \cdot 10^{-16}$ in the following active layer interface. All twenty-nine assembly inequalities and all seven margins are strictly positive in the exact certificate.

Proposition 32 (Layer using source frames and data corners). *Fix $\tau = 1 - 1816/10^9$ and $\sigma = 1 - 18179/10^{10}$. Let $c, \epsilon, \lambda, \lambda', \beta, \zeta$ be fixed positive rational numbers with $0 < \beta < 1$, and put*

$$C_1 = 6/5 - \beta/5 + \zeta, \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

Require

$$\max\{\tau, \sigma, \chi\} < \lambda < \lambda' < 1, \quad \max\{\sigma + \beta(1 - \sigma), 1 - c, 0\} < \lambda', \quad \epsilon C_1 < 1.$$

Fix positive constants $a_d \leq b_d$ and $a_r \leq b_r$, and fix positive constants $C_M, C_{\text{desc}}, C_w$ with $C_w > 1$. For these fixed choices there are a cutoff p_0 , one fixed finite-alphabet multitape procedure, and an implicit time constant with the following guarantee. The cutoff and time constant may depend on the fixed rational parameters and these constants. For every choice of positive integers p, d, r with $p \geq p_0$ and

$$a_d p^\epsilon \leq d \leq b_d p^\epsilon, \quad 2^{a_r p/d} \leq r \leq 2^{b_r p/d},$$

put $K = \lfloor d^c \rfloor$, and let $1 \leq D \leq d$, $P, S \geq 1$, and $0 \leq \rho < K$ be integers. Let $w \geq 1$ be an integer component width. The input supplies on a descriptor tape the self-delimiting header

$$\mathcal{E} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(D)\Gamma(P)\Gamma(S)\Gamma(K)\Gamma(\rho + 1)\Gamma(w),$$

using the integer code from (1). Assume that this header is valid with the stated values and that the accompanying array is in lexicographic order on the complete address set

$$[P] \times [2^K]^D \times [S], \quad M = PS 2^{KD},$$

with the following record format. For every address $x \in [P] \times [2^K]^D \times [S]$, write its record as

$$z(x) = \sum_{j=0}^{r-1} z_{x,j} y^j,$$

stored in coefficient order. For every x and $0 \leq j < r$, assume $|z_{x,j}| \leq 1$ and $\Re z_{x,j}, \Im z_{x,j} \in 2^{-p}\mathbb{Z}$. Each component is 2^{-p} times a signed w -bit two's-complement integer, written most significant bit first; the real component precedes the imaginary component. Require the pointwise bounds

$$\log_2(2M) \leq C_M p, \quad |\mathcal{E}| \leq C_{\text{desc}} p, \quad p + 2 \leq w \leq C_w p,$$

where $|\mathcal{E}|$ is the header's bit length. The condition $w \geq p + 2$ allows every permitted disk-grid component, including -1 and 1 . The logical bit volume is $V = 2Mrw = \Theta(Mrp)$.

On each of the D consecutive K -bit address chunks, select bit ρ , numbered from the least significant bit, and let $\mathcal{H}$ be the tensor product of

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

on those selected bits. It acts coefficientwise and fixes every unselected address bit and both spectator fields. Let Q_p truncate the real and imaginary parts of every coefficient toward zero to $2^{-p}\mathbb{Z}$.

The procedure returns $Q_p(\mathcal{H}z)$ in the same array, coefficient order, and component format in $O(Vd^{\lambda'})$ time, uniformly over all these choices of $p, d, r, D, P, S, \rho, w$ and z . Every returned coefficient belongs to the complex unit disk, and, in the maximum coefficient norm,

$$\|Q_p(\mathcal{H}z) - \mathcal{H}z\|_\infty < \sqrt{2} 2^{-p}.$$

The procedure evaluates the entire layer exactly before this final truncation. Put

$$E = 64(W_c + m + 1)^3, \quad C_{\text{dep}} = 1000(E + 16m + 1), \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil,$$

where m, W_c, s_c are the fixed constants of Proposition 14. The procedure uses at most $p + \lceil C_0 d^{C_1} \rceil$ fractional bits and $\lceil C_0 d^{C_1} \rceil + 3$ integer and sign bits per real component internally; these widths are $O(p)$. The time includes address changes, deterministic repair, row padding and its removal, and fixed-tape workspace cleanup. Its tape count is fixed, and its time constant is independent of the allowed array lengths and coefficient values.

Proof. The procedure reads the current parameters and component width from $\mathcal{E}$. Parsing this header and forming the initial collapsed descriptors take polynomial time in p , already charged in the descriptor work above. The whole-residual construction of Section 5.8 replaces the three selected classes by one child per edge. Its mixed-width time moment at σ is strictly below one by Section 5.14. For $e = mf + r_0$, $0 \leq r_0 < m$, handle the remainder axes individually; the true children have widths f , $21896f$, $21168f$, or $21141f$ according to their type. Section 5.10 supplies complete role streams of exactly $1/W_c$ of the parent's logical volume. The compact-control selected-bit construction supplies each binary basis change using Lemma 25. We verify that its cost is uniform over the displayed bands for d, r . Put $\Delta = \lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, a common cutoff gives $\Delta \leq p$. The guarded record width is then $R_g = 2r(p + 2\Delta + 3) \leq 12rp$; also $R_g \geq 2rp$. Padding at most doubles the original record count M , and row splitting only decreases it. Thus a packed child with M_{child} records has the following value A_* of its rectangular width parameter:

$$\begin{aligned} A_* &= \lceil \log_2(2M_{\text{child}}R_g) \rceil \\ &\leq \lceil \log_2(4MR_g) \rceil \leq C_A p, \quad C_A = C_M + b_r + 7. \end{aligned}$$

The last inequality uses $\log_2(2M) \leq C_M p$, $\log_2 r \leq b_r p/d \leq b_r p$, and $\log_2 p \leq p$.

Uniformly for large p , the bands give $f \leq d \leq p$ and

$$K \geq \frac{1}{2} a_d^c p^{\epsilon c}, \quad r \geq 2^{(a_r/b_d)p^{1-\epsilon}}.$$

Thus $K/\log p \rightarrow \infty$, r dominates every fixed polynomial, and $G = 4\lceil \log_2 p \rceil + 6$ eventually obeys $K \geq G + 4\lceil \log_2 p \rceil + 10$. Every role stream retains both complete front fields and the complete back

field outside its row index. The above $A_* \leq C_A p$ bound applies to each compact rectangle. For its repair,

$$\frac{A_*^3}{R_g} \leq \frac{C_A^3 p^2}{2r}, \quad \delta A_* \left(1 + \frac{A_*}{R_g}\right) \leq \frac{5C_A}{128p^2} \left(1 + \frac{C_A}{2r}\right).$$

The wider-control rotation bound is also uniform, because every polynomial offset charge divided by R_g tends to zero. Hence Proposition 28 costs $O(M_{\text{child}} R_g ((f \log p)^\tau + 1))$ with a common constant. Reservation preprocessing costs $O(V(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1))$ and is included by the strict condition $1 - c < \lambda'$. The integer row construction and the weighted-tree analysis in Section 5.10 give total time $O(Vd^{\lambda'})$ on fixed tapes, including all padded and auxiliary streams, mixed-width children, and bounded local remainders. Equation (23) converts the resulting exact C -layer to $\mathcal{H}$.

The dependency-path calculation in Section 5.9 bounds both denominator and magnitude growth by $\lceil C_0 d^{C_1} \rceil$. Since $d \leq b_d p^\epsilon$ and $\epsilon C_1 < 1$, this is $o(p)$ uniformly over the allowed d , so every intermediate coefficient is represented exactly with the stated $O(p)$ width. The initial scan removes or adds only redundant leading sign bits and appends the reserved fractional zeros to put each supplied component in this guarded format. To truncate toward zero at the end, shift its absolute numerator right by the number of added fractional bits, restore its sign, and write the result in w bits. These scans cost $O(V)$. Finally, every row of $\mathcal{H}$ has absolute row sum one. Thus $\mathcal{H}$ maps disk arrays to disk arrays, and componentwise truncation preserves that disk and has error less than $\sqrt{2}2^{-p}$: truncation cannot increase the absolute value of either component, and changes each by less than 2^{-p} . The truncated numerators have magnitude at most 2^p , so the supplied component width can hold them. $\square$

6 Synthetic transforms and their tape layout

We now implement the normalized Fourier transforms needed for convolution. Their entries will be short polynomials in a quotient where the required roots of unity act by exact signed permutations of coefficient lists. We also specify where each frequency entry is stored, because the opposite transform consumes that same order after a pointwise product.

The polynomial roots and radix-two butterflies belong to the fast polynomial-transform method of Nussbaumer and Quandalle [21, 20]; we use the form developed in [15, Section 2.4 and Lemma 3.2]. The work here is to apply the simultaneous layers on a suitable tape layout and retain the resulting frequency order through convolution.

For an integer $r \geq 2$ that is a power of two, put

$$\mathcal{R}_r = \mathbb{C}[y]/(y^r + 1), \quad \left\| \sum_{k=0}^{r-1} z_k y^k \right\|_\infty = \max_{0 \leq k < r} |z_k|.$$

We always use representatives of degree less than r . Multiplication by y^a , for any integer a , permutes the coefficients and changes some signs; consequently it is an isometry for this norm. For arrays over $\mathcal{R}_r$, the same notation denotes the maximum over all coefficients in all entries. A disk array has norm at most one. A p -bit grid array has real and imaginary coefficient parts in $2^{-p}\mathbb{Z}$. For any array, let Q_p truncate each real and imaginary coefficient part toward zero to this grid. Then

$$\|Q_p z - z\|_\infty < \sqrt{2}2^{-p}, \quad \|Q_p z\|_\infty \leq \|z\|_\infty. \quad (34)$$

6.1 Roots, butterflies, and stored frequency order

For a power of two t dividing $2r$, define $\omega_t = y^{2r/t}$. For $z \in \mathcal{R}_r^t$ and $0 \leq j < t$, set

$$(F_t^- z)_j = \frac{1}{t} \sum_{k=0}^{t-1} \omega_t^{-jk} z_k, \quad (F_t^+ z)_j = \frac{1}{t} \sum_{k=0}^{t-1} \omega_t^{jk} z_k.$$

Every power of ω_t is an isometry, and each transform entry is an average of t such images. Thus both transforms are contractions. The ring has zero divisors, so we verify character cancellation directly instead of dividing by $\omega_t^j - 1$.

Lemma 33 (Synthetic character cancellation). *If j is an integer, then*

$$\sum_{k=0}^{t-1} \omega_t^{jk} = \begin{cases} t, & t \mid j, \\ 0, & t \nmid j. \end{cases} \quad F_t^+ F_t^- = t^{-1} I.$$

Proof. Only the second case needs proof. Write $j = 2^v j_0$ with j_0 odd and $0 \leq v < \log_2 t$. Replacing k by $k + t/2^{v+1}$ modulo t permutes the summands and multiplies their sum by $y^{r j_0} = -1$. Hence the sum is zero, since the additive group of $\mathcal{R}_r$ has no two-torsion. Expanding the composite transforms and applying this identity proves the final formula. $\square$

Write

$$H_0 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

The length-one procedure is the identity. For $t \geq 2$, the decimation-in-frequency step pairs z_k with $z_{k+t/2}$ for $0 \leq k < t/2$. The two outputs of H_0 occupy these same slots, whose high address bit is $b = 0$ or $b = 1$, respectively. Multiply the $b = 1$ output by ω_t^{-k} , and then apply the length- $t/2$ procedure separately within the two branches. This computes F_t^- because, for $b \in \{0, 1\}$,

$$(F_t^- z)_{2h+b} = \frac{1}{t/2} \sum_{k=0}^{t/2-1} (\omega_t^2)^{-hk} \omega_t^{-bk} \frac{z_k + (-1)^b z_{k+t/2}}{2}. \quad (35)$$

To follow the recursion, name the original address-bit slots of an axis of length $t = 2^a$ by $a-1, \dots, 0$, from most to least significant. For $0 \leq s \leq a$, after s levels the values of the first s named slots are branch bits $b_0, \dots, b_{s-1}$. Equation (35) identifies these as frequency bits $0, \dots, s-1$, numbered from the least significant bit. The remaining slots encode an index in $[0, u)$, where $u = t/2^s$. When $s < a$, the next level uses slot $h = a-1-s$ to distinguish the paired inputs and then to hold b_s . Writing x_v for the bit in a still unprocessed slot v , the index used by its twiddle is

$$k = \sum_{v=0}^{h-1} 2^v x_v, \quad 0 \leq k < u/2.$$

Previously written branch bits do not enter k . The same description holds in each shorter branch by (35), which proves the assertion inductively.

After all a rounds, the named slots from high to low therefore contain the frequency bits from low to high. If $\text{rev}_a(j)$ reverses the a binary digits of j , the entry $(F_t^- z)_j$ occupies slot $\text{rev}_a(j)$. Let B_t denote this record permutation: it moves the entry at j to $\text{rev}_a(j)$. The recursion writes its outputs directly in this order; it does not move whole frequency indices.

The procedure consuming this stored order traverses the recursion in reverse: first process the two smaller branches, then apply the inverse monomial to the second branch, and finally apply

H_0 . Each forward level is invertible, since both its butterfly and its monomial are invertible. As $H_0^2 = I/2$, the reversed level is half the inverse of the corresponding forward level. There are $a = \log_2 t$ levels, so the reverse procedure composed with $B_t F_t^-$ is $2^{-a} I = I/t$. Lemma 33 gives the same composite for $F_t^+ B_t^{-1}$ with $B_t F_t^-$. The forward procedure is invertible, so these two reverse operators agree. Thus the two exact procedures implement

$$B_t F_t^-, \quad F_t^+ B_t^{-1}, \quad (36)$$

respectively.

For D axes of lengths $t_1, \dots, t_D$, put

$$M = \prod_{i=1}^D t_i, \quad F_t^\pm = \bigotimes_{i=1}^D F_{t_i}^\pm, \quad B = \bigotimes_{i=1}^D B_{t_i}.$$

A round applies one current H_0 on each participating axis. The twiddle for axis i depends only on that axis's address bits. The two entries paired by another axis's butterfly have the same axis- i address, so that twiddle commutes with the other butterfly. Consequently the individual axis steps may be grouped as a parallel butterfly $\mathcal{H}$, followed by one recordwise monomial operation J_E . At each address, J_E multiplies the record by

$$y^E, \quad E = - \sum_{i \text{ participating}} \frac{2r}{u_i} b_i k_i \pmod{2r}. \quad (37)$$

Here u_i is that axis's current recursive length, b_i its newly written branch bit, and $0 \leq k_i < u_i/2$ is encoded by its still unprocessed named slots. The exponent depends on the address, not on coefficient values. In the opposite direction the grouped round is $\mathcal{H} J_E^{-1}$: apply y^{-E} first and the parallel butterfly second. Each $\mathcal{H}$ is a contraction and each J_E is an isometry, so every completed round is a contraction.

6.2 A layout on which the parallel layers apply

To apply Proposition 32 in a transform round, we must place the selected bits of the participating axes at a common offset in consecutive K -bit chunks. We now construct that layout, keeping each polynomial's coefficients in a contiguous suffix. The layout uses the single-bit moves of Lemma 3 and the chunk interchanges of Lemma 25; each movement is included in the transform cost. The layer then supplies one truncated contraction per round, and its temporary rows and work streams are removed before the next round begins.

Lemma 34 (Transform layout and cost). *Fix the rational parameters and positive comparison constants $a_d \leq b_d$, $a_r \leq b_r$ of Proposition 32. Fix constants $C_\ell > 0$ and $C_w > 1$, and choose the layer constants once as*

$$C_M = C_\ell + 1, \quad C_{\text{desc}} = 32(C_\ell + C_w + 1).$$

Use these constants and C_w in Proposition 32. These fixed choices determine a cutoff p_1 , one fixed-tape procedure, and an implicit time constant with the following guarantee. For every choice of positive integers p, d, r with $p \geq p_1$ and

$$a_d p^\epsilon \leq d \leq b_d p^\epsilon, \quad 2^{a_r p/d} \leq r \leq 2^{b_r p/d},$$

suppose $d \geq 2$ and $r = 2^\ell$ with an integer $2 \leq \ell < p$, and require

$$d\ell \leq C_\ell p, \quad 1 \leq K = \lfloor d^c \rfloor \leq \ell - 1.$$

Let $t_i \in \{2^{\ell-1}, 2^\ell\}$ for $1 \leq i \leq d-1$, put $M = \prod_{i=1}^{d-1} t_i$ and $T = rM$, and let w be an integer with $p+2 \leq w \leq C_w p$. The cutoff and time constant are independent of p, d, r , these axis choices, the round headers constructed from them, and w within the displayed bounds.

The input supplies the self-delimiting header

$$\mathcal{T} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(K)\Gamma(w) \prod_{i=1}^{d-1} \Gamma(t_i),$$

where the product is concatenation in axis order and Γ is the integer code of (1). The supplied d determines the number of axis entries, and $\ell = \log_2 r$ is obtained from the supplied power of two r . Assume that the header is valid with the stated values and that the input records use the two's-complement component format of Proposition 32 with width w .

The procedure constructs an explicit positional layout permutation L of record addresses, preserving their polynomial suffixes, and transforms disk grid arrays of M contiguous degree-less-than- r polynomial records. Its exact counterpart is $LB\mathcal{F}_t^-$. Its opposite procedure consumes that order and has exact counterpart $\mathcal{F}_t^+ B^{-1} L^{-1}$. Each procedure has error less than $\sqrt{2}\ell 2^{-p}$, returns disk grid coefficients in the same component format, and costs

$$O\left(Tp \left[dK + pK^{\tau-1} + \ell d^\lambda\right]\right). \quad (38)$$

Both layout construction and final restoration are charged in this bound. The forward procedure may retain its output layout for a pointwise product and the following opposite procedure.

Proof. Put $D = d - 1$ and write $a_i = \log_2 t_i$. The supplied encoding has $2w = \Theta(p)$ bits per coefficient, so the logical bit volume is $\Theta(Tp)$. Call an axis long if $a_i = \ell$ and short if $a_i = \ell - 1$. Name the original bit slots of axis i by (i, h) , $a_i - 1 \geq h \geq 0$. Initially these slots are in axis order and, within each axis, in decreasing h . The coefficients of each polynomial form a contiguous suffix. Put

$$b = (\ell - 1) \bmod K, \quad q = \lfloor (\ell - 1)/K \rfloor.$$

Form a list of the slots to move to a prefix. First list $(i, \ell - 1)$ for the length- 2^ℓ axes in increasing i . Then list (i, h) for $h = b - 1, \dots, 0$, and within each h in increasing i ; this second list is empty if $b = 0$. Move each listed slot immediately after the prefix already formed, leaving all other slots in their relative order. There are at most $D(b+1) = O(dK)$ listed slots, so the moves cost $O(TpdK)$. A descriptor records the physical position of every named slot. The part of each axis left outside the prefix now consists of q consecutive K -bit chunks.

Number the axes by i and their chunks, from high to low, by j . The remaining physical order changes from

$$(1, 1), (1, 2), \dots, (1, q), (2, 1), \dots, (D, q) \quad \text{to} \quad (1, 1), (2, 1), \dots, (D, 1), (1, 2), \dots, (D, q).$$

To obtain this order, scan its target positions from left to right. At each position except the last, exchange the chunk currently there with the required chunk if they differ. This uses at most $Dq - 1$ exchanges of width- K address chunks, even when the chunks are nonadjacent. Each costs $O(TpK^\tau)$, so all these exchanges cost

$$O(Tp Dq K^\tau) = O(Tp p K^{\tau-1}).$$

These prefix moves and chunk exchanges define L . We use the same letter for its induced record permutation: a record at address x moves to address Lx , while its polynomial suffix stays intact. Recording the inverse operations in reverse order gives L^{-1} with the same cost.

For a small illustration of this positional permutation, take $\ell = 6$, $K = 2$, and one long axis followed by one short axis. Then $b = 1$ and $q = 2$. Write $[i : h, h']$ for the chunk containing the two named slots $(i, h), (i, h')$, in that order. After L the slots are

$$\underbrace{(1, 5), (1, 0), (2, 0)}_{\text{prefix}}; \quad [1 : 4, 3] [2 : 4, 3] [1 : 2, 1] [2 : 2, 1].$$

For example, the round at named position $h = 4$ uses offset one in each of the first two chunks. It writes frequency bit one on the long axis and frequency bit zero on the short axis. The named position specifies the same physical pairing rule on both axes even though the frequency significance differs. This example illustrates only the slot permutation; the parameter bounds of the lemma govern its asymptotic use.

The header has $O(p)$ bits: its axis entries use $O(\sum_i a_i + d) = O(p)$ bits, and its remaining entries have $O(p)$ bits in total. The layout lists have binary length polynomial in p . Parsing the header and generating these lists by elementary integer arithmetic take $\text{poly}(p) = O(Tp)$ time. Indeed the fixed comparison bands give $T \geq r \geq 2^{(a_r/b_d)p^{1-\epsilon}}$, uniformly over the permitted d, r , so T dominates every fixed power of p .

First process the round at named position $h = \ell - 1$ on the long axes, if there are any. Every axis then has current recursive length $2^{\ell-1}$. Next process $h = \ell - 2, \dots, b$ in the qK rounds on the remaining chunks, and finally the bottom positions $h = b - 1, \dots, 0$, if $b > 0$. Thus all participating axes in a common round have current length $u_i = 2^{h+1}$. The branch bit written at position h is frequency bit $a_i - 1 - h$; its frequency significance can differ between long and short axes. The descriptor locates that named slot after L , so the physical move has not changed which pair the current butterfly uses. The opposite procedure executes this schedule in reverse order and, when present, ends with the extra leading round on the long axes.

For a position h in the remaining chunks, the selected bit has offset $(h - b) \bmod K$, counted from the least significant bit, in each of the D consecutive chunks with the same chunk index j . Let n_P and n_S be the numbers of record-address bits before and after that group. Every combination of those bits occurs, so the layer sees the complete address set

$$[P] \times [2^K]^D \times [S], \quad P = 2^{n_P}, \quad S = 2^{n_S}, \quad n_P + DK + n_S = \sum_i a_i \leq D\ell = O(p).$$

Thus its record count is $PS2^{DK} = M$ and $\log_2(2M) \leq (C_\ell + 1)p = C_M p$. The current transform parameters supply p, d, r, K, w , and the round determines D, P, S and the selected offset ρ . Encode these values as the header $\mathcal{E}$ of Proposition 32. The same fixed descriptor bound covers every axis choice and round. Indeed $|\Gamma(v)| \leq 2\log_2 v + 1$, and $\ell + n_P + n_S \leq d\ell \leq C_\ell p$. Also $d, D \leq \max\{C_\ell, 1\}p$, $K, \rho + 1 \leq p$, and $w \leq C_w p$. Adding the nine code lengths therefore gives

$$|\mathcal{E}| \leq 2C_\ell p + 12\log_2 p + 4\log_2 \max\{C_\ell, 1\} + 2\log_2 C_w + 9 \leq C_{\text{desc}} p.$$

The last inequality uses $p \geq 1$, $\log_2 p \leq p$, and $\log_2 x \leq x$ for $x \geq 1$. This is the descriptor passed to the layer; the longer table locating named slots remains separate for the transform's address calculations. The simultaneous-layer procedure therefore applies at cost $O(Tpd^{\lambda'})$ per round and $O(Tp\ell d^{\lambda'})$ in total.

In a round on prefix slots, process the participating bits individually. Temporarily move one such bit immediately before the polynomial suffix. In each now adjacent pair, buffer the first record

and scan it with the second, writing the half-sums and half-differences to two work tapes. Append the half-sum record and then the half-difference record, and reverse the bit move after all pairs are processed. Including head returns, the work is $O(rp)$ per pair and $O(Tp)$ for the moves and scans at coefficient width $O(p)$. That width suffices without truncating between axes: after $s \leq D$ individual H_0 factors, the coefficients are exact dyadics on the $2^{-(p+s)}$ grid and still lie in the disk. A temporary sum has modulus at most two and needs only a constant number of extra integer bits. Hence $p + D + O(1) = O(p)$ bits per component suffice, since $D \leq d$ and $d\ell = O(p)$. Complete the combined monomial exactly and apply Q_p once at the end of the round. Write the resulting disk-grid numerators in the supplied w -bit component format. Their magnitudes are at most 2^p , so this takes one linear scan within the same bound. In an opposite round on prefix slots the inverse monomial comes first and preserves the grid and disk, so the same width bound holds for its individual factors. There are at most $b + 1 \leq K$ rounds on prefix slots, giving cost $O(TpdK)$.

For each output record of a forward butterfly round, the descriptor locates the new branch bit b_i and the lower named slots that encode k_i . Read their current values to compute (37). Before an opposite round at position h , all lower positions have already been processed in reverse. Those slots again index the lower inputs of that level, while slot h still gives b_i . They therefore encode the same k_i needed by the inverse monomial before its butterfly. The binary descriptions of the addresses, exponent, and layout have polynomial length in p . Ordinary fixed-tape arithmetic therefore computes the exponent in $\text{poly}(p)$ time per polynomial record. Each record contains $\Theta(rp)$ bits, and r dominates every fixed power of p , so this work fits within its scan cost. The same bound covers scans of the polynomially many round and layout descriptors. Growing lists reside on tapes; a fixed number of counter and work tapes suffices for every number of axes.

To implement the monomial, write $E = er + s$ with $0 \leq s < r$. Split the coefficient list before position $r - s$, put its tail before its head, negate the moved tail, and apply the common sign $(-1)^e$. This is multiplication by y^E modulo $y^r + 1$. There are always two pieces. Splitting them to two tapes and merging in that order costs $O(rp)$ per record, including head returns; $s = 0$ is an ordinary signed copy. The inverse twiddle uses the same split and merge with exponent $-E$, and no coefficient sort is required.

For a forward round on a group of D chunks, the layer call returns $Q_p(\mathcal{H}z)$ and the next operation is J_E . Truncation toward zero is odd and acts separately on every real and imaginary part. Since J_E is a signed coefficient permutation, it follows that

$$J_E Q_p(\mathcal{H}z) = Q_p(J_E \mathcal{H}z).$$

Thus the actual forward call sequence returns exactly the result of truncating once after the exact contraction round. In the opposite direction, J_E^{-1} first preserves the disk grid, and the layer call returns $Q_p(\mathcal{H}J_E^{-1}z)$ directly. The rounds on prefix slots have the same forms, with one truncation, by their exact evaluation above. A contraction does not increase the norm of the difference between the computed and exact arrays; the single truncation then adds less than $\sqrt{2}2^{-p}$ by (34). Both the contraction and truncation preserve the disk. Starting with zero error, at most ℓ rounds therefore give the claimed error and disk bounds. The layout permutations are isometries and add no numerical error.

It remains to identify the order of the full procedures. Before relocating slots, write $\mathcal{D}_- = BF_t^-$ and $\mathcal{D}_+ = F_t^+ B^{-1}$ for the exact logical forward and opposite procedures. Executing each operation on named slots after L conjugates it by L . The forward procedure first moves the records by L , whereas the opposite procedure restores them by L^{-1} after its reverse arithmetic. Their full exact operators are

$$(LD_-L^{-1})L = LBF_t^-, \quad L^{-1}(LD_+L^{-1}) = F_t^+ B^{-1} L^{-1}.$$

In particular, the forward entry with frequency address j is stored at $L(Bj)$. The reverse arithmetic consumes the frequency branch coordinates and outputs the normalized opposite transform indexed by the original axis coordinates, which are still in layout L . The factor B^{-1} is realized by that arithmetic; the final charged movement L^{-1} restores axis-major physical order. $\square$

Pointwise multiplication commutes with applying the same record permutation LB to both operands. We may therefore multiply the forward outputs in their retained order and pass that order directly to the opposite procedure. The following polynomial product implements this operation with an established integer multiplier.

7 Exact normalized polynomial products

To multiply the polynomial records, we use the fixed-tape integer multiplier of Harvey and van der Hoeven [15, Theorem 1.1], with cost $O(N \log(2N))$ on N -bit inputs. This is an independent subroutine. Its inputs below have $O(rp)$ bits, and we include the conversions between coefficient streams and signed integers in the cost. The reduction is a signed form of Kronecker substitution; compare [15, Lemma 2.5]. We specify the packing and recovery of coefficients to include negative values and the required tape format.

Lemma 35 (Signed packing and normalized ring multiplication). *Let $r = 2^\ell < 2^p$ with integers $\ell \geq 1$ and $p \geq 2$, and fix $C_w > 1$. Let w be an integer with $p + 2 \leq w \leq C_w p$. The input supplies $\Gamma(p)\Gamma(r)\Gamma(w)$ and two disk grid polynomials $f, g \in \mathcal{R}_r$ in the component format of Proposition 32 with width w . One can compute the exact Gaussian integer numerators of fg , and then return $Q_p(fg/r)$ in that same format, in time $O(rp \log(rp))$ on a fixed number of tapes. The output is a disk grid polynomial with error less than $\sqrt{2}2^{-p}$ from fg/r .*

Proof. The supplied power of two r determines $\ell = \log_2 r$. Write

$$f = 2^{-p}(a + ib), \quad g = 2^{-p}(c + ie),$$

where $a, b, c, e \in \mathbb{Z}[y]$ have degree less than r and coefficients of absolute value at most 2^p . Set $B_0 = 2^{4p}$ and, for $u \in \{a, b, c, e\}$, define the signed evaluation integer

$$A_u = u(B_0) = \sum_{j=0}^{r-1} u_j B_0^j.$$

The bound

$$|A_u| \leq 2^p \frac{B_0^r - 1}{B_0 - 1} < B_0^r/2$$

shows that each magnitude fits in $4pr$ bits. Multiply the four pairs of absolute values using the established multiplier and restore each product's sign. The resulting integers are $A_a A_c, A_b A_e, A_a A_e, A_b A_c$. For example, if $h = ac$ is the ordinary product in $\mathbb{Z}[y]$, then

$$A_a A_c = (ac)(B_0) = \sum_{j=0}^{2r-2} h_j B_0^j.$$

The other three evaluation products have the same form. Every coefficient in any of these ordinary polynomial products satisfies

$$|h_j| \leq r2^{2p} < 2^{3p} < B_0/2. \tag{39}$$

After the lower coefficients have been removed from an evaluation, this strict bound identifies the next coefficient by its centered residue modulo B_0 .

To form A_u , start with carry $\gamma_0 = 0$ and write

$$d_j \equiv u_j + \gamma_j \pmod{B_0}, \quad 0 \leq d_j < B_0, \quad \gamma_{j+1} = \frac{u_j + \gamma_j - d_j}{B_0}.$$

Since $-B_0 < u_j + \gamma_j < B_0$ whenever $\gamma_j \in \{-1, 0\}$, the carry stays in $\{-1, 0\}$. Telescoping the recurrence gives

$$A_u = \sum_{j=0}^{r-1} d_j B_0^j + \gamma_r B_0^r.$$

The r blocks are therefore a fixed-width two's-complement encoding of A_u , with $\gamma_r = -1$ exactly when $A_u < 0$. In that case complement and increment the blocks to obtain $|A_u|$; otherwise retain them. Every block has $4p$ bits and every carry has constant size, so a constant number of scans costs $O(rp)$. Endianness changes required by the multiplier are linear-time reversals on a work tape.

Now let Z be one of the four signed product integers, and let h be its corresponding ordinary polynomial product. Put $J = 2r$ and pad that product by $h_{J-1} = 0$. The bound on the evaluation integers gives $|Z| < B_0^J/4$, so Z fits in J blocks of width $4p$. Convert its sign and magnitude to sign-extended two's complement:

$$Z = \sum_{j=0}^{J-1} q_j B_0^j + \varepsilon B_0^J, \quad 0 \leq q_j < B_0, \quad \varepsilon \in \{-1, 0\}.$$

Zero-fill the unused high blocks; for a negative sign, complement and increment within this fixed width. These conversions take a constant number of scans. Put $\rho_0 = 0$. At block j , let $\hat{h}_j$ be the unique integer in $[-B_0/2, B_0/2)$ congruent to $q_j + \rho_j$ modulo B_0 , and put

$$\rho_{j+1} = \frac{q_j + \rho_j - \hat{h}_j}{B_0}.$$

The carry belongs to $\{0, 1\}$: if ρ_j has this property, then $0 \leq q_j + \rho_j \leq B_0$, and centering either leaves this value unchanged or subtracts B_0 .

The meaning of the carry is the following residual identity, after coefficients below j have been correctly extracted:

$$\sum_{k=j}^{J-1} h_k B_0^{k-j} = \sum_{k=j}^{J-1} q_k B_0^{k-j} + \rho_j + \varepsilon B_0^{J-j}.$$

At $j = 0$ this is the equality between the two representations of $Z = h(B_0)$, with $\rho_0 = 0$. For $j < J$, reduction modulo B_0 gives $h_j \equiv q_j + \rho_j \pmod{B_0}$. By (39), including the padded value $h_{J-1} = 0$, the unique centered representative is h_j itself. Thus $\hat{h}_j = h_j$. Subtracting this coefficient and dividing the residual identity by B_0 gives the next identity, with exactly the displayed update for ρ_{j+1} . At $j = J$ the identity is $0 = \rho_J + \varepsilon$. Hence a negative product has final carry 1, canceled by its sign extension $\varepsilon = -1$; a nonnegative product has final carry zero. The machine implements the induction by reading one block and updating this constant-size carry at each step, so extraction costs $O(rp)$.

The real and imaginary numerator streams are respectively $ac - be$ and $ae + bc$. To reduce them modulo $y^r + 1$, subtract the coefficient at $j + r$ from the one at j for $0 \leq j < r$, treating a

missing coefficient as zero. Splitting the two halves to tapes aligns these terms in one further scan. For each first input index $j_1 \in \{0, \dots, r-1\}$, exactly one second index j_2 in that range satisfies $j_1 + j_2 \equiv j \pmod{r}$. Its sign is positive when $j_1 + j_2 = j$ and negative when $j_1 + j_2 = j + r$. Thus each resulting Gaussian integer coefficient H_j is a sum of exactly r signed products of Gaussian integer input coefficients of modulus at most 2^p . Consequently

$$|H_j| \leq r2^{2p}.$$

The same triangle inequality before imposing the disk bound gives $\|fg/r\|_\infty \leq \|f\|_\infty \|g\|_\infty$; in particular the normalized product of disk polynomials lies in the disk. All intermediate coefficient additions fit in $O(p)$ bits. The width- $4p$ packing fields and the exact multiplier output prevent overflow or ambiguity between adjacent fields.

Finally the p -bit grid numerator of the normalized product is

$$\text{trunc}_0 \left(\frac{\Re H_j}{2^{p+\ell}} \right) + i \text{trunc}_0 \left(\frac{\Im H_j}{2^{p+\ell}} \right).$$

To implement it, shift the absolute magnitude right by $p + \ell$ bits and restore its sign, at cost $O(p)$ per coefficient. The returned coefficient is 2^{-p} times this numerator, namely $Q_p(2^{-2p}H_j/r)$; (34) proves its error and disk bounds. Each returned component numerator has magnitude at most 2^p , so write it as a signed w -bit two's-complement word, most significant bit first, with the real word before the imaginary word and coefficient indices in increasing order. This costs $O(rw) = O(rp)$ and supplies the format used by the opposite transform. Four exact integer products dominate the linear packing, extraction, combination, wrapping, and truncation costs. $\square$

7.1 The normalized convolution interface

For arrays on $\prod_i \mathbb{Z}/t_i\mathbb{Z}$ with values in $\mathcal{R}_r$, let $*$ denote cyclic convolution in the displayed axes and define

$$\mathcal{M}_r(f, g) = \frac{f * g}{rM} = \frac{f * g}{T}.$$

For a fixed output coefficient, each of the M first cyclic addresses determines one second address, and each of the r first polynomial indices determines one second index modulo r . The negacyclic wrap supplies the sign. Thus that coefficient averages rM signed scalar products, and $\mathcal{M}_r$ is a bilinear contraction:

$$\|\mathcal{M}_r(f, g)\|_\infty \leq \|f\|_\infty \|g\|_\infty.$$

If the other operand is a disk array, replacing one operand changes the output by at most the norm distance between its two values.

Proposition 36 (Convolution from the retained transform layout). *Under the hypotheses of Lemma 34, $\mathcal{M}_r$ on disk grid operands has an approximation with error less than*

$$\sqrt{2} M(3\ell + 1)2^{-p} \tag{40}$$

and cost

$$O \left(Tp \left[dK + pK^{\tau-1} + \ell d^{\lambda'} + \log(rp) \right] \right).$$

Put $m_M = \log_2 M$ and $w_M = w + m_M$. The returned grid array is in the original array and coefficient order, with the same two's-complement component format at width w_M . The procedure supplies $\Gamma(w_M)$ on the output descriptor tape, so the caller knows every component boundary from this width and the retained shape parameters. If one operand has norm at most $\rho < 1$ and the bound in (40) is at most $1 - \rho$, the returned grid array is a disk array.

Proof. Perform both forward transforms and retain their common LB order. Pass the supplied p, r, w to Lemma 35 for corresponding polynomial records. Its output keeps that record order and uses the component format required by the opposite procedure. Apply the opposite procedure to that order, and multiply its result exactly by $M = T/r$.

To check normalization, expansion of the finite sums gives

$$F_t^-(f * g) = M(F_t^- f)(F_t^- g).$$

As $F_t^+ F_t^- = I/M$, it follows that

$$MF_t^+ \left(\frac{(F_t^- f)(F_t^- g)}{r} \right) = \frac{f * g}{rM}.$$

Let $P(u, v)$ denote the exact pointwise product uv/r , and put $\mathcal{T}_+ = F_t^+ B^{-1} L^{-1}$. Since LB permutes records and leaves their polynomial suffixes intact, $P(LBu, LBv) = LBP(u, v)$. Thus $\mathcal{T}_+$ applied to the pointwise product of the exact stored forward outputs equals $F_t^+ P(F_t^- f, F_t^- g)$. The displayed normalization identity identifies its exact scaling by M with $\mathcal{M}_r(f, g)$.

For the error estimate, put $\delta = \sqrt{2}\ell 2^{-p}$ and $\eta = \sqrt{2} 2^{-p}$. Let u, v be the exact stored forward outputs and $\tilde{u}, \tilde{v}$ the returned ones. All four are disk arrays, and each forward error is less than δ . The exact map P is bilinear and satisfies $\|P(x, y)\|_\infty \leq \|x\|_\infty \|y\|_\infty$ by the coefficient bound in Lemma 35. Hence

$$\begin{aligned} P(\tilde{u}, \tilde{v}) - P(u, v) &= P(\tilde{u} - u, \tilde{v}) + P(u, \tilde{v} - v), \\ \|P(\tilde{u}, \tilde{v}) - P(u, v)\|_\infty &< 2\delta, \end{aligned}$$

where the first term uses the disk bound on $\tilde{v}$ and the second uses it on u . Write $\tilde{z} = Q_p(P(\tilde{u}, \tilde{v}))$ and $z_* = P(u, v)$. The product lemma makes $\tilde{z}$ a disk grid array and gives $\|\tilde{z} - z_*\|_\infty < 2\delta + \eta$. If $\tilde{h}$ is the returned opposite output, its approximation bound applies to $\tilde{z}$. Since $\mathcal{T}_+$ is a contraction,

$$\|\tilde{h} - \mathcal{T}_+ z_*\|_\infty \leq \|\tilde{h} - \mathcal{T}_+ \tilde{z}\|_\infty + \|\mathcal{T}_+ (\tilde{z} - z_*)\|_\infty < 3\delta + \eta.$$

The final exact multiplication by M therefore gives error less than $M(3\delta + \eta) = \sqrt{2} M(3\ell + 1)2^{-p}$, as claimed. If one original operand has norm at most ρ , then $\|\mathcal{M}_r(f, g)\|_\infty \leq \rho$. Under the stated error margin, the returned norm is less than $\rho + (1 - \rho) = 1$.

Every computed output before the final scaling is a disk grid array. The layer's internal dyadic values and the integer product numerators use the larger exact widths stated in their respective procedures. The scale $M = 2^{m_M}$ is a power of two, so it preserves the p -bit grid. Compute $m_M = \sum_i \log_2 t_i$ from the supplied axis lengths. For each signed w -bit numerator returned by the opposite procedure, append exactly m_M zero bits at its least significant end. This is the exact two's-complement encoding of the scaled numerator in $w_M = w + m_M$ bits, including for negative values. The real word still precedes the imaginary word and the coefficient order is unchanged. Since $w \leq C_w p$ and $m_M \leq C_\ell p$, this known output width is $O(p)$. Writing $\Gamma(w_M)$ takes $O(p)$ time.

There are $M = T/r$ pointwise polynomial products. Their total cost is $O(Tp \log(rp))$; the three transform costs are as stated, and the last exact scaling is linear in Tp . All data movement uses the previously specified scans, swaps, and layout restoration. $\square$

8 Resampling with the permutations left in the transform

The transform lengths needed for the integer product are pairwise coprime, whereas the synthetic transforms use powers of two. Gaussian resampling connects these two shapes. Its usual implementation sorts twice on each coordinate. We retain both permutations in the transform formula and

compute that formula directly. This section proves the resulting interface, including its tape cost when the number of coordinates grows.

For an integer $q \geq 1$, write

$$(F_q u)_k = \frac{1}{q} \sum_{j=0}^{q-1} u_j e^{-2\pi i j k / q}, \quad (F_q^+ u)_k = \frac{1}{q} \sum_{j=0}^{q-1} u_j e^{2\pi i j k / q}.$$

All subscripts on a vector of length q are taken modulo q . The vector norm is the maximum modulus of its coordinates. All operator norms are induced by this norm: $\|L\| = \sup_{\|u\| \leq 1} \|Lu\|$. Each normalized Fourier transform above is a contraction, since each output averages inputs multiplied by complex numbers of modulus one. Coordinate permutations preserve the norm. For an integer precision $p \geq 1$, put

$$\mathbb{D}_p = \{2^{-p}(a + ib) : a, b \in \mathbb{Z}, a^2 + b^2 \leq 2^{2p}\}.$$

For storage between completed scalar routines, put $w_c = p + 2$. Represent $2^{-p}(a + ib) \in \mathbb{D}_p$ by a signed w_c -bit two's-complement word for a , followed by one for b , with each word written most significant bit first. A coordinate list is the concatenation of these fixed-length records in its stated order. Since $|a|, |b| \leq 2^p < 2^{w_c-1}$, this format includes both signs of every permitted numerator, including -2^p and 2^p . An approximation $\tilde{L} : \mathbb{D}_p^a \rightarrow \mathbb{D}_p^b$ to a linear contraction $L : \mathbb{C}^a \rightarrow \mathbb{C}^b$ has error E if $2^p \|\tilde{L}u - Lu\| < E$ for every input in $\mathbb{D}_p^a$. The approximation itself need not be linear. Composing such algorithms adds their errors: at each step, insert the exact map at the computed input and use its contraction bound on the accumulated input error.

We use the established $O(N \log N)$ integer multiplier of Harvey and van der Hoeven [15, Theorem 1.1] for the shorter integer records that occur below. We also use their Lemma 2.12: for integer precision $p > 100$ and integers $k > 1$, $0 \leq j < k$, one can compute $\tilde{\zeta} \in \mathbb{D}_p$ with $2^p |\tilde{\zeta} - e^{2\pi i j / k}| < 2$ in time $O(\max(p, \log k)^{1+\delta})$, for every fixed $0 < \delta < 1/8$. Throughout this section, numbered citations to that paper refer to its author-hosted 45-page manuscript.

8.1 The one-dimensional interface

Lemma 37 (Resampling without executing the two permutations). *Fix $0 < \delta < 1/8$. Let $p > 100$, $2 \leq s < t < 2^p$ and $2 \leq \alpha < \sqrt{p}$ be integers, with $\gcd(s, t) = 1$. Put $\sigma = t/s$ and $\theta := \sigma - 1$, and assume*

$$\alpha^{-2} \leq \theta \leq \frac{1}{4}, \quad n_{\alpha, \theta} := \min \left\{ \left\lceil \frac{p}{\alpha^2 \theta} \right\rceil, \left\lceil \frac{p+1}{4\alpha^2} + \frac{1}{\theta} \right\rceil \right\}.$$

Define coordinate permutations by

$$(P_s u)_j = u_{tj}, \quad (P_t v)_k = v_{-sk}.$$

There are linear contractions $A : \mathbb{C}^s \rightarrow \mathbb{C}^t$ and $B_0 : \mathbb{C}^t \rightarrow \mathbb{C}^s$ satisfying

$$P_s F_s = 2^{2\alpha^2} B_0 P_t F_t A. \tag{41}$$

There are fixed multitape algorithms for approximations $\tilde{A} : \mathbb{D}_p^s \rightarrow \mathbb{D}_p^t$ and $\tilde{B}_0 : \mathbb{D}_p^t \rightarrow \mathbb{D}_p^s$, each with error less than p^2 , using respectively

$$O((t+p)p^{1+\delta}) \quad \text{and} \quad O(n_{\alpha, \theta}(t+p)p^{1+\delta})$$

time. The constants are uniform in p, s, t, α . Inputs and outputs use the scalar component format above and occur in increasing coordinate order.

Proof. The Gaussian resampling argument uses two maps $\mathbf{S}, \mathbf{T} : \mathbb{C}^s \rightarrow \mathbb{C}^t$. For $0 \leq k < t$, they are

$$(\mathbf{S}u)_k = \frac{1}{\alpha} \sum_{j \in \mathbb{Z}} e^{-\pi\alpha^{-2}(j-sk/t)^2} u_j, \quad (\mathbf{T}u)_k = \sum_{j \in \mathbb{Z}} e^{-\pi\alpha^2(tj/s-k)^2} u_j.$$

Both sums converge absolutely. For $0 \leq j < s$, define

$$q_j = \left\lfloor \frac{tj}{s} + \frac{1}{2} \right\rfloor, \quad \beta_j = \frac{tj}{s} - q_j.$$

Thus q_j is an integer nearest to tj/s , with ties rounded upward. Let $C : \mathbb{C}^t \rightarrow \mathbb{C}^s$ select these coordinates, and let $\mathbf{D} : \mathbb{C}^s \rightarrow \mathbb{C}^s$ be the diagonal map

$$(Cv)_j = v_{q_j}, \quad (\mathbf{D}u)_j = e^{\pi\alpha^2\beta_j^2} u_j.$$

Set $N = C\mathbf{T}\mathbf{D}$. In $(Nu)_j$, the summand with index j is exactly u_j : the factor in $\mathbf{D}$ cancels the Gaussian factor at the selected coordinate q_j . The other summands form $((N - I)u)_j$. The exact Gaussian identity and inverse estimate proved in [15, Theorem 4.2 and Lemma 4.6] give

$$\mathbf{T}P_sF_s = P_tF_t\mathbf{S}, \quad \|N - I\| < 2.01e^{-\pi\alpha^2\theta/2} < 2^{-\alpha^2\theta}. \quad (42)$$

The first identity uses the negative Fourier sign in our definition of F_q ; its right-hand permutation is consequently multiplication of the frequency by $-s$. The cited Lemma 4.6 assumes only $\alpha^2\theta \geq 1$, which holds here; it gives $\|N - I\| < 2.01e^{-\pi/2} < 0.42$. The proof of the cited Proposition 4.7(i) uses only this bound, $\alpha \geq 2$ and the cited Lemma 4.5, so its norm bounds below also hold. Thus $\|N - I\| < 1/2$, so N is invertible by the Neumann series. Put

$$\mathbf{S}' = \mathbf{S}/2, \quad \mathbf{J}' = N^{-1}/2, \quad \mathbf{D}' = 2^{-(2\alpha^2-2)}\mathbf{D}.$$

The proof of Proposition 4.7(i) in the cited paper gives

$$\|\mathbf{S}'\| < \frac{3}{4}, \quad \|\mathbf{J}'\| < \frac{7}{8}, \quad \|\mathbf{D}'\| < 1.$$

Now set $A = \mathbf{S}'$ and $B_0 = \mathbf{D}'\mathbf{J}'C$. Because $\|C\| = 1$, both are contractions. The diagonal map $\mathbf{D}$ is invertible, and $C\mathbf{T} = N\mathbf{D}^{-1}$, so

$$(\mathbf{D}N^{-1}C)\mathbf{T} = \mathbf{D}N^{-1}(N\mathbf{D}^{-1}) = I.$$

Applying this left inverse to the first identity in (42) gives $\mathbf{D}N^{-1}CP_tF_t\mathbf{S} = P_sF_s$. The normalizations in $\mathbf{D}'$, $\mathbf{J}'$, and $\mathbf{S}'$ have product $2^{-(2\alpha^2-2)}2^{-1}2^{-1} = 2^{-2\alpha^2}$. Hence

$$B_0P_tF_tA = 2^{-2\alpha^2}\mathbf{D}N^{-1}CP_tF_t\mathbf{S} = 2^{-2\alpha^2}P_sF_s,$$

which is (41).

Ordered numerical maps. Lemma 4.8 of [15] supplies $\tilde{\mathbf{D}}'$. Lemma 40 below supplies $\tilde{\mathbf{S}}'$ and an approximation $\tilde{E}$ of $E = N - I$ by chirped correlations. For $\tilde{\mathbf{J}}'$ we use the Neumann evaluation of [15, Lemma 4.12] with this $\tilde{E}$ and with $n_{\alpha,\theta}$ terms. Its error analysis uses only $\|E\| < 1/2$, $\varepsilon(\tilde{E}) < p/3$, $n \leq p$, and the bound $\|E^n\| \leq 2^{-p}$, which replaces $\|E\|^n \leq 2^{-p}$; the tail uses $\|E^{n+k}\| \leq \|E^n\|\|E\|^k$. Corollary 39 supplies $n \leq p$ and $\|E^n\| \leq 2^{-p}$, so the error bound $3p^2/4$ of that lemma is unchanged. Inputs and outputs are in increasing coordinate order, and every output

is in the disk grid. The record conversions and tape movement needed in our format are accounted for below. The guarantees are

map	error bound	time bound
$\tilde{S}'$	$16p$	$O((t+p)p^{1+\delta})$
$\tilde{J}'$	$3p^2/4$	$O(n_{\alpha,\theta}(t+p)p^{1+\delta})$
$\tilde{D}'$	4	$O(tp^{1+\delta})$.

For C , observe that $q_0 = 0$, $q_{s-1} < t$, and $q_{j+1} > q_j$, because $t/s > 1$. Thus C selects an ordered subsequence. A forward scan, with the current input coordinate stored on a work tape, copies a record exactly when that coordinate equals the next q_j .

The selected indices can themselves be generated in order using only $O(p)$ work per index. Precompute $t = hs + \rho$ with $0 \leq \rho < s$. Start with $f_0 = e_0 = 0$, and maintain $tj = sf_j + e_j$ with $0 \leq e_j < s$. At index j , first use $q_j = f_j + \mathbf{1}_{2e_j \geq s}$. If $j < s - 1$, after copying that record form the values for index $j + 1$ by

$$c_j = \mathbf{1}_{e_j + \rho \geq s}, \quad e_{j+1} = e_j + \rho - c_j s, \quad f_{j+1} = f_j + h + c_j.$$

Here $e_j + \rho < 2s$, so c_j is the only possible carry; the displayed update gives $t(j+1) = sf_{j+1} + e_{j+1}$. All these counters have $O(p)$ bits. Elementary division computes h, ρ in $O((\log t)^2) = O(tp)$ time. The record scan and its coordinate counter also cost $O(tp)$, so C takes $O(tp)$ time with zero numerical error.

Consequently we may take

$$\tilde{A} = \tilde{S}', \quad \tilde{B}_0 = \tilde{D}' \tilde{J}' C.$$

Contraction of the exact maps yields errors less than $16p < p^2$ and $4 + 3p^2/4 < p^2$, respectively. The costs add to the claimed bound.

The original proof of Proposition 4.7(ii) implements $B = P_s^{-1} \tilde{D}' \tilde{J}' C P_t$. Its two label-and-sort operations implement P_t and P_s^{-1} , each costing $O(tp \log t)$. In (41) these permutations remain in the transform identity. The algorithms for A and B_0 use C , the quoted map $\tilde{D}'$, and the maps of Lemma 40, whose truncations contain those of the cited Lemmas 4.9 and 4.11.

Record conversion and tape movement. We specify how these scalar records meet the cited numerical interfaces. Sections 2.1–2.2 of the cited paper use standard binary integers, represent $2^{-p}(a+ib) \in \mathbb{D}_p$ by the exact integer pair (a, b) with p known, and store vectors as ordered coordinate lists. The cited exponential and Gaussian routines return grid values in this representation. Their computed numerator pairs are exact representations of the approximations; the numerical errors remain those stated for the approximated analytic maps. The source leaves the sign code, digit order, padding and field boundaries open, so we make a concrete choice when implementing its constructive algorithms.

At each scalar input or output boundary of a cited routine, use one sign symbol, the ordinary binary magnitude in most-significant-bit-first order, and a fixed separator for each integer. Use one zero digit and the nonnegative sign for zero, put the real field before the imaginary field, and concatenate pairs in coordinate order. Each grid numerator magnitude has at most $p + 1$ digits. To convert one field to our w_c -bit word, buffer its magnitude, right-align it in a zero-filled block of that width, and, for a negative sign, complement the block and add one from right to left. Copy the block from left to right to the output. The inverse conversion reads one complete w_c -bit word, remembers its leading sign, complements and adds one if negative, and emits the signed magnitude after removing leading zeroes. In both directions the represented integer is unchanged.

A unary ruler of w_c cells is prepared once per call by starting with an empty ruler and reading the binary digits of p from most to least significant, successively doubling the current unary prefix and appending one cell when the next digit is one, then adding two cells. This takes $O(p)$ time. Reusing the ruler, a constant number of sweeps of the current field and work blocks performs each conversion, including head returns and cleanup, in $O(p)$ time on fixed tapes. Separators identify the next source field, and the ruler identifies the next fixed-width word; the input head finishes after the consumed field and the output head after the appended field. The inverse adapter supplies the Gaussian input vectors, and the forward adapter formats their outputs. Lemma 2.12 instead receives the ordinary binary integers j, k and needs only the output adapter. These value-preserving conversions add no error. On the Gaussian line lengths $s < t$, their $O((s+t)p) = O(tp)$ work, and their $O(p)$ work per exponential output, are included in the stated bounds.

Tape movement inside $\tilde{S}'$ and $\tilde{E}$, including the cyclic boundary, is included in Lemma 40.

The Neumann evaluation also keeps its computed records bounded. Put $k = n_{\alpha, \theta}$. Lemma 40 gives, for every $z \in \mathbb{D}_p^s$,

$$\tilde{E}z \in \mathbb{D}_p^s, \quad 2^p \|\tilde{E}z - Ez\| < p/3,$$

and its proof establishes the rounded output bound

$$\|\tilde{E}z\| < 0.42\|z\| + (p/3)2^{-p} < 1.$$

For $u \in \mathbb{D}_p^s$, Lemma 4.12 rounds the real and imaginary parts of each coordinate of $u/2$ toward zero at precision p , obtaining $\tilde{v}^{(0)} \in \mathbb{D}_p^s$. It then computes $\tilde{v}^{(j)} = \tilde{E}\tilde{v}^{(j-1)}$ for $1 \leq j < k$. Induction therefore keeps every iterate in $\mathbb{D}_p^s$. The approximation is $\tilde{J}'u = \sum_{j=0}^{k-1} (-1)^j \tilde{v}^{(j)}$. Since $k \leq p$, each component of any signed partial numerator sum has magnitude at most $k2^p$ and needs $p + \lceil \log_2(k+1) \rceil + O(1) = O(p)$ bits. These partial sums are accumulated exactly on separate tapes; only completed iterates are fed into $\tilde{E}$.

The $k-1$ calls of $\tilde{E}$ cost $O(k(t+p)p^{1+\delta})$, including their tape movement, by Lemma 40. The current iterate, next iterate, and running sum use a fixed number of tapes holding $O(p)$ -bit records. Outer accumulation costs $O(ksp)$, within the same bound. All truncations contain those of the cited algorithms, and the tape count and constants remain uniform in the parameters. $\square$

8.2 Faster Gaussian maps

This subsection supplies the two ingredients used in the proof of Lemma 37. We keep the notation of that proof and put $\sigma = t/s$, $\theta = \sigma - 1$. Define q_j and β_j by the same formulas for every integer j , so that $q_{j+s} = q_j + t$ and $\beta_{j+s} = \beta_j$. Write $E = N - I$. By [15, Section 4.2],

$$(Eu)_\ell = \sum_{h \neq 0} e^{-\pi\alpha^2 X_{\ell,h}} u_{\ell+h}, \quad X_{\ell,h} = (\sigma h + \beta_\ell)^2 - \beta_{\ell+h}^2,$$

with indices of u taken modulo s , and every $X_{\ell,h}$ is nonnegative.

Lemma 38 (Powers of the correction). *Assume $\alpha \geq 2$ and $0 < \theta < 1$. For every integer $n \geq 1$,*

$$\|E^n\| \leq e^{\pi\alpha^2(1/(4\theta)+1/2)} (2.01 e^{-\pi\alpha^2\sigma})^n.$$

Proof. Expanding the product, $(E^n u)_\ell$ is the sum over $h_1, \dots, h_n \in \mathbb{Z} \setminus \{0\}$ of $e^{-\pi\alpha^2 X} u_{j_n}$, where $j_0 = \ell$, $j_i = j_{i-1} + h_i$ and $X = \sum_i X_{j_{i-1}, h_i}$. Moving each subtracted square to the following step,

$$X = \sum_{i=1}^n c(j_{i-1}, h_i) + \beta_{j_0}^2 - \beta_{j_n}^2, \quad c(j, h) = (\sigma h + \beta_j)^2 - \beta_j^2 = \sigma h(\sigma h + 2\beta_j).$$

Put $\Phi(j) = \sigma(\beta_j^2 + 1/4)/\theta$. For one step (j, h) let $y = \beta_j + h\theta$ and $w = q_{j+h} - q_j - h \in \mathbb{Z}$. Then $\beta_{j+h} = \beta_j + \sigma h - (q_{j+h} - q_j) = y - w$, so $|y - w| \leq 1/2$. Substitute $\beta_j = y - h\theta$ in $c(j, h)$ and in $\Phi(j + h) - \Phi(j) = \sigma((y - w)^2 - (y - h\theta)^2)/\theta$. Expanding and using $\sigma - \theta = 1$ gives

$$c(j, h) - \Phi(j + h) + \Phi(j) = \sigma h^2 + \frac{\sigma}{\theta} w(2y - w) \geq \sigma h^2,$$

because $w(2y - w) = w^2 + 2w(y - w) \geq w^2 - |w| \geq 0$. Sum over the path. Since $\sigma/(4\theta) \leq \Phi \leq \sigma/(2\theta)$ and $\beta_{j_n}^2 \leq 1/4$,

$$X \geq \sigma \sum_i h_i^2 - \frac{\sigma}{4\theta} - \frac{1}{4} = \sigma \sum_i h_i^2 - \frac{1}{4\theta} - \frac{1}{2}.$$

Consequently

$$\|E^n\| \leq e^{\pi\alpha^2(1/(4\theta)+1/2)} \left(\sum_{h \neq 0} e^{-\pi\alpha^2\sigma h^2} \right)^n.$$

As $\pi\alpha^2\sigma > 4\pi$, the inner sum is less than $2e^{-\pi\alpha^2\sigma}(1 + 2e^{-3\pi\alpha^2\sigma}) < 2.01e^{-\pi\alpha^2\sigma}$. $\square$

The original estimate $\|E\|^n < 2^{-\alpha^2\theta n}$ gains only $\alpha^2\theta$ bits per term. The lemma gains about $\pi\alpha^2 \log_2 e$ bits per term after a burn-in of order α^2/θ bits.

Corollary 39. *Assume $\alpha \geq 2$, $\alpha^2\theta \geq 1$ and $\theta < 1$, and let $n_{\alpha,\theta}$ be as in Lemma 37. Then $n_{\alpha,\theta} \leq p$ and $\|E^{n_{\alpha,\theta}}\| \leq 2^{-p}$.*

Proof. If $n_{\alpha,\theta} = \lceil p/(\alpha^2\theta) \rceil$, then $n_{\alpha,\theta} \leq p$ and $\|E^n\| \leq \|E\|^n < 2^{-\alpha^2\theta n} \leq 2^{-p}$ by (42). Otherwise $n = n_{\alpha,\theta} \geq (p+1)/(4\alpha^2) + 1/\theta$ and $n < \lceil p/(\alpha^2\theta) \rceil \leq p$. Put $\kappa_0 = \pi \log_2 e$, so $4.531 < \kappa_0 < 4.534$, and note $\log_2 2.01 < 1.01$. Taking base-two logarithms in Lemma 38,

$$\log_2 \|E^n\| \leq \kappa_0 \alpha^2 \left(\frac{1}{4\theta} + \frac{1}{2} \right) - n(\kappa_0 \alpha^2 \sigma - 1.01).$$

Since $\sigma > 1$ and $\alpha \geq 2$, the last bracket is at least $4\alpha^2$. Also $4.534(1/(4\theta) + 1/2) \leq 4/\theta$ because $\theta < 1$. Hence $\log_2 \|E^n\| \leq 4\alpha^2/\theta - 4\alpha^2 n \leq -p - 1$. $\square$

Lemma 40 (Gaussian maps by chirped correlation). *Assume the hypotheses of Lemma 37. There are fixed multitape algorithms computing approximations $\tilde{S}' : \mathbb{D}_p^s \rightarrow \mathbb{D}_p^t$ of S' and $\tilde{E} : \mathbb{D}_p^s \rightarrow \mathbb{D}_p^s$ of E , with errors less than $16p$ and $p/3$ respectively, outputs in the disk grid, and time $O((t+p)p^{1+\delta})$ each. Inputs and outputs occur in increasing coordinate order. The constants are uniform in p, s, t, α .*

Proof. Chirp identity. For a rational y and integers a, b , put $\hat{a} = a + y/\sigma$. Expanding both sides and using $\sigma - \theta = 1$,

$$(\sigma\hat{a} - b)^2 = \sigma\theta\hat{a}^2 + \sigma(\hat{a} - b)^2 - \theta b^2.$$

Hence, for rational $\lambda > 0$,

$$e^{-\pi\lambda(\sigma a + y - b)^2} = e^{\pi\lambda\theta b^2} e^{-\pi\lambda\sigma\theta(a + y/\sigma)^2} e^{-\pi\lambda\sigma(a - b + y/\sigma)^2}. \quad (43)$$

For inputs f_a , $0 \leq a < L_A$, the sums $F_b = \sum_a e^{-\pi\lambda(\sigma a + y - b)^2} f_a$, $0 \leq b < L_B$, are therefore $F_b = e^{\pi\lambda\theta b^2} c_b$, where

$$c_b = \sum_a g_a K_{a-b}, \quad g_a = e^{-\pi\lambda\sigma\theta(a + y/\sigma)^2} f_a, \quad K_r = e^{-\pi\lambda\sigma(r + y/\sigma)^2}.$$

This is one correlation of finite sequences; $K_r \in (0, 1]$ and $|g_a| \leq |f_a|$.

One block. Let $P \geq p$ be an integer, $|f_a| \leq F$ with $F \geq 1$, and $e^{\pi\lambda\theta b^2} \leq 2^\Gamma$ for $0 \leq b < L_B$, with $\Gamma \leq 2P$. Suppose approximations $\tilde{f}_a$ with $|\tilde{f}_a - f_a| \leq 2^{4-P}F$ are available. All exponents below are π times rationals whose numerators and denominators have $O(p)$ bits. Lemmas 2.13 and 2.14 of [15], at precision P , give the decaying factors and kernel values, and the numbers $2^{-\Gamma}e^{\pi\lambda\theta b^2}$, each with absolute error at most 2^{1-P} . Multiply each $\tilde{f}_a$ by its factor and round to precision P . Represent every resulting real or imaginary part as an integer multiple of 2^{-P} , split it into its nonnegative and negative parts, and pack each sequence into one integer with slots of $2P + \lceil \log_2(FL_A) \rceil + 8$ bits. A constant number of products by the established multiplier [15, Theorem 1.1] then gives every rounded c_b exactly, because no slot overflows. Multiply by the approximate factor $2^{-\Gamma}e^{\pi\lambda\theta b^2}$, shift by Γ , and round to precision p .

The rounded g_a and K_r have errors at most $2^{5-P}F$ and 2^{1-P} , so the rounded correlation differs from c_b by at most $2^{7-P}FL_A$, while $|c_b| \leq FL_A$. After the shift, the final factor has absolute error at most $2^{\Gamma+1-P}$. Therefore the computed value differs from F_b by at most $2^{\Gamma+9-P}FL_A$, which is at most 2^{-p-2} whenever

$$P \geq p + \Gamma + \lceil \log_2(FL_A) \rceil + 11.$$

The block uses $O(L_A + L_B)$ exponentials and products at precision P and integer products on $O((L_A + L_B)(P + \log(FL_A)))$ bits. When $P = O(p)$ and $\log(FL_A) = O(p)$, its cost is $O((L_A + L_B)p^{1+\delta})$.

The map S' . Here $(S'u)_k = (2\alpha)^{-1} \sum_j e^{-\pi\lambda(\sigma j - k)^2} u_j$ with $\lambda = (\sigma\alpha)^{-2}$. Let $m = \lceil \sqrt{p} \rceil \alpha$, the window of [15, Lemma 4.9]. Partition $0 \leq k < t$ into consecutive blocks of at most m outputs. For a block starting at k_0 , take inputs $j = j_0 + a$, $0 \leq a < L_A = 3m + 3$, where $j_0 = \lfloor k_0/\sigma \rfloor - m$, with the periodic indices of u , and put $k = k_0 + b$, $y = \sigma j_0 - k_0$. These inputs contain every j with $|j - k/\sigma| < m$ for every k in the block. The block sums therefore omit only terms omitted by the truncated sums of that lemma, so they differ from $(S'u)_k$ by less than 2^{-p} . Here $f_a = u_{j_0+a}$ is exact, $F = 1$, and for $b < m$, $\pi\lambda\theta b^2 \leq \pi\theta(\sqrt{p} + 1)^2 \leq (\pi/4)(1.21p)$ because $\theta \leq 1/4$ and $p > 100$, so $\Gamma = \lceil 1.4p \rceil$ suffices. Take $P = 3p$, and apply the factor $(2\alpha)^{-1}$ by one more product at precision P . Each output then has error less than $2^{-p}(1 + 1/4 + 1) < 3 \cdot 2^{-p}$, and its norm is less than $\|S'\| + 3 \cdot 2^{-p} < 1$.

The map E . With $f_j = e^{\pi\alpha^2\beta_j^2} u_j$ and $\lambda = \alpha^2$, the definition of N gives

$$(Nu)_\ell = \sum_j e^{-\pi\lambda(\sigma j - q_\ell)^2} f_j,$$

and its term $j = \ell$ is exactly u_ℓ . The inputs f_j are formed by Lemma 2.14 of [15] for $2^{-\lceil 1.14\alpha^2 \rceil} e^{\pi\alpha^2\beta_j^2}$, a shift, and one product with u_j . Put $B = \lceil 1.14\alpha^2 \rceil$ and $F = 2^B$. The shift multiplies the generator's absolute error by 2^B ; including the product and rounding, the input errors are at most $2^{4-P}F$. This same F also bounds $|f_j|$. Let $m = \lceil \sqrt{p}/(2\alpha) \rceil$, the window of [15, Lemma 4.11]. Partition $0 \leq \ell < s$ into blocks of at most m indices. For a block starting at ℓ_0 , take inputs $j = j_0 + a$, $0 \leq a < L_A = 3m + 1$, with $j_0 = \ell_0 - m$, and evaluate the correlation at every integer $b = q_{\ell_0} + b'$, $0 \leq b' < L_B = 2m + 2$; these include all q_ℓ of the block because $\sigma < 2$. Put $y = \sigma j_0 - q_{\ell_0}$. The choice $F = 2^{\lceil 1.14\alpha^2 \rceil} \leq 2^{1.15p}$ uses $\alpha^2 < p$ and $p > 100$. In particular it bounds the shifted approximation error as well as the exact chirp magnitude. Moreover,

$$\pi\lambda\theta b'^2 \leq \pi\theta\alpha^2(2m + 2)^2 \leq \frac{\pi}{4}(\sqrt{p} + 4\alpha)^2 < \frac{25\pi}{4}p,$$

so $\Gamma = 29p$ suffices, using $(\sqrt{p} + 4\alpha)^2 < 25p$ and $\theta \leq 1/4$. For $\alpha \geq 2$ and integer $p > 100$, $L_A = 3\lceil \sqrt{p}/(2\alpha) \rceil + 1 \leq p$ and $\lceil \log_2 L_A \rceil \leq p - 1$. Hence the block precision requirement is

$$p + \Gamma + \lceil \log_2(FL_A) \rceil + 11 \leq 32.14p + 11 < 34p.$$

Thus $P = 34p$ still suffices with the enlarged error enclosure. Select the outputs $b = q_\ell$ in order, subtract the exact u_ℓ , and round. The block contains every term $0 < |h| \leq m$ of the truncated sum in that lemma, so the computed $(\tilde{E}u)_\ell$ differs from $(Eu)_\ell$ by less than $(3 + 1 + 1)2^{-p} < (p/3)2^{-p}$. Its norm is less than $\|E\| + 5 \cdot 2^{-p} < 1$.

Cost and tape movement. The selected indices q_ℓ , the numerators $s\beta_\ell$, and the block offsets are generated by the counters already used for C , in $O(p)$ time per index. Before a line, copy u with $3m + 3$ periodic records appended at each end, in $O((s + m)p)$ time; then every block reads one consecutive interval. Consecutive input intervals overlap in $O(m)$ records, so all reads, packing and unpacking cost $O((t + m)p)$. There are $O(t/m + 1)$ blocks, each costing $O(mp^{1+\delta})$; in both cases $m < 2p$. This gives the time bound $O((t + p)p^{1+\delta})$. $\square$

8.3 Growing dimension and the padded array

For the tensor product, apply the one-dimensional maps to one coordinate line at a time. The chunk-swap algorithm supplies the movement of the physical axis fields, including its cost when the number of axes grows.

Lemma 41 (Tensor interface). *Suppose every pair (s_i, t_i) , $1 \leq i \leq d$, satisfies Lemma 37 with the same p, α, δ . Put $T = \prod_i t_i$, $\gamma = 2d\alpha^2$, and*

$$R = \bigotimes_i P_{s_i}, \quad Q = \bigotimes_i P_{t_i}, \quad \mathcal{A} = \bigotimes_i A_i, \quad \mathcal{B}_0 = \bigotimes_i B_{0,i}.$$

Then

$$RF_s = 2^\gamma \mathcal{B}_0 Q F_t \mathcal{A}, \quad F_q := \bigotimes_i F_{q_i}. \quad (44)$$

Also write $F_q^+ = \bigotimes_i F_{q_i}^+$. Both $\mathcal{A}$ and $\mathcal{B}_0$ are contractions. The linewise implementations of $\mathcal{A}$ and $\mathcal{B}_0$ each have error less than dp^2 . Put $n_* = \max_i n_{\alpha, \theta_i}$. If every $t_i \geq p$, the total time spent in the one-dimensional algorithms for both maps is $O(dTp^{1+\delta}(1 + n_*))$, with a constant independent of d .

If each $t_i \in \{r/2, r\}$ with $r = 2^\ell$, the array is kept in the full padded box of T entries, and each entry occupies $O(p)$ bits, then exposing and restoring all the lines costs

$$O(Tpd(1 + \ell^\tau))$$

using the equal-width chunk-swap bound $O(Vu^\tau)$ and the fixed-digit movement primitive. Any fixed polynomial in p of setup per line is negligible compared with Tp when r grows faster than every such polynomial and d is polynomially bounded in p .

Proof. Tensoring (41) proves (44). To implement $\mathcal{A}$, first embed its input in the padded box by writing zero outside $\prod_i [0, s_i)$. Expand the axes one at a time. After the axes in a set I have been expanded, the valid entries have coordinate ranges $[0, t_i)$ for $i \in I$ and $[0, s_i)$ otherwise; every other entry is zero. Expose the next axis i as the innermost address field. On a line whose other coordinates are valid, read its first s_i values and apply $\tilde{A}_i$, then write the resulting t_i values. On every other line write t_i zeros. The new valid set is the one for $I \cup \{i\}$.

For compression, start with all coordinates in $[0, t_i)$ valid. After a set J of axes has been compressed, validity means $j_h < s_h$ for $h \in J$ and $j_h < t_h$ otherwise. For the next axis i , apply $\tilde{B}_{0,i}$ to each t_i -line whose other coordinates are valid. Write its s_i outputs followed by $t_i - s_i$ zeros. A line invalid on another coordinate is already zero; consume it and write a zero line without calling $\tilde{B}_{0,i}$. The box therefore keeps its full size, and the new valid set is the one for $J \cup \{i\}$.

For a fixed axis, the exact map acts independently on its valid lines and is zero on the other lines. Taking the maximum over the line norms shows that this axis map is a contraction; taking

the maximum of the line errors gives an error less than p^2 for the whole axis. Composing the d axis maps gives error less than dp^2 . At most T/t_i lines invoke the one-dimensional algorithm for axis i , so their cost is bounded by

$$\sum_{i=1}^d \frac{T}{t_i} O(n_*(t_i + p)p^{1+\delta}) = O(dTp^{1+\delta}(1 + n_*)),$$

using $t_i \geq p$. Initial padding costs $O(Tp)$, and scanning and zeroing the remaining lines costs $O(Tp)$ per axis. Both are within the same bound. All calls reuse one fixed collection of work tapes. Loop indices and the growing shape descriptor reside on tapes; they are not extra heads.

The descriptor records which named coordinate each physical address field holds, together with its width and valid range. Before processing axis i , interchange its entire address field with the innermost field; after processing its lines, undo precisely that interchange. If it is already innermost, no interchange is needed. Thus each axis starts and ends in the original field order, and all axes together require at most $2(d-1)$ field interchanges.

These interchanges need not be adjacent. Lemma 25 permits an arbitrary intervening field $[G]$, obtained by collapsing the intervening axes, and a suffix containing the complete coefficient word. For widths that differ by one, use the explicit nonadjacent construction following that lemma. Suppressing the prefix and suffix, its two cases are

$$\begin{aligned} (e, x, G, y) &\mapsto (e, y, G, x) \mapsto (y, G, e, x), \\ (x, G, e, y) &\mapsto (e, x, G, y) \mapsto (e, y, G, x), \end{aligned}$$

where x, y have equal bit width and e is one bit. Each uses one equal-width interchange and one single-bit move. Empty equal-width pieces require no swap. The intervening axes keep their internal order, and each complete record keeps its bit order. The cost of a full field interchange is $O(Tp(1 + \ell^\tau))$, including workspace cleanup.

At a completed interchange the descriptor moves the coordinate names, widths, and validity bounds with their fields. A processed line therefore still runs through the same named coordinate in increasing numerical order. Other named coordinates retain their values; their physical order is immaterial to the validity test. After the line algorithm, the inverse interchange restores the original field order. The padded box keeps its full physical size throughout. Invalid entries remain zero at the completed axis-map boundaries specified above; temporary bit moves and the internal arithmetic of a swap need not preserve an intermediate validity mask. The completed permutations and their inverses are exact, so the contraction and error arguments do not change.

The list and collapsed descriptors are stored on tapes, not in new heads. Here $t_i \geq 2$, so $d \leq \log_2 T$. With V the physical bit volume, each collapsed length is at most V , and the list of axis names, ranges, and widths has length polynomial in $\log(2V)$. Ordinary integer arithmetic therefore prepares each call descriptor in $\text{poly}(\log(2V)) = O(V)$ time. Each primitive reuses its fixed work tapes. Thus all exposure and restoration costs $O(Tpd(1 + \ell^\tau))$.

Since $t_i \geq r/2$, each tensor map invokes at most $\sum_i T/t_i \leq 2dT/r$ lines. For any fixed setup degree c , work $O(p^c)$ per line totals $O(dTp^c/r) = o(Tp)$: after division by Tp , the ratio is $O(dp^{c-1}/r)$, which tends to zero under the stated growth assumptions. This estimate includes local descriptor copying, validity tests per line, and preparing line boundaries. $\square$

8.4 The middle transform and the final permutation

The factor QF_t in (44) can be computed from a cyclic convolution and two pointwise phase multiplications. The chirp identity below performs the frequency change in Q without applying Q as a

separate permutation of the array. The second identity shows how the retained source permutation R cancels when computing a convolution. Here $*$ denotes cyclic convolution on the product of the coordinate groups and a dot denotes coordinatewise multiplication.

Lemma 42 (Chirp and permutation cancellation). *With R, Q, s_i, t_i as in Lemma 41, assume every t_i is even. For $\mathbf{j} \in \mathbb{Z}^d$, define*

$$a_{\mathbf{j}} = \exp \left(-\pi i \sum_{i=1}^d \frac{s_i j_i^2}{t_i} \right).$$

Then a has period t_i in coordinate i , and

$$QF_{\mathbf{t}}u = \bar{a} \cdot \frac{a * (\bar{a} \cdot u)}{T}. \quad (45)$$

If $S = \prod_i s_i$, then

$$(RF_s^+)[(RF_s u) \cdot (RF_s v)] = \frac{u * v}{S^2}. \quad (46)$$

Complex conjugation of the input and output of an algorithm for RF_s gives an algorithm for RF_s^+ .

Proof. Replacing j_i by $j_i + t_i$ changes the exponent defining a by $-\pi i s_i(2j_i + t_i)$, an integral multiple of $2\pi i$. The coefficient of $u_{\mathbf{k}}$ in the right side of (45) is

$$\frac{1}{T} \bar{a}_{\mathbf{j} - \mathbf{k}} \bar{a}_{\mathbf{k}} = \frac{1}{T} \exp \left(2\pi i \sum_i \frac{s_i j_i k_i}{t_i} \right).$$

This is precisely the coefficient in $QF_{\mathbf{t}}$, because Q reads frequency $(-s_i j_i)_i$ from the negative-sign transform.

For a unit c modulo q , let $(R_c z)_j = z_{cj}$. Substituting $h = ck$ in the Fourier sum gives

$$(F_q^+ R_c z)_j = \frac{1}{q} \sum_{h=0}^{q-1} z_h e^{2\pi i h c^{-1} j / q} = (R_c^{-1} F_q^+ z)_j.$$

Each t_i is a unit modulo s_i , so tensoring this identity gives $F_s^+ R = R^{-1} F_s^+$ and hence $RF_s^+ R = F_s^+$. Also $(RF_s u) \cdot (RF_s v) = R(F_s u \cdot F_s v)$, because a coordinate permutation commutes with pointwise products. Therefore

$$RF_s^+ [R(F_s u \cdot F_s v)] = F_s^+ (F_s u \cdot F_s v).$$

The normalized convolution rule is $F_s(u * v) = S(F_s u) \cdot (F_s v)$, while $F_s^+ F_s = I/S$. The two normalizations give

$$F_s^+ (F_s u \cdot F_s v) = \frac{1}{S} F_s^+ F_s (u * v) = \frac{u * v}{S^2},$$

proving (46). Finally, conjugation changes the negative Fourier sign to the positive one and commutes with the real permutation R . It preserves both the disk grid and the norm, so conjugating the input and output also preserves an algorithm's error bound. $\square$

9 Exact integer multiplication and its cost

We now use the preceding algorithms to multiply two n -bit integers. An equal-width address swap on logical bit volume V costs $O(Vu^\tau)$ for width u . A completed parallel normalized butterfly layer costs $O(Vd^{\lambda'})$ and has coefficient error less than $\sqrt{2}2^{-p}$. The ordered Gaussian resampling maps are

contractions; their disk-grid approximations have error less than $p^2 2^{-p}$ and cost $O(n(t+p)p^{1+\delta})$ on a line of length t , where n is the Neumann count of Lemma 37. Address permutations and intermediate dyadic arithmetic are exact. We choose compatible parameters for these procedures, recover the integer coefficients from their approximations, and account for the remaining tape operations.

9.1 A fixed rational choice

Use Lemma 25 and Proposition 32. Their rank moments, not the retained uniform recurrences, certify the following exponents. Choose

$$\begin{aligned} \tau &= 1 - \frac{1816}{10^9}, \quad \sigma = 1 - \frac{18179}{10^{10}}, \quad \beta = \frac{1}{1000}, \quad \zeta = \frac{1}{10000}, \\ C_1 &= \frac{6}{5} - \frac{\beta}{5} + \zeta = \frac{11999}{10000}, \quad \epsilon = \frac{499999}{1000000}, \quad c = 1, \quad \delta = \frac{1}{10^{10}}, \\ \lambda &= \tau + \frac{1}{10^{16}}, \quad \lambda' = \tau + \frac{2}{10^{16}}, \quad \kappa = \frac{90799}{100000000000} > 2^{-23}. \end{aligned} \tag{47}$$

Since $\sigma < \tau$, the internal exponent is $\chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\} = \tau$. The leaf exponent is $\sigma + \beta(1 - \sigma) = 1 - 18160821/10^{13} < \tau$. The stopping test is $e^{1000} < d$. Section 5.9 proves the displayed guard exponent and its fixed C_0 . The reservation exponent $\max\{1 - c, 0\}$ is zero; in particular $c = 1$ is permitted. The integer mixed-width row depth has size $O(\log d)$ and does not change this exponent.

Here are all twenty-nine strict parameter slacks used in the exact certificate. Every entry in the right column is positive. In the table, χ has the value just given and $h_{\text{leaf}} = \sigma + \beta(1 - \sigma)$.

Condition slack	Exact value
τ	$\frac{124999773}{125000000}$
$1 - \tau$	$\frac{227}{125000000}$
σ	$\frac{9999981821}{10000000000}$
$1 - \sigma$	$\frac{18179}{10000000000}$
c	1
ϵ	$\frac{499999}{1000000}$
β	$\frac{1}{1000}$
$1 - \beta$	$\frac{999}{1000}$
$\lambda - \tau$	$\frac{1}{10000000000000000}$
$\lambda - \sigma$	$\frac{19000001}{10000000000000000}$
$1 - \lambda$	$\frac{18159999999}{10000000000000000}$
$\lambda - \chi$	$\frac{1}{10000000000000000}$
$\lambda' - \lambda$	$\frac{1}{10000000000000000}$
$\lambda' - h_{\text{leaf}}$	$\frac{410501}{5000000000000000}$
$1 - \lambda'$	$\frac{9079999999}{5000000000000000}$
$1 - \epsilon C_1$	$\frac{4000511999}{10000000000}$
$(1 - \epsilon)(1 - \tau)$	$\frac{113500227}{12500000000000000}$
$1 - \epsilon(1 + c)$	$\frac{1}{500000}$
$1 - \delta - \epsilon$	$\frac{5000009999}{10000000000}$
δ	$\frac{1}{10000000000}$
$1/8 - \delta$	$\frac{1249999999}{10000000000}$

Condition slack	Exact value
$1 - 2\epsilon$ (prime intervals)	$\frac{1}{500000}$
$1 - \epsilon(1 + c)$ ($K = o(\ell)$)	$\frac{1}{499999}$
ϵc	$\frac{1000000}{500001}$
$1 - \epsilon$	$\frac{1000000}{90799}$
κ	$\frac{100000000000}{19999}$
$1 - \delta - 2\epsilon$	$\frac{1}{10000000000}$
$1 - 2\epsilon$ ($\alpha^2\theta$)	$\frac{1}{4999999920000001}$
$\lambda' - \max\{1 - c, 0\}$	$\frac{1}{5000000000000000}$

9.2 Input and transform sizes

The input is the promised string $x\#y$, with both operands written most significant bit first. In one left-to-right pass, copy x and y to two fixed work tapes and count the bits of x with a binary counter. The promise gives both operands this counted length n . The counter bound in Section 2 and the two sequential copies give cost $O(n)$, including all leading zeroes. The work-copy heads finish just after the least significant bits, ready to move leftward when the digits are formed.

9.2.1 The padded box and working precision

After obtaining n , compute the size parameters below with the separate setup costs stated in this subsection. For sufficiently large n , put

$$b = \lceil \log_2 n \rceil, \quad p = 6b, \quad d = \lfloor b^\epsilon \rfloor, \quad K = \lfloor d^\epsilon \rfloor = d.$$

Take the power of two T in $[4n/b, 8n/b)$, and let $r = 2^{\lceil (\log_2 T)/d \rceil}$ and $\ell = \log_2 T$. Choose $d\ell - \log_2 T$ of the d axis lengths to be $r/2$ and the others to be r . This gives

$$t_i \in \{r/2, r\}, \quad \prod_i t_i = T, \quad t_d = r.$$

Indeed $d\ell - \log_2 T$ is an integer in $[0, d)$, so at least one axis has length r ; choose it as the last axis. These definitions give

$$Tp = \Theta(n), \quad d = \Theta(p^\epsilon), \quad K = \Theta(p^{\epsilon c}), \quad \ell = \Theta(p^{1-\epsilon}), \quad r = 2^{\Theta(p^{1-\epsilon})}. \quad (48)$$

The comparison constants can be fixed for every input length at once. For sufficiently large b , the bounds on n, T give $b/2 \leq \log_2 T \leq b$. Since $p = 6b$, $d = \lfloor b^\epsilon \rfloor$ and $d \leq b$, they imply

$$\frac{1}{12}p^\epsilon \leq d \leq p^\epsilon, \quad \frac{p}{12d} \leq \ell \leq \frac{p}{3d}.$$

Thus we fix $a_d = a_r = 1/12$, $b_d = 1$, $b_r = 1/3$, and $C_\ell = 1$ in the layer and transform interfaces. Their pointwise bounds cover every computed pair (d, r) , including all choices arising at the same p . These bounds also give that r exceeds every fixed power of p , $K/\log p \rightarrow \infty$, and $\ell/K \rightarrow \infty$. There are eventually enough axes for the $O(\log d + d^{1-\epsilon} \log p)$ reserved row and compact-control axes, which form $o(d)$ axes, and enough bits for all work blocks.

Set

$$\alpha = \left\lfloor \sqrt{b/(8d)} \right\rfloor, \quad \gamma = 2d\alpha^2, \quad \eta = \frac{1}{4d}.$$

Then

$$\gamma \leq b/4, \quad \alpha^2 < p. \quad (49)$$

For large n also $b \geq 96d$; then $\sqrt{b/(8d)} \geq 2\sqrt{3}$, so $\alpha \geq 2$ and $\alpha^2 \geq b/(16d)$.

9.2.2 Prime source lengths

The distinct prime lengths s_i must be close to the powers of two t_i : their product must occupy a fixed fraction of the padded box, while each ratio t_i/s_i must leave enough separation for resampling. We verify both requirements for the parameters just chosen.

The prime-interval result [15, Lemma 5.1], based on explicit estimates of Rosser and Schoenfeld [23], gives the following bound: for $0 < \eta < 1/4$ and $x > e^{2/\eta}$, the interval

$$(1 - 2\eta)x < q \leq (1 - \eta)x$$

contains at least $\eta x / (2 \log x)$ primes. Here $\log$ is the natural logarithm. The size bounds in (48) give $d = \Theta(p^\epsilon)$ and $\log r = \Theta(p/d)$. For large n we have $d \geq 2$, so $0 < \eta < 1/4$, and

$$\frac{\log(r/2)}{d} = \Theta(p^{1-2\epsilon}) \rightarrow \infty.$$

Thus both $x = r/2$ and $x = r$ satisfy $\log x > 8d = 2/\eta$. For either choice, the stated lower bound for the number of primes is $x/(8d \log x)$ and exceeds d : indeed, $\log x = \Theta(p^{1-\epsilon})$, whereas $\log(8d^2 \log x) = O(\log p) = o(\log x)$, so $x > 8d^2 \log x$ eventually. The two intervals are disjoint because $(1 - \eta)r/2 < (1 - 2\eta)r$ for $\eta < 1/4$, and both intervals lie above 2. We may therefore choose distinct odd primes satisfying

$$(1 - 2\eta)t_i < s_i \leq (1 - \eta)t_i.$$

Every $\gcd(s_i, t_i)$ equals one because t_i is a power of two. Put $S = \prod_i s_i$. Multiplying the lower bounds on s_i and applying strict Bernoulli inequality gives

$$\frac{S}{T} > \left(1 - \frac{1}{2d}\right)^d > \frac{1}{2} \quad (d \geq 2).$$

Together with $s_i < t_i$, this gives $T/2 < S < T$. Writing $\theta_i = t_i/s_i - 1$, the bounds on s_i give $\eta < \theta_i < 2\eta/(1 - 2\eta) = 1/(2d - 1) \leq 1/4$ once $d \geq 3$. Therefore

$$\alpha^2 \theta_i > \frac{b}{16d} \cdot \frac{1}{4d} = \frac{b}{64d^2} \geq 1$$

once $b \geq 64d^2$, which holds for large n because $d^2 \leq b^{2\epsilon}$ and $2\epsilon < 1$. The Neumann counts of Lemma 37 satisfy

$$n_{\alpha, \theta_i} \leq \frac{p+1}{4\alpha^2} + \frac{1}{\theta_i} + 1 \leq \frac{(6b+1)16d}{4b} + 4d + 1 \leq 30d.$$

We also have $t_i \leq r \leq T < 8n/b < 2^{6b} = 2^p$ and $t_i \geq r/2 > p$. Thus every hypothesis of Lemma 37 holds for all sufficiently large input lengths. The superpolynomial growth of r established above also supplies the setup condition in Lemma 41.

The machine finds these primes by scanning the two intervals deterministically and testing each candidate by trial division through its square root. Testing every candidate would visit $O(r)$ candidates and at most $O(\sqrt{r})$ trial divisors per candidate. Elementary division on $O(\log r)$ -bit integers and management of a list of at most d selected primes give the conservative bound $O(dr^{3/2} \text{poly}(p))$. Its logarithm is

$$O(\log p + \log r) = O(\log p + p/d) = o(p).$$

Since $p = 6\lceil \log_2 n \rceil = \Theta(\log n)$, the search takes $n^{o(1)} = o(n)$ time. All remaining scalar setup, including comparisons for fixed rational powers, modular inverses, prime products and shape descriptors, has cost polynomial in p . For example, $d = \lfloor b^{499999/1000000} \rfloor$ is found by binary search using exact comparisons $d^{1000000} \leq b^{499999}$; the exponent is fixed. Here $K = d$ exactly; the same comparison method computes all other fixed rational powers. All primes and length parameters are therefore generated by the machine; they are not advice.

9.2.3 The radix-digit polynomials

Write the two inputs as polynomials in 2^b , each with $q = \lceil n/b \rceil$ nonnegative digits. Their product has degree at most

$$2q - 2 < 2n/b \leq T/2 < S.$$

Thus convolution modulo $x^S - 1$ has no nonzero wraparound. Every product coefficient obeys

$$0 \leq c_j \leq q(2^b - 1)^2 < 2^{3b}, \quad (50)$$

since $q \leq n \leq 2^b$.

9.3 An explicit cyclic-convolution layout

To use the source tensor, we first place the ordinary coefficient stream in its tensor order. Agarwal and Cooley used the Chinese remainder theorem to convert one cyclic convolution into multidimensional cyclic convolution [1]. The following map uses that algebraic split, with coordinate powers that make the change of order a sequence of controlled cyclic shifts.

Let $s_1, \dots, s_d$ be the pairwise coprime source lengths, let $S = \prod_i s_i$, and put

$$P_i = \prod_{j < i} s_j, \quad \mu_i = P_i^{-1} \pmod{s_i}.$$

Here P_i denotes an integer prefix product, not a Fourier-coordinate permutation. The assignment

$$\Phi : \mathbb{C}[x]/(x^S - 1) \longrightarrow \mathbb{C}[x_1, \dots, x_d]/(x_1^{s_1} - 1, \dots, x_d^{s_d} - 1), \quad x \longmapsto \prod_i x_i^{\mu_i}$$

respects $x^S = 1$, so it defines an algebra map. The target monomial group has size S , and the image of x has order S , because each μ_i is a unit and the s_i are pairwise coprime. Its powers therefore run through all target basis monomials exactly once. Hence Φ is an isomorphism and preserves cyclic convolution.

Every ordinary coefficient index has a unique mixed-radix expression

$$k = \sum_{j=1}^d a_j P_j, \quad 0 \leq a_j < s_j.$$

The coordinate of $\Phi(x^k)$ in axis i is

$$b_i = a_i + \mu_i \sum_{j < i} a_j P_j \pmod{s_i}. \quad (51)$$

Indeed terms with $j > i$ are divisible by s_i , and $\mu_i P_i = 1$ modulo s_i . Numerical order of k is lexicographic order of $(a_d, \dots, a_1)$. Insert the padding in one nested scan of the full box $\prod_i [0, t_i)$ in this lexicographic order, with a_d the outermost counter and a_1 the innermost. At an address

satisfying $a_i < s_i$ for every i , copy the next original coefficient record; at every other address, write a zero record. The valid addresses occur in the original coefficient order, so each record is copied to its intended mixed-radix address. To delete the padding after the inverse axis reorder, use the same enumeration, copying the record at each valid address and skipping the record at each invalid address.

Store the nested binary counters consecutively on a fixed work tape. Their total width is $\sum_i \log_2 t_i = \log_2 T = O(p)$, and the stored bounds s_i also have total width $O(p)$. At each of the T addresses, updating the counters, testing validity, and handling the $O(p)$ -bit record therefore cost $O(p)$ steps, including counter-head returns. Each insertion or deletion scan costs $O(Tp)$.

Now reorder the complete padded axes to $(a_1, \dots, a_d)$. In this physical order, the coordinates with indices below i precede coordinate i . Update the targets in the order $i = d, d-1, \dots, 1$ using (51). When target i is updated, none of its lower-index controls has yet changed, so the offset is computed from the original digits a_j required by the formula.

To specify the action on padded addresses, let

$$\mathcal{P} = \prod_i [0, t_i), \quad \mathcal{V} = \prod_i [0, s_i)$$

be sets of integer addresses. On a fixed control prefix, rotate the valid target interval $[0, s_i)$ by the offset in (51) and fix $[s_i, t_i)$. If any lower-index control is invalid, use the identity. This defines a bijection of $\mathcal{P}$: for fixed controls and spectators it is a cyclic permutation of the valid target interval and the identity elsewhere. Each step preserves $\mathcal{V}$ and its complement. Consequently zero padding remains zero after every step. An inverse is explicit: process $i = 1, \dots, d$, subtracting the same expression formed from the already recovered lower coordinates; then undo the axis reorder and perform the deletion scan just described.

Each triangular update is a prefix-controlled cyclic shift. The controls precede its target in the current physical order. For each prefix, prepare the offset, split the valid target interval into its two consecutive pieces, and merge those pieces in the opposite order. Every target position carries its full spectator suffix as a block; the invalid interval is copied unchanged. There are at most T/t_i control prefixes for target i . For any fixed setup degree C , all offsets therefore cost $O(dTp^C/r) = o(Tp)$ to prepare, since $t_i \geq r/2$ and r exceeds every fixed power of p .

The axis widths are ℓ or $\ell-1$. Reverse the axis order by interchanging physical positions j and $d+1-j$ for $j = 1, \dots, \lfloor d/2 \rfloor$. Every pair is disjoint, so this uses $\lfloor d/2 \rfloor$ full-field interchanges, not a sequence of adjacent moves. Each is covered by Lemma 25 and its subsequent one-bit-width extension, with all intervening fields collapsed into its spectator gap. Its cost is $O(Tp(1 + \ell^\tau))$. Keep each axis name and width attached to that field in the descriptor. The final order is exactly $(a_1, \dots, a_d)$, as required by the triangular updates above. Their target order and their requirement that controls precede targets are unchanged. The inverse uses the reversed interchange schedule after recovering the digits.

The completed reversal and its inverse preserve complete records and their internal bit order. Padding is inserted before the forward reversal and removed only after the inverse reversal, so the insertion/deletion enumeration is unchanged. Since $d \leq \log_2 T$, the named-axis list and all collapsed lengths have polynomial length in $\log(2Tp)$. Each call descriptor can thus be prepared in $\text{poly}(\log(2Tp)) = O(Tp)$ time. The d controlled rotations and the padding scans cost $O(dTp)$ as above. Therefore Φ , its inverse, and their padding operations cost

$$O(Tpd(1 + \ell^\tau)).$$

This construction describes payload movement from address a to its new address. It is distinct

from the operator convention $(R_c z)_j = z_{cj}$ used for Fourier dilations, which moves a payload from address j to $c^{-1}j$.

9.4 Complex convolution by a coefficient twist

The power-of-two complex convolution used in the chirp identity is obtained from the synthetic convolution by a coefficient twist. Put $\mathcal{R} = \mathcal{R}_r = \mathbb{C}[y]/(y^r + 1)$ and $\zeta = e^{\pi i/r}$. The map

$$\Theta : \mathbb{C}[x_d]/(x_d^r - 1) \longrightarrow \mathcal{R}, \quad x_d \longmapsto \zeta y$$

is well defined, since $(\zeta y)^r = (-1)(-1) = 1$. On the bases $(x_d^k)_{0 \leq k < r}$ and $(y^k)_{0 \leq k < r}$ it is diagonal, multiplying coefficient k by the nonzero number ζ^k . It is therefore an isomorphism, with inverse multiplying that coefficient by ζ^{-k} . Both exact maps preserve the coefficient sup norm. For an explicit check on the wrap, write $k + l = j + hr$, where $h \in \{0, 1\}$. A product term after twisting and negacyclic reduction has factor

$$\zeta^{k+l}(-1)^h = \zeta^j(\zeta^r)^h(-1)^h = \zeta^j.$$

Untwisting therefore recovers the cyclic coefficient with the correct sign. We next specify how this linewise map meets the ring-convolution interface. Put $M = \prod_{i < d} t_i = T/r$. For a complex array f on $\prod_{i=1}^d \mathbb{Z}/t_i \mathbb{Z}$, with $t_d = r$, define

$$(\iota f)_j = \sum_{k=0}^{r-1} f_{j,k} y^k, \quad j \in \prod_{i < d} \mathbb{Z}/t_i \mathbb{Z}.$$

This identifies the scalar tensor with an array of M records in $\mathcal{R}$. As maps of values, ι and ι^{-1} preserve norms and grids. Write $(D_{\pm} f)_{j,k} = \zeta^{\pm k} f_{j,k}$. The linewise extension of Θ is ιD_+ , with inverse $D_- \iota^{-1}$. Consequently, with $\mathcal{M}_r$ from Section 7, normalized complex cyclic convolution is

$$\mathcal{M}_{\mathbb{C}}(f, g) := D_- \iota^{-1} \mathcal{M}_r(\iota D_+ f, \iota D_+ g) = \frac{f * g}{T}.$$

Here $\mathcal{M}_r$ convolves the first $d - 1$ axes over $\mathcal{R}$, whereas $*$ on the right convolves all d scalar axes. The division by T is already part of $\mathcal{M}_r$. The maps $D_{\pm}$ and $\iota^{\pm 1}$ are isometries, so $\mathcal{M}_{\mathbb{C}}$ is also a bilinear contraction.

9.4.1 Computed complex convolution

For the tape representation, use the scalar format of Section 8, with $w_c = p + 2$ bits per component. For $p \geq 2$, this satisfies $p + 2 \leq w_c \leq 2p$, so fix $C_w = 2$ and instantiate the synthetic component width as $w = w_c$. Both representations then concatenate exactly the same real and imaginary words in the same coefficient order: the last coordinate is already the contiguous polynomial suffix. Hence ι and ι^{-1} change only the shape descriptor and require no payload movement when these component formats agree.

For the synthetic transform calls, the machine writes the header

$$\mathcal{T} = \Gamma(p)\Gamma(d)\Gamma(r)\Gamma(K)\Gamma(w_c) \prod_{i=1}^{d-1} \Gamma(t_i).$$

For each normalized polynomial product it supplies $\Gamma(p)\Gamma(r)\Gamma(w_c)$. These are precisely the headers of Lemmas 34 and 35, formed from the computed parameters. Their lengths are $O(p)$; forming the

transform header costs $\text{poly}(p) = o(Tp)$, and copying a product header for each of the $M = T/r$ polynomial records costs $O(Mp) = O(Tp)$.

For $0 \leq k < r$, obtain disk-grid approximations $\tilde{\theta}_k^+$ and $\tilde{\theta}_k^-$ to ζ^k and ζ^{-k} , each with error less than $2 \cdot 2^{-p}$. The exponential routine uses denominator $2r$ and costs $O(p^{1+\delta})$ per factor. Write $\tilde{\mathcal{M}}_r$ for the procedure in Proposition 36. For disk-grid inputs f, g , one of norm at most .7, the computed complex convolution is the following sequence, with the twist arrays repeated on every last-coordinate line:

$$\begin{aligned} f' &= Q_p(\tilde{\theta}^+ \cdot f), \\ g' &= Q_p(\tilde{\theta}^+ \cdot g), \\ h' &= \tilde{\mathcal{M}}_r(\iota f', \iota g'), \\ \tilde{\mathcal{M}}_{\mathbb{C}}(f, g) &= Q_p(\tilde{\theta}^- \cdot \iota^{-1} h'). \end{aligned}$$

Each phase product is computed exactly as a dyadic product before Q_p truncates its two components. The factors and disk operands have numerators of magnitude at most 2^p ; four signed integer products and their sums therefore use $O(p)$ bits. Truncation shifts the absolute component numerators right by p places and restores their signs, then writes the resulting disk-grid value in the w_c -bit component format. It costs $O(p^{1+\delta})$ per coefficient. Generating every factor anew therefore still gives total twist cost $O(Tp^{1+\delta})$. The precision calculation below proves that these calls return disk-grid arrays.

At the h' step, Proposition 36 returns its computed grid numerators after the exact scale M , in the known width $w_M = w_c + \log_2 M$. The disk margin proved below gives $\|h'\| < .701$ in every use here, so each component numerator has magnitude at most 2^p . Before applying ι^{-1} , scan each w_M -bit word and remove only leading sign-extension bits, writing the same integer in w_c bits. The retained sign bit agrees with every removed bit by that magnitude bound. This exact normalization costs $O(Tw_M) = O(Tp)$ and supplies the matching scalar format for the output untwist; it introduces no numerical truncation.

9.5 Chirp and source-transform procedures

We next use the complex convolution to evaluate the middle factor QF_t in (44), and then compose it with the Gaussian maps. For the chirp array a in (45), the phase numerator at tensor address j is computed modulo $2r$ as

$$-\sum_i s_i j_i^2 (r/t_i) \pmod{2r}.$$

All integers have $O(p)$ bits and $r/t_i \in \{1, 2\}$. Scan the stored axis descriptors and compute the d squares and products with the previously established integer multiplier. Additions, reductions modulo a power of two, and restarting local counters cost $O(p)$ per term; multiplication costs $O(p \log p) = O(p^{1+\delta})$. Exponential evaluation then has the same $O(p^{1+\delta})$ bound. Thus chirp construction costs $O(Tdp^{1+\delta})$ in total. In particular, this cost includes the scan of the axis descriptors at every coefficient.

Let $\tilde{a}$ be the resulting disk-grid array, so $\|\tilde{a} - a\|_\infty < 2 \cdot 2^{-p}$. For a disk-grid target tensor x of norm at most .6, the computed chirp transform forms the arrays

$$\begin{aligned} x_0 &= Q_p(\tilde{a} \cdot x), \\ y_0 &= \tilde{\mathcal{M}}_{\mathbb{C}}(\tilde{a}, x_0), \\ \tilde{\mathcal{C}}(x) &= Q_p(\tilde{a} \cdot y_0). \end{aligned}$$

Thus $\tilde{a}$ is the convolution kernel and x_0 is the data operand after the first endpoint phase. Replacing the approximations by exact phases and exact convolution gives $\bar{a} \cdot \mathcal{M}_{\mathbb{C}}(a, \bar{a} \cdot x) = QF_t x$ by (45). Below we show that $\|x\|_{\infty} \leq .6$ implies $\|x_0\|_{\infty} \leq .6$, which supplies the smaller operand required by the computed convolution.

9.5.1 The source transform

For a source disk-grid tensor of norm at most .5, write $\widetilde{\mathcal{F}}$ for the following procedure: apply the linewise approximation to $\mathcal{A}$, apply the computed chirp transform $\widetilde{\mathcal{C}}$, apply the linewise approximation to $\mathcal{B}_0$, and multiply the grid numerators exactly by 2^{γ} . Its exact counterpart is

$$\mathcal{F} = 2^{\gamma} \mathcal{B}_0 QF_t \mathcal{A} = RF_s.$$

Here an exact counterpart replaces every numerical subroutine by the exact map it approximates and omits truncations. The source arrays are stored in the padded power-of-two box, with zeros outside the valid source coordinates. The compressed scalar output uses w_c -bit components. Appending γ zero bits to each word gives the exact scaled numerator in the known signed width $w_c + \gamma$. For the stated input norm, the disk estimate below proves that every scaled coordinate still belongs to $\mathbb{D}_p$. Its component numerators therefore have magnitude at most 2^p , so a scan removes only leading sign-extension bits and returns the same integers in w_c bits. This scan costs $O(T(w_c + \gamma)) = O(Tp)$ and leaves the padded zero records zero. Define the opposite-sign procedure by

$$\widetilde{\mathcal{F}}^+(z) = \overline{\widetilde{\mathcal{F}}(\bar{z})};$$

its exact counterpart is RF_s^+ by Lemma 42.

9.6 The multiplication algorithm

The source-transform procedures now supply the three transforms needed for the radix-digit convolution. Read each stored operand backward from its least significant end in groups of b bits. This produces the radix- 2^b digits $a_0, \dots, a_{q-1}$ in increasing coefficient order. Buffer each group on a fixed work tape and read it back most significant bit first, reversing its least-significant-first input order; pad the final short group with leading zeroes. For digit a_j , write the exact grid numerator $a_j 2^{p-b-2}$ in a signed w_c -bit real word, followed by a zero w_c -bit imaginary word. This exponent is nonnegative because $p = 6b$, and the represented value is exactly $a_j 2^{-b-2}$. Append $S - q$ zero coefficient records. Buffering and ordering the groups costs $O(n)$, including the one partial group's padding, and writing the S records costs $O(Sp)$. Together with the initial count and copies, this input conversion costs $O(n + Sp) = O(Tp)$; all n input bits, including leading zeroes, are retained before the final group is padded.

With these streams prepared, perform the following operations.

1. Apply the coefficient-address map Φ , with its padded layout, to obtain source arrays u, v . Their coefficients are exact p -bit grid values of norm less than $1/4$.
2. Compute $\widetilde{\mathcal{F}}(u)$ and $\widetilde{\mathcal{F}}(v)$.
3. Multiply corresponding complex coefficients exactly and truncate each real and imaginary part toward zero to p fractional bits, writing the result in the common w_c -bit component format. Denote the resulting source array by z .

4. Compute $\widetilde{\mathcal{F}}^+(z)$ and multiply its grid numerators exactly by S , retaining the denominator 2^p and writing each component in signed width $w_c + b$. Denote the represented dyadic array by $\widetilde{w}$.
5. Multiply the grid numerators of $\widetilde{w}$ exactly by $2^{2b+4}S$, again retaining the denominator 2^p , and write each component in signed width $w_c + 4b + 4$. Round the represented real values to the nearest integers. At each valid address, write the rounded coefficient in one signed $(3b+1)$ -bit word, most significant bit first; at every padded address, write a zero word of that width. Supply this local payload width to Φ^{-1} and apply it, including removal of the address padding, to restore the ordinary coefficient order.
6. Propagate radix- 2^b carries in increasing coefficient order. Write the resulting bits from least to most significant on a work tape, then reverse that stream and output exactly $2n$ bits in most-significant-first order.

The two occurrences of S in this algorithm undo different normalizations. By (46), the exact counterpart after the three transforms is $(u * v)/S^2$, so after the first scale it is $w = (u * v)/S$. The second scale gives $2^{2b+4}(u * v)$, whose inverse image under Φ is the original integer coefficient convolution. The next subsection verifies the disk bounds needed by every call and proves that the final error is less than $1/2$; it also justifies the rounding, carries, and output length.

9.7 Precision and exact recovery

Proposition 32 uses $C_1 = 11999/10000$, so $\epsilon C_1 < 1$ makes its coefficient width $O(p)$. Lemma 34 therefore applies to every synthetic transform here, with error less than $\sqrt{2}\ell 2^{-p} < L2^{-p}$, where $L = \log_2 T$, for all sufficiently large n .

9.7.1 Scales and disk margins

Put $h_p = 2^{-p}$, and use the coefficient sup norm throughout this subsection. An error of a will mean absolute error less than ah_p . A linear contraction propagates an input error without increasing it. For a bilinear contraction $\mathcal{M}$, replacing its operands gives

$$\|\mathcal{M}(f', g') - \mathcal{M}(f, g)\| \leq \|f' - f\| \|g'\| + \|f\| \|g' - g\|.$$

When all four operands are in the disk, their two errors therefore add.

We also need the norm as well as the error of a computed phase product. Let e be a unit phase, let $\tilde{e}$ be a disk-grid approximation with $|\tilde{e} - e| < 2h_p$, and let x be a disk operand. The truncation bound in (34) gives

$$\|Q_p(\tilde{e}x) - ex\| < (2\|x\| + \sqrt{2})h_p < 4h_p, \quad \|Q_p(\tilde{e}x)\| \leq \|x\|.$$

The second inequality follows from $|\tilde{e}| \leq 1$ and the fact that Q_p does not increase the norm. It preserves every smaller operand norm through a twist or an endpoint phase multiplication. The same bounds hold coordinatewise for arrays of phases.

For disk-grid ring arrays f, g , put $M = T/r$. Two normalized synthetic forwards, pointwise polynomial multiplication divided by r , and the normalized opposite transform give $(f * g)/(rM^2)$. The exact multiplication by M in Proposition 36 then gives $\mathcal{M}_r(f, g) = (f * g)/T$. Before that multiplication, the three transform errors and the polynomial-product truncation total less than $3L + 2$. Thus the error of the returned normalized convolution is bounded by

$$E_{\text{ring}} := M(3L + 2) < 5TL.$$

If one operand has norm at most .7, bilinear contraction gives $\|\mathcal{M}_r(f, g)\| \leq .7$. Its computed grid output consequently has norm less than $.7 + 5TLh_p < .701$ for large n . This proves the disk margin after the scale M , without clipping the output.

Consider now $\widetilde{\mathcal{M}}_{\mathbb{C}}(f, g)$ for disk grid operands, one of norm at most .7. Its two computed input twists f', g' are disk grid arrays, and the smaller one still has norm at most .7 by the phase inequality. The identification ι preserves these bounds. The preceding ring margin therefore gives $\|h'\| < .701$. Neither ι^{-1} nor the computed output untwist increases this norm. In particular, $\widetilde{\mathcal{M}}_{\mathbb{C}}(f, g)$ is a disk grid array.

To compare it with $\mathcal{M}_{\mathbb{C}}(f, g)$, insert the exact normalized $\mathcal{M}_r$ at $\iota f'$ and $\iota g'$. Its bilinear contraction bound propagates the two input twist errors, each less than 4. Its approximation contributes E_{ring} , and the output untwist contributes less than 4. All these comparisons are at the normalized $\mathcal{M}_r$ interface, after its factor M . They give

$$\|\widetilde{\mathcal{M}}_{\mathbb{C}}(f, g) - \mathcal{M}_{\mathbb{C}}(f, g)\| < (E_{\text{ring}} + 4 + 4 + 4)h_p < 6TLh_p.$$

For a disk-grid chirp input x with $\|x\| \leq .6$, the previously defined data operand $x_0 = Q_p(\bar{a} \cdot x)$ has norm at most .6. The kernel $\bar{a}$ is a disk grid array. Thus this pair satisfies the smaller-operand condition just proved: $\|y_0\| < .701$, and the last endpoint multiplication gives $\|\tilde{\mathcal{C}}(x)\| < .701$. In particular, the chirp output is a legal disk-grid input to the compression maps.

For its error, the exact operands a and $\bar{a} \cdot x$ are in the disk because every entry of a has modulus one. The kernel approximation contributes less than 2 and the first endpoint multiplication contributes less than 4 when inserted into the exact bilinear contraction $\mathcal{M}_{\mathbb{C}}$. The computed convolution contributes less than $6TL$, and the last endpoint multiplication less than 4. Equation (45) identifies the exact result, so

$$\|\tilde{\mathcal{C}}(x) - QF_t x\| < (6TL + 2 + 4 + 4)h_p < 8TLh_p.$$

Finally let the input to a source transform have norm at most .5. The computed expansion has error less than dp^2 and norm less than $.5 + dp^2h_p < .51$. It is therefore a permitted chirp input. The chirp output is in the disk by the preceding argument, so each compression call is defined and returns another disk-grid array. Insert the exact contractions $\mathcal{A}$, QF_t , and $\mathcal{B}_0$ at the computed inputs. The error before the source scale is less than $2dp^2 + 8TL$. Multiplication by 2^γ therefore gives error less than $2^\gamma(2dp^2 + 8TL)h_p$, which is bounded by $E_s h_p$ with

$$E_s = 9 \cdot 2^\gamma TL, \tag{52}$$

because $2dp^2 < TL$ eventually. The exact source transform RF_s is a contraction, so the scaled approximation has norm at most $.5 + E_s h_p < .501$. Its exact numerator scaling therefore also leaves a disk-grid output without clipping. Conjugation preserves the grid and norm, so the opposite procedure has the same error and disk guarantee. All these margins may be enforced by one fixed cutoff. For example, $b \geq 2^{40}$, together with the other eventual construction conditions, makes dp^2h_p , $8TLh_p$, $E_s h_p$ and $28 \cdot 2^\gamma T^2 L h_p$ smaller than 10^{-3} .

9.7.2 The final integer coefficients

The scaled radix digits in u, v are exact p -bit grid values of norm strictly less than $1/4$. Put

$$\begin{aligned} U &= RF_s u, & V_1 &= RF_s v, \\ \tilde{U} &= \tilde{\mathcal{F}}(u), & \tilde{V}_1 &= \tilde{\mathcal{F}}(v). \end{aligned}$$

The exact arrays have norm less than $1/4$, and the computed arrays have norm less than $1/4 + E_s h_p < .26$. Hence $z = Q_p(\tilde{U} \cdot \tilde{V}_1)$ has norm at most $.26^2 < .5$. To estimate its error, write

$$\tilde{U} \cdot \tilde{V}_1 - U \cdot V_1 = (\tilde{U} - U) \cdot \tilde{V}_1 + U \cdot (\tilde{V}_1 - V_1).$$

The factors $\tilde{V}_1$ and U are in the disk. The two terms have total norm less than $2E_s h_p$, and truncation adds less than $2h_p$. Thus $\|z - U \cdot V_1\| < (2E_s + 2)h_p$.

Let $G := RF_s^+$ be the exact opposite contraction. Lemma 42, applied to $U = RF_s u$ and $V_1 = RF_s v$, gives

$$G(U \cdot V_1) = (u * v)/S^2.$$

The computed opposite transform may be applied to z because $\|z\| < .5$. Insert G at that computed input:

$$\begin{aligned} \|\widetilde{\mathcal{F}}^+(z) - G(U \cdot V_1)\| &\leq \|\widetilde{\mathcal{F}}^+(z) - Gz\| \\ &\quad + \|G(z - U \cdot V_1)\| \\ &< (3E_s + 2)h_p. \end{aligned}$$

The first exact numerator multiplication by S gives $\tilde{w}$, approximating $w = (u * v)/S$ with error less than $E_w h_p$, where

$$E_w := S(3E_s + 2) < 28 \cdot 2^\gamma T^2 L.$$

Each coefficient of w is an average of S products of coefficients of u and v , so $\|w\| < 1/16$. The earlier cutoff therefore gives

$$\|\tilde{w}\| < 1/16 + E_w h_p < 1/16 + 10^{-3} < 1.$$

This proves the disk bound for the computed first scaled result.

In tensor order, the unscaled integer coefficient array is $2^{2b+4}Sw$. This second occurrence of S reverses the normalization still present in w , and 2^{2b+4} reverses the two input scales. Both multiplications act exactly on grid numerators. Since $S \leq T < 2^b$ and $L < b$ eventually, the final absolute error is less than

$$2^{2b+4}S \cdot 28 \cdot 2^\gamma T^2 L \cdot 2^{-6b} < 448b \cdot 2^{-b+\gamma} \leq 448b \cdot 2^{-3b/4} < \frac{1}{2}. \quad (53)$$

The true coefficients are real integers. If H_j is the real numerator after the second scaling, its represented value is $H_j/2^p$. The strict error below $1/2$ makes the nearest integer to that value the unique correct coefficient.

These exact operations use known fixed record widths. If N is a signed w_0 -bit integer, then $-2^{w_0-1} \leq N < 2^{w_0-1}$. Since $0 < S < 2^b$, its product satisfies $-2^{w_0+b-1} < SN < 2^{w_0+b-1}$ and therefore fits in signed width $w_0 + b$. The first product by S starts at width w_c and is written in width $w_c + b$. The next product by S is written in width $w_c + 2b$; appending $2b + 4$ least significant zero bits then gives the exact second scale in width $w_c + 4b + 4$. Each complex record still places its most-significant-bit-first real word before its imaginary word. The known widths determine every boundary in the next scan and are $O(p)$ because $p = 6b$.

The established integer multiplier performs the two products by S in $O(p \log p)$ time per entry. To serialize each exact signed product at its stated width, zero-pad its magnitude and complement and increment for a negative sign. The proved bounds ensure that no significant bit is discarded. These conversions and the power-of-two shift use linear scans. To round $H_j/2^p$, recover the real magnitude, inspect its low p bits, and apply the rounding carry with the sign restored. The strict error bound makes the result the correct nonnegative coefficient, and (50) puts it in $[0, 2^{3b})$. It

therefore fits in one signed $(3b + 1)$ -bit two's-complement word, written most significant bit first. Scan the padded box in the physical axis order recorded by its shape descriptor, using the same $O(p)$ -bit nested counters and validity tests as above. Consume the imaginary word and write this real coefficient word at each valid address, with an all-zero word at each padded address. This serialization costs $O(Tp)$ in total. It keeps the tensor address order and supplies payload width $3b + 1$ to the local descriptors for Φ^{-1} , so the address routine receives a stream of known fixed-length records.

Lemma 35 proves exact signed packing, centered extraction and negacyclic reduction. Its only numerical error is the stated truncation after division by r , already included above. Undo the coefficient-address isomorphism Φ before propagating radix carries.

To propagate carries, let $Q_0 = 2^b$. The recovered ordinary coefficients are the c_j in (50); they are zero for $2q - 2 < j < S$. With $q = \lceil n/b \rceil$ as above, use the recurrence for $0 \leq j < S$:

$$k_0 = 0, \quad d_j = (c_j + k_j) \bmod Q_0, \quad k_{j+1} = \left\lfloor \frac{c_j + k_j}{Q_0} \right\rfloor.$$

The carries satisfy $0 \leq k_j \leq q(Q_0 - 1) < Q_0^2$. If this holds at j , then

$$c_j + k_j \leq q(Q_0 - 1)^2 + q(Q_0 - 1) = qQ_0(Q_0 - 1),$$

which proves the next carry bound. The defining relation $c_j + k_j = d_j + Q_0 k_{j+1}$ also gives the exact invariant

$$\sum_{h < j} c_h Q_0^h = \sum_{h < j} d_h Q_0^h + k_j Q_0^j.$$

It holds for $j = 0$, and the defining relation proves its next instance. The unprocessed terms are multiples of Q_0^j , so the identity fixes the first j radix digits of the full product. The value k_j is the carry from the processed coefficients into the remaining positions.

After Φ^{-1} the coefficients occur in increasing ordinary index, so this carry scan proceeds from least to most significant. Carry storage and work per coefficient are $O(b)$ bits and steps. Write each digit's bits from low to high on a work tape, and drain the final $O(b)$ -bit carry in the same order. Copy the completed stream backward once to obtain most-significant-first order. The stream has length $O(Sb) = O(Tp)$, including chunk padding and the final carry, so this reversal is a charged linear tape pass. The exact product is less than 2^{2n} ; all bits above position $2n - 1$ are therefore zero. Remove only those zero padding bits and retain leading zeros as needed to produce exactly $2n$ bits. Zero inputs obey the same invariant and output rule.

9.8 The complete time bound

For the cost calculation put $V = Tp = \Theta(n)$. The fixed-width coefficient formats use $\Theta(p)$ bits per coefficient, so a full padded array has $\Theta(V)$ stored bits. Table 2 lists costs after division by V ; a fixed number of repeated transforms and convolutions is included in its constants.

The seven dominant powers in Table 2 have the following margins below one:

$$\begin{aligned} g_1 &= 1 - \epsilon(1 + c), & g_2 &= \epsilon c(1 - \tau), & g_3 &= \epsilon(1 - \lambda'), \\ g_4 &= (1 - \tau)(1 - \epsilon), & g_5 &= 1 - \delta - 2\epsilon, \\ g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned} \tag{54}$$

Operation	Normalized cost	Power of p
Prefix-slot moves and individual butterfly rounds	dK	$\epsilon(1 + c)$
Chunk exchanges for transform layout	$pK^{\tau-1}$	$1 - \epsilon c(1 - \tau)$
Simultaneous butterfly rounds	$\ell d^{\lambda'}$	$1 - \epsilon(1 - \lambda')$
CRT and axis layouts	$d(1 + \ell^\tau)$	$\tau + \epsilon(1 - \tau)$
Gaussian line maps	$d^2 p^\delta$	$2\epsilon + \delta$
Chirps, twists and scalar products	dp^δ	$\epsilon + \delta$
Packed polynomial products	$\log(rp)$	$1 - \epsilon$
Final scaling and rounding	p^δ	absorbed above
Input, padding, carries and output	1	absorbed above

Table 2: Costs divided by Tp ; each coefficient representation has $O(p)$ bits.

The seven margins have the following exact values:

$$\begin{array}{l|l}
g_1 & \frac{1}{500000} \\
g_2 & \frac{113499773}{125000000000000} \\
g_3 & \frac{4539990919500001}{500000000000000000000000} \\
g_4 & \frac{113500227}{1250000000000000} \\
g_5 & \frac{19999}{10000000000} \\
g_6 & \frac{5000009999}{10000000000} \\
g_7 & \frac{499999}{1000000}
\end{array}$$

Exact substitution in the unchanged seven-term assembly accounting gives

$$G_* := \min_i g_i = g_3 = \epsilon(1 - \lambda') = \frac{4539990919500001}{500000000000000000000000} > \frac{90799}{100000000000} = \kappa.$$

The final strict absorption gap is

$$G_* - \kappa = \frac{40919500001}{500000000000000000000000} > 0.$$

The new reservation, compact movement and repair costs are all included in the completed-layer term g_3 . Put $\rho = G_* - \kappa > 0$; this fixed strict gap absorbs all remaining fixed powers of $\log p$. The guard is $O(d^{11999/10000}) = o(p)$ because

$$\epsilon C_1 = \frac{5999488001}{10000000000} < 1, \quad 1 - \epsilon C_1 = \frac{4000511999}{10000000000}.$$

Furthermore, $\alpha = \Theta(p^{500001/2000000})$ and $\gamma \leq b/4$. Also $K = \Theta(p^{499999/1000000}) = o(\ell)$, $K/\log p \rightarrow \infty$, and $\ell = \Theta(p^{500001/1000000})$. The prime-interval ratio grows as $p^{1/500000}$, and so does b/d^2 . All construction and precision conditions hold beyond a common fixed cutoff.

Here d is absorbed by $d\ell^\tau$, and $\log(rp) = O(p^{1-\epsilon} + \log p) = O(p^{1-\epsilon})$. The first three rows are the terms in (38). The prefix-slot moves and individual butterfly rounds in Lemma 34 use $O(dK)$ linear scans. Its positional layout uses $O(p/K)$ exchanges of width- K chunks, because $d\ell = O(p)$. Each exchange costs $O(VK^\tau)$ by Lemma 25, so the normalized total is

$$O(p/K) K^\tau = O(pK^{\tau-1}).$$

For the at most ℓ simultaneous rounds, Proposition 32 supplies $O(Vd^{\lambda'})$ per round. Its bound already includes the binary basis changes, deterministic repair, row padding and removal, all mixed-width children, and fixed-tape cleanup.

Axis exposure and restoration require $O(d)$ nonadjacent full-field interchanges by Lemma 41; the CRT reversal also uses $O(d)$ such interchanges. Each costs $O(V(1 + \ell^\tau))$, using the arbitrary spectator gap and the one-bit-width extension of Lemma 25. Thus this row has exponent $\epsilon + \tau(1 - \epsilon)$ rather than $2\epsilon + \tau(1 - \epsilon)$. The d CRT rotations copy valid and invalid intervals in $O(dV)$ time. Their offset setup was bounded above by $O(dTp^C/r) = o(Tp)$, using at most T/t_i control prefixes for target i .

For Gaussian resampling, axis i has at most T/t_i processed lines, each of length $t_i > p$. Since every Neumann count is at most $30d$, the line-machine costs sum to

$$\sum_{i=1}^d \frac{T}{t_i} O(30dt_i p^{1+\delta}) = O(d^2 T p^{1+\delta}).$$

Dividing by $V = Tp$ gives the fifth row.

Chirp numerators have $O(p)$ bits and can be computed modulo $2r$ as sums of d terms $s_i j_i^2(r/t_i)$. Computing their squares and products with the previously established $O(N \log(2N))$ multiplier on N -bit inputs, then generating the exponentials, costs $O(dp^{1+\delta})$ per entry. The smaller twist and scalar pointwise-product costs fit the same row: they use a fixed number of exact products on $O(p)$ -bit component numerators, followed by linear truncation scans. The two final products by S also cost $O(p \log p)$ per entry, so their normalized cost is $O(\log p) = O(p^\delta)$ as listed separately. There are T/r polynomial records in each pointwise ring product. Four integer products on $O(rp)$ bits per record therefore cost $O(Tp \log(rp))$ in total. Signed packing, centered extraction, negacyclic wrapping, and copying are linear stream passes. These calls use the previously established multiplier on records of length $O(rp) = o(n)$; they do not call the improved algorithm being constructed. The exact source scaling by 2^γ is a shift of $O(p)$ -bit numerators by $\gamma = O(p)$ places, so it is a linear scan per entry and fits the final-scaling row. The scalar-format adapters and the exact removal of sign-extension bits after the ring and source scales also use linear scans of their $O(p)$ -bit components, within the listed scan costs.

Setup and exceptional streams across all calls. We next separate setup at recursion nodes from work repeated for every polynomial record or every Gaussian line. Here a visit means a node of the layer or address-layout recursions, or one of their round and group invocations. The internal recursion of the established integer multiplier and the iterations of the quoted Gaussian algorithms are already included in their separately charged running times. The layer and address-layout recursions have constant branching and depth $O(\log p)$, and the number of their rounds and root invocations is polynomial in p . Choose a fixed exponent A so that their total number of visits is at most p^A , and a fixed C bounding descriptor setup performed once per visit by p^C steps. This setup then costs at most $p^{A+C} = o(Tp)$.

At a layer or synthetic-transform visit that processes polynomial records, there are at most $O(T/r)$ such records, including the bounded row padding. Every relevant child keeps the whole length- r polynomial suffix. Consequently $O(p^C)$ descriptor or address work per record, over these visits, costs at most $O(Tp^{A+C}/r) = o(Tp)$. Setup before each Gaussian line call has a different count: there are at most $2dT/r$ lines per tensor map and only a fixed number of such maps, so $O(p^C)$ setup per line costs $O(dTp^C/r) = o(Tp)$. These estimates all use that r exceeds every fixed power of p .

The general address-swap subroutine also covers one-bit payloads, where no polynomial suffix is available. For root volume V_0 and current child volume V_j , the descriptor estimate in the proof of Section 4.8 gives $(\log(2V_0))^{O(1)} = O(V_j)$. It uses the invariant that both original chunk ranges remain present in every child, and therefore includes polynomial descriptor work even for

a one-bit payload. Elementary counter updates use local length copies, so they do not rescan a growing descriptor for each data bit. The chirp computation does scan the axis descriptors at every coefficient; its $O(Tdp^{1+\delta})$ cost was included in the chirp row.

For a compact-control change of basis of current volume V_{call} , the exceptional fraction is at most $5/(128p^3)$ beyond the common cutoff. Sorting its keyed exceptional records costs

$$O(V_{\text{call}}\delta p(1 + 1/r)) = O(V_{\text{call}}p^{-2}).$$

Extraction and reinsertion cost $O(V_{\text{call}})$, and polynomial predicate and key work per record is absorbed by the whole polynomial record suffix. These are already charged at *each node* in the compact-control recurrence. Their sum is bounded by that volume-weighted recurrence and therefore by the completed-layer cost in the table. No claim that their global total is $o(V)$ is needed, and the old global estimate using an exponentially small 2^{-K} bad fraction is not used for this new gadget.

Finally, all role, arithmetic, and stack tapes form fixed finite families. A child parks and restores only its parent's streams, with the stack head at the active top; copying and cleanup are charged to that logical volume. No operation repeatedly scans parked ancestors. Growing axis lists and call states are stored data, not extra tapes or alphabet symbols. Global setup and prime search cost $o(V)$, as proved earlier.

Each of the seven displayed powers is at most $1 - G_*$ by (54). Any fixed power of $\log p$ is at most p^ρ eventually. The complete large-input bound is consequently

$$O(Tp p^{1-\kappa}) = O(n(\log n)^{1-\kappa}).$$

Fix one cutoff beyond all the size, precision, margin and layout conditions, and use schoolbook multiplication below it. The finitely many smaller lengths are included by enlarging the implicit constant. With $\lg n = \max\{\lceil \log_2 n \rceil, 1\}$, the resulting single fixed-alphabet, fixed-tape deterministic machine has worst-case time $O(n(\lg n)^{1-\kappa})$ for every input length and produces precisely the required $2n$ -bit product.

10 Exact division and integer square root

Let n be the supplied input width, let a be the nonnegative dividend or radicand, and let $b > 0$ be the divisor for the division task, with the values and output conventions of Corollary 2. Here $p \geq 1$ denotes an integer target precision for a Newton routine.

Proof of Corollary 2. We use the classical Newton reductions as presented by Brent [5, Lemmas 2.1–2.4], with their tape costs made explicit. Set $B(p) = p(\lg p)^{1-\kappa}$. A linear scan handles a zero dividend or radicand and finds the significant bits of a positive input. For division write $b = 2^h c$, where $0 \leq h < n$ and $1 \leq c < 2$. For square root write $a = 2^{2e} c$, where $0 \leq e < n/2$ and $1 \leq c < 4$. Copy the normalized digits most significant first to a tape with a marked beginning. This initial scan also records the relevant exponent.

For an integer $t \geq 1$, let $R_t(z) = 2^{-t} \lfloor 2^t z \rfloor$ and put $\delta = 2^{-t}$. The prefix $c_t = R_t(c)$ has $0 \leq c - c_t < \delta$. Each product below is computed exactly on integer significands and then rounded by the indicated R_t ; additions and shifts are exact until a stated rounding. When precision increases, the stored dyadic iterate is extended by zero fractional bits.

For the reciprocal iteration $x \leftarrow x(2 - cx)$, compute

$$u = R_t(c_t x), \quad x^+ = R_t(x(2 - u)).$$

Put $\varepsilon = 1 - cx$. If $|\varepsilon| \leq 1/2$, then $0 < x \leq 3/2$. Writing $\eta_1 = u - c_t x$ and $\eta_2 = x^+ - x(2 - u)$ gives $|\eta_i| < \delta$ and

$$1 - cx^+ = \varepsilon^2 - c(c - c_t)x^2 + cx\eta_1 - c\eta_2.$$

Since $cx^2 = (1 - \varepsilon)^2/c \leq 9/4$, $cx \leq 3/2$, and $c < 2$,

$$|1 - cx^+| \leq \varepsilon^2 + \frac{23}{4}\delta < \varepsilon^2 + 6\delta. \quad (*)$$

The fixed seed $x = 5/8$ has $|\varepsilon| \leq 3/8$ throughout $1 \leq c < 2$. One step with $t = 6$ therefore gives residual less than $9/64 + 6/64 = 15/64 < 1/4$.

For a requested target $p \geq 1$, begin with $m = 1$ after this warm-up. While $m < p$, set $m' = \min\{2m, p\}$ and work at $t = m' + 5$ fractional bits. If the residual is at most 2^{-m-1} , then $(*)$ gives

$$|1 - cx^+| \leq 2^{-2m-2} + 6 \cdot 2^{-m'-5} \leq \frac{7}{16} 2^{-m'} < 2^{-m'-1}.$$

Set $m = m'$ and continue. The reciprocal routine at target p thus returns a dyadic x satisfying $|1 - cx| \leq 2^{-p-1}$ and $|x - 1/c| \leq 2^{-p-1}$. These errors are measured against the true c at every stage, although the prefix c_t changes.

For the root iteration $x \leftarrow (x + c/x)/2$, first suppose $1/2 \leq x \leq 5/2$. Normalize this stored dyadic exactly as $x = 2^j d$, where $1 \leq d < 2$ and $j \in \{-1, 0, 1\}$. The reciprocal routine at target t returns z with $|1 - dz| \leq 2^{-t-1}$. Put $v = 2^{-j}z$. Because $d \geq 1$ and $2^{-j} \leq 2$, we have $|v - 1/x| \leq 2^{-t} = \delta$. Compute the quotient and root update by

$$q_t = R_t(c_t v), \quad x^+ = R_t((x + q_t)/2).$$

Here $c_t < 4$ and $1/x \leq 2$, so the quotient has the uniform error

$$|q_t - c/x| \leq \delta + c_t |v - 1/x| + \frac{|c_t - c|}{x} \leq 7\delta.$$

The exact Newton identity gives

$$|x^+ - \sqrt{c}| \leq \frac{(x - \sqrt{c})^2}{2x} + \delta + \frac{1}{2}|q_t - c/x| \leq \frac{(x - \sqrt{c})^2}{2x} + \frac{9}{2}\delta.$$

The fixed seed $x = 3/2$ has $|x - \sqrt{c}| \leq 1/2$. At this seed the first term is at most $1/12$, so one step with $t = 6$ has error less than $1/12 + 5/64 < 1/4$. Thereafter the invariant $|x - \sqrt{c}| \leq 2^{-m-1}$, with $m \geq 1$, gives $3/4 \leq x \leq 9/4$ and hence supplies the denominator bound for the next step. Since $1/(2x) \leq 1$ and $9/2 < 6$, the same capped doubling schedule with five guard bits gives $|x - \sqrt{c}| \leq 2^{-p-1}$ at target p .

These operations use a fixed number of scalar registers. In a reciprocal step, $x \leq 3/2$ and $u \leq c_t x \leq 3/2$ whenever $(*)$ is used. In a root step, $x \leq 5/2$, $v \leq 2 + \delta < 3$, and $q_t < 12$. A fixed number of integer guard bits therefore suffices. At working precision t , each retained significand has $t + O(1)$ bits, including the fixed additional precision of the inner reciprocal, and each exact product has $2t + O(1)$ bits. Theorem 1 applies to the padded significands. The streaming conventions give $O(t)$ cost for additions, shifts, copying, truncation and head returns.

Each stage starts at the marked beginning of its normalized input, reads only the required prefix, and returns over that prefix. If the finite input ends first, the remaining fractional bits are zero. A root stage forms d once in $O(t)$ time; each stage of its inner reciprocal routine likewise reads only its own shorter prefix of d . Thus no early stage rescans an entire longer input.

For the capped doubling targets m_j ending at p , $\sum_j (m_j + 5) = O(p)$, and monotonicity of $\lg$ gives

$$\sum_j B(m_j + 5) \leq (\lg(p + 5))^{1-\kappa} \sum_j (m_j + 5) = O(B(p)).$$

The fixed warm-up adds constant cost. A reciprocal stage uses two products, so the complete reciprocal routine at target p costs $O(B(p))$. A root stage of working precision t uses a reciprocal at target t and one further product, and hence costs $O(B(t))$. The outer precision sum is again $O(B(p))$. The scalar scan costs sum to $O(p)$ at both levels.

The outer registers and the reused reciprocal and multiplier tapes form one fixed finite tape set with fixed subroutine depth. Marking the ends of each call's visited intervals on fixed tracks permits those intervals to be cleared and their heads returned in a constant multiple of the call's running time. No random-access simulation is used.

Take $P = 2n + 4$. At this target there are $O((\lg n)^2)$ outer and inner stage calls. Their $O(\lg n)$ -bit length and precision descriptors incur a fixed polynomial in $\lg n$ of total work, hence $O(n)$. Together with the initial normalization scan, this accounts for the fixed-tape implementation. For division, the reciprocal routine gives $|x - 1/c| \leq 2^{-P}$. Form $a2^{-h}x$ by an exact significand product and an exact shift of the binary point, retaining its fractional bits. Since $a < 2^n$,

$$\left| a2^{-h}x - \frac{a}{b} \right| \leq a2^{-h}2^{-P} < 2^{n-P} < \frac{1}{4}.$$

Its floor q_0 differs from $q = \lfloor a/b \rfloor$ by at most one. Compute $r_0 = a - bq_0$ exactly. If $r_0 < 0$, decrement q_0 and add b ; if $r_0 \geq b$, increment q_0 and subtract b . For square root, an approximation to $\sqrt{c}$ within 2^{-P} becomes an approximation to $\sqrt{a}$ within $2^{e-P} < 1/4$ after the exact shift by 2^e . Its floor s_0 differs from $\lfloor \sqrt{a} \rfloor$ by at most one. Decrement it if $s_0^2 > a$, and increment it if $(s_0 + 1)^2 \leq a$; otherwise retain it. The remainder and square tests cover exact quotients and perfect squares. All final significands, trial answers and exact test products have $O(n)$ bits; their computation costs $O(B(n))$, and output padding is linear. Leading zeroes affect neither normalization bound, since n is the supplied input length. $\square$

11 A consequence for matrix transposition

For positive integers n_1, n_2, b , a row-major $n_1 \times n_2$ array of b -bit strings stores entry (i, j) , where $0 \leq i < n_1$ and $0 \leq j < n_2$, at position $n_2i + j$; transposition places this string at position $n_1j + i$. Harvey and van der Hoeven prove that any multiplication machine with worst-case cost $M(m)$ yields a fixed multitape transposition machine with cost [16, Corollary 6.2]

$$O(M(m) + m \lg \lg \max(n_1, n_2)), \quad n_1 n_2 b \leq m.$$

The cited transposition interface receives m as well as n_1, n_2, b and the matrix stream [16, Section 1.1]. For the matrix problem defined here, compute $m = n_1 n_2 b$ from the three binary descriptors and supply all four parameters. The four descriptors have total length $O(\log(2m))$, and the two schoolbook products that compute m cost $O(\log^2(2m)) = O(m)$ steps.

The cited multiplication interface likewise receives m explicitly and two nonnegative operands smaller than 2^m ; it uses standard binary representations of the operands and product. On each such call, read the supplied m and form two words of exactly m bits, most significant bit first, by adding leading zeroes. Feed their $x\#y$ concatenation to Theorem 1. Its final output pass writes the $2m$ -bit product consecutively. Scan this block to remove leading zeroes, emitting one 0 bit if every bit is zero. This fixes a canonical binary representation of the product. If a tape orientation

requires reversal, read the stored word backward while writing it forward on a spare tape. Padding, normalization, and copying therefore cost $O(m)$ steps on a fixed number of tapes.

Suppose the multiplication bound is $O(m(\lg m)^{1-\kappa})$ for a fixed $0 < \kappa < 1$. Positivity of the dimensions and entry width gives $\max(n_1, n_2) \leq n_1 n_2 b \leq m$. Hence

$$\lg \lg \max(n_1, n_2) \leq \lg \lg m = o((\lg m)^{1-\kappa}).$$

The final relation uses $1 - \kappa > 0$; the bounded cases are absorbed using the stated convention for $\lg$. Thus the cited transposition cost is $O(m(\lg m)^{1-\kappa})$. Taking $m = n_1 n_2 b$ expresses this bound in matrix bits. In particular, an $n \times n$ binary matrix can be transposed in $O(n^2(\lg n)^{1-\kappa})$ time. Dimensions equal to one cause no exception, since our convention gives $\lg 1 = 1$.

A Packed additions on grouped rectangles

The simultaneous-layer construction uses the rectangular layout of Lemma 26. We record here a more general version in which a supplied descriptor lists groups with different spectator lengths. Each group is a complete rectangle, so the full rectangular algorithm, including its deterministic repair, can be applied to one group at a time. The only additional work is reading the group descriptors and passing consecutive groups through the same local tapes.

Definition 43 (Grouped packed stream). Let $M, R \geq 1$, $K \geq 6$, $f \geq 1$, $L = fK$, and $0 \leq \rho < K$. Fix an ordered template of $t \geq 3$ two-bit tags: 00 denotes a spectator field, and 01, 10, 11 denote x, y, z , respectively. Each chunk tag occurs exactly once. The template is common to all $J \geq 1$ groups of this input, but its length may vary between inputs. Group ν has a positive row count n_ν and a positive length $a_{\nu,h}$ at each spectator tag h ; at a chunk tag put $a_{\nu,h} = 2^L$. Define

$$Q_\nu = \prod_{h=1}^t a_{\nu,h}, \quad M_\nu = n_\nu Q_\nu, \quad C_\nu = \sum_{\mu < \nu} M_\mu, \quad V_\nu = M_\nu R,$$

and require $M = \sum_{\nu=1}^J M_\nu$.

The stream consists of M records of R bits. In group order, their addresses form the consecutive lexicographic rectangles

$$[n_\nu] \times \prod_{h=1}^t [a_{\nu,h}].$$

The first coordinate is a row ordinal. Every row contains the full Cartesian suffix displayed above: no suffix length depends on a coordinate within that row. The suffix lengths may differ between groups. Every finite sequence of per-row spectator length vectors in the fixed template is representable, using groups of count one when necessary.

Use the integer code Γ from (1). The supplied descriptor is exactly

$$\begin{aligned} \mathcal{D} = & \Gamma(M)\Gamma(R)\Gamma(K)\Gamma(f)\Gamma(\rho+1)\Gamma(t) T_1 \cdots T_t \Gamma(J) \\ & \cdot \prod_{\nu=1}^J \left(\Gamma(n_\nu) \prod_{h:T_h=00} \Gamma(a_{\nu,h}) \right), \end{aligned}$$

where T_h is the tag at position h , and juxtaposition and products mean bitstring concatenation in increasing index order. The header, tag count, and group count delimit the entire list. A grouped packed stream is valid when this descriptor and its accompanying records have exactly the parameters, shapes, and orders just specified.

Lemma 44 (Packed selected-bit addition for grouped rectangles). *Let a valid grouped packed stream of Definition 43 be supplied, and let $A \geq 2$ bound the actual total bit length of its descriptor $\mathcal{D}$, including the header and the entry for R . At the positions $j_i = \rho + iK$, numbered from the least significant bit, one fixed multitape procedure performs*

$$y_{j_i} \longleftarrow y_{j_i} \oplus x_{j_i} \quad (0 \leq i < f)$$

in every group. It fixes the group, row ordinal, spectator coordinates, all other chunk bits, and every record payload; in particular, the chunk z is restored. For $V = MR$, its time is

$$O\left(V(L^\tau + 1) + MA^3 + M \min\{1, 80(f-1)2^{-K}\} A(R+A)\right). \quad (55)$$

When $A = O(p)$, R exceeds every fixed polynomial in p , $f \leq d \leq p$, and $K/\log p \rightarrow \infty$, this is $O(V(L^\tau + 1))$, uniformly over the group counts, shapes, and payloads whose supplied descriptors satisfy that bound.

Proof. Write $s = t - 3$ for the number of spectator tags. Each group explicitly stores $s + 1$ positive integer codes, so $J(s + 1) \leq A$ and $Jt = J(s + 3) \leq 3A$. Also $J \leq M$, because every group is nonempty. The header includes $\Gamma(M)$ and $\Gamma(R)$, so $\log_2 M + \log_2 R = O(A)$. Every field length, partial suffix product, row count and group record count is at most M . These integers, as well as $V_\nu = M_\nu R$, therefore have $O(A)$ bits. In particular,

$$A_\nu := \lceil \log_2(2V_\nu) \rceil = O(A)$$

with a constant independent of the number and shapes of the groups.

Parse the descriptor in forward group order, keeping the common template on a separate tape and saving the current group's length vector locally. Collapse the row ordinal and the spectator fields before the first chunk into one prefix field. Collapse each spectator interval between the chunks and after the last chunk in the same way; an empty interval has length one. The resulting four positive lengths N_0, N_1, N_2, N_3 , together with the current order of the names x, y, z , give exactly the rectangle in Lemma 26, with M_ν records and volume V_ν . Fixing a collapsed coordinate fixes every one of its mixed-radix factors. Hence the rectangular algorithm preserves the row ordinal and all original spectator coordinates, as the grouped claim requires.

For each group, reading the template and length vector, forming the four products, and writing the local descriptor with its fixed number of entries take $O(A^3)$ time: there are $O(A)$ field visits and schoolbook operations, each on $O(A)$ -bit integers. The descriptor supplied to the rectangular routine contains only its four collapsed lengths and R, K, f, ρ , in canonical binary, with the finite permutation of the three chunk names. The routine does not receive or rescan the outer group table.

Use separate outer input and output tapes with forward cursors. Copy the V_ν bits of group ν into a local area, apply the complete rectangular algorithm there, append its output to the outer output tape, and clear the visited local area. Keep the outer cursors at their group boundaries during the local call. Reuse the same local and primitive work tapes for every group. The copy, append, positioning and cleanup cost $O(V_\nu)$; one final outer rewind costs $O(V)$. The inner routine restores its local work tapes as part of its stated bound. Thus the tape count is fixed, and no local call scans an earlier or later group.

Put $\delta_{\text{bad}} = \min\{1, 80(f-1)2^{-K}\}$. Within group ν , Lemma 26 performs all the required XORs, restores z for every initial value, and preserves each payload. Its exceptional records are extracted,

sorted and reinserted entirely within that group. Their destination ranks therefore require only the local A_ν -bit address bound; no ordering across groups is needed. The local time is

$$O\left(V_\nu(L^\tau + 1) + M_\nu A_\nu^3 + \delta_{\text{bad}} M_\nu A_\nu(R + A_\nu)\right).$$

Summing these bounds and the descriptor work, using $\sum_\nu M_\nu = M$, $\sum_\nu V_\nu = V$, $A_\nu = O(A)$ and $J \leq M$, gives

$$O\left(V(L^\tau + 1) + MA^3 + \delta_{\text{bad}} MA(R + A)\right),$$

which is (55). This sum includes the whole record payload on every local exceptional-sort pass. The rectangular lemma also covers $f = 1$, when its rotations and repair are empty.

After division by $V = MR$, the two additional terms are bounded by A^3/R and $80(f - 1)2^{-K}A(1 + A/R)$. Under the displayed family bounds, the first tends to zero, and the second is $O(p^2 2^{-K}(1 + A/R)) = o(1)$. These bounds depend on the supplied groups only through M, R, A , proving the asserted uniformity. $\square$

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